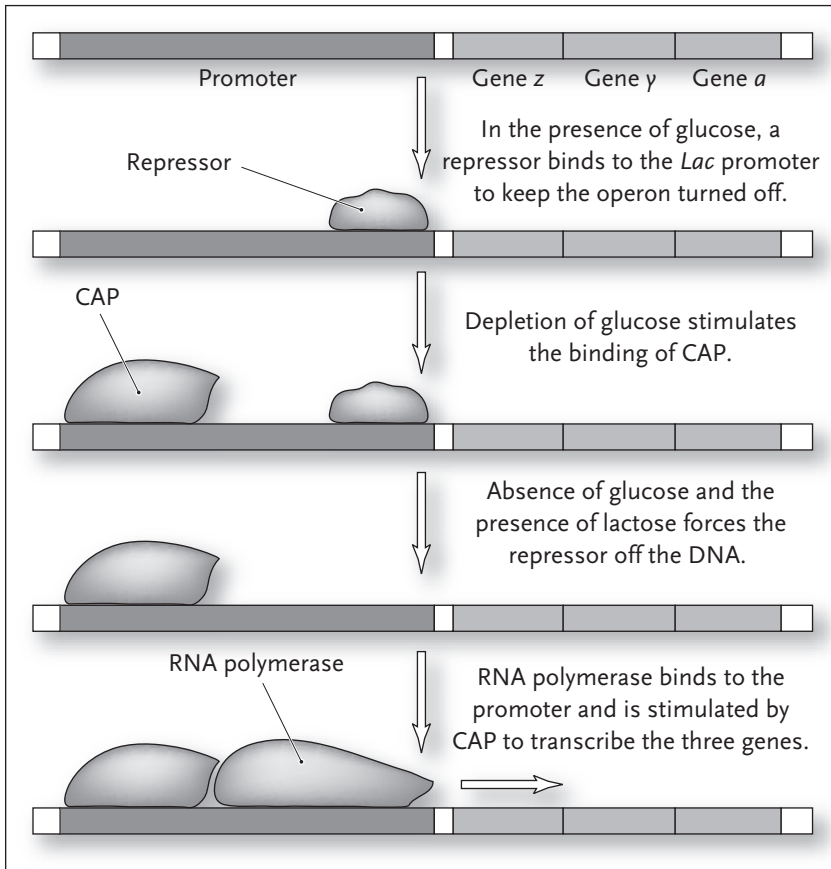


that cause neurological disorders, cardiovascular disease, diabetes, metabolic disorders, and aging.

Turning Genes On and Off

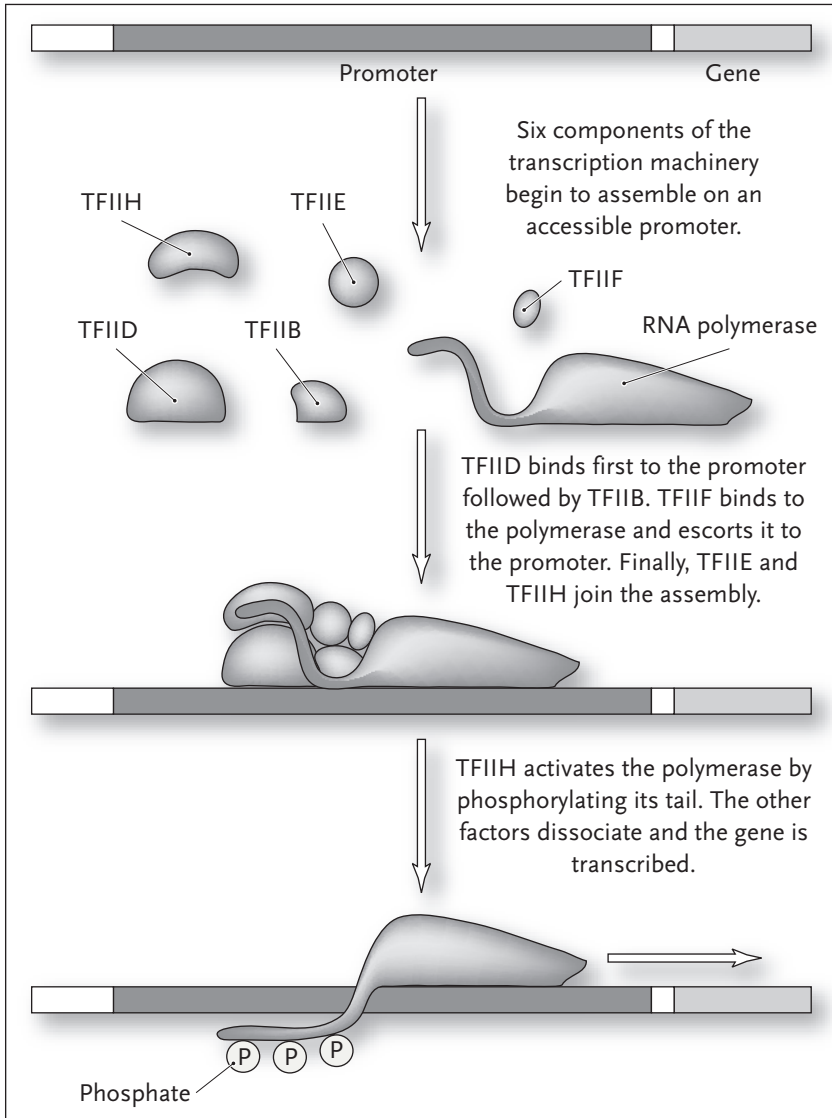
All genes come equipped with a controlling region, called the promoter, that serves the same function as a light switch and provides the binding site for the RNA polymerase that transcribes the gene into RNA. The switching function of the promoter is regulated by other proteins, known generally as transcription factors, that bind to the promoter in a way that either blocks the polymerase (the “off” position) or activates it (the “on” position). In some cases, two or more genes are controlled by the same promoter, so when the promoter switches on, all the genes are activated simultaneously (i.e., transcribed into RNA). This arrangement, in which a single promoter is linked to two or more genes, is known as an operon, and it is very common among prokaryotes. Indeed, it was the study of the bacterial *Lac* operon that gave geneticists their first insights into the control of gene expression. The *Lac* operon has three structural genes that code for proteins needed for the import and processing of the sugar lactose, from which this operon gets its name. By convention, these genes are known as *z*, *y*, and *a*. The *z* gene codes for β -galactosidase, an enzyme that hydrolyzes lactose to galactose and glucose. The *y* gene codes for a permease, which facilitates the entry of lactose into the bacteria, and the *a* gene codes for a transacetylase, which metabolizes lactoselike compounds.

A simplified scheme for controlling the *Lac* operon is shown in the figure on page 85. A repressor protein binds to the promoter, in a region called the operator, thus preventing the binding of the polymerase. This happens whenever glucose, the preferred sugar, is available. As long as the repressor is bound to the operator, the operon is turned off. Depletion of glucose stimulates the binding of a protein called catabolite activator protein (CAP) to another area of the promoter, called the regulator, or CAP site. If lactose happens to be present when the glucose is depleted, the repressor is forced off the promoter, allowing the binding of the polymerase. Once bound, CAP activates the polymerase and the *Lac* operon is transcribed.



The *Lac* operon. This operon consists of three structural genes (*z*, *y*, and *a*) under the control of a single promoter. The repressor binds to a portion of the promoter, called the operator. The catabolite activator protein (CAP) binds at the opposite end of the promoter, in a region known as the regulator, or CAP site. The *Lac* operon is designed to stay off as long as glucose is present.

The control of gene expression in eukaryotes is more complex, but the basic logic remains the same. In this case, control occurs in three steps. First, special proteins expose the gene's promoter; second, transcription factors assemble on the promoter; and third, regulatory proteins activate the transcriptional machinery. Eukaryote regulatory proteins belong to three families represented by helix-loop-helix, zinc finger, and leucine zipper proteins. The various members of these families are able to



The control of eukaryote gene expression. RNA polymerase and five transcription factors assemble on an exposed promoter. Once these components are in place, transcription factor IIF (TFIIF) is activated by a eukaryote-specific enhancer, analogous to the prokaryote CAP. In this case, the enhancer is located some distance—sometimes thousands of base pairs—upstream (i.e., to the left) from the polymerase. Only a portion of the gene is shown.

recognize specific DNA sequences that are associated with the promoter of every gene. The interaction between the regulatory protein and the DNA is not passive. Rather than simply binding to the DNA, the protein changes the conformation of the local neighborhood by twisting the DNA into a more opened helical structure. The conformational change that the DNA undergoes exposes the promoter to the transcription machinery. This process is complex and not fully understood, but is known to involve a relaxation of the DNA helix as well as alterations to the local chromatin conformation.

After the promoter is exposed, six components of the transcription machinery, including RNA polymerase and five transcription factors, quickly assemble. Once the transcription machinery is assembled, the regulatory protein activates a transcription factor, called IIH, which in turn activates the polymerase to permit transcription of the gene.

For a prokaryote, the decision to turn a particular gene on or off is relatively straightforward, and usually involves signals that it receives directly from the environment. If we could read the mind of a bacterium, its thought processes might go something like this:

Is there any glucose around? If not, is there any lactose? If there is plenty of glucose on hand, keep the Lac operon off, because there is no point in running it otherwise. If we run out of glucose but detect lactose, turn the glucose operon off and the Lac operon on. If we run out of glucose and lactose, activate auxiliary operons that are capable of processing other sugars, such as maltose or arabinose. If there are no sugars around, activate the genes responsible for processing proteins, fats, and anything else that carries calories.

The control of gene expression in eukaryotes generally adheres to the logic established by the prokaryotes. However, in eukaryotes, the details of the process are more involved, requiring a greater number of steps and molecular participants. Eukaryotes are much more careful about controlling spurious and inappropriate gene expression. This is particularly true for multicellular eukaryotes, for which sloppy control of gene expression can lead to cancerous growths. The desire for precise control can be seen in the number of factors that are needed just to establish a transcription crew in eukaryotes as opposed to prokaryotes (see figure on page 86). A process that has many components automatically provides many pathways for controlling that process. This is

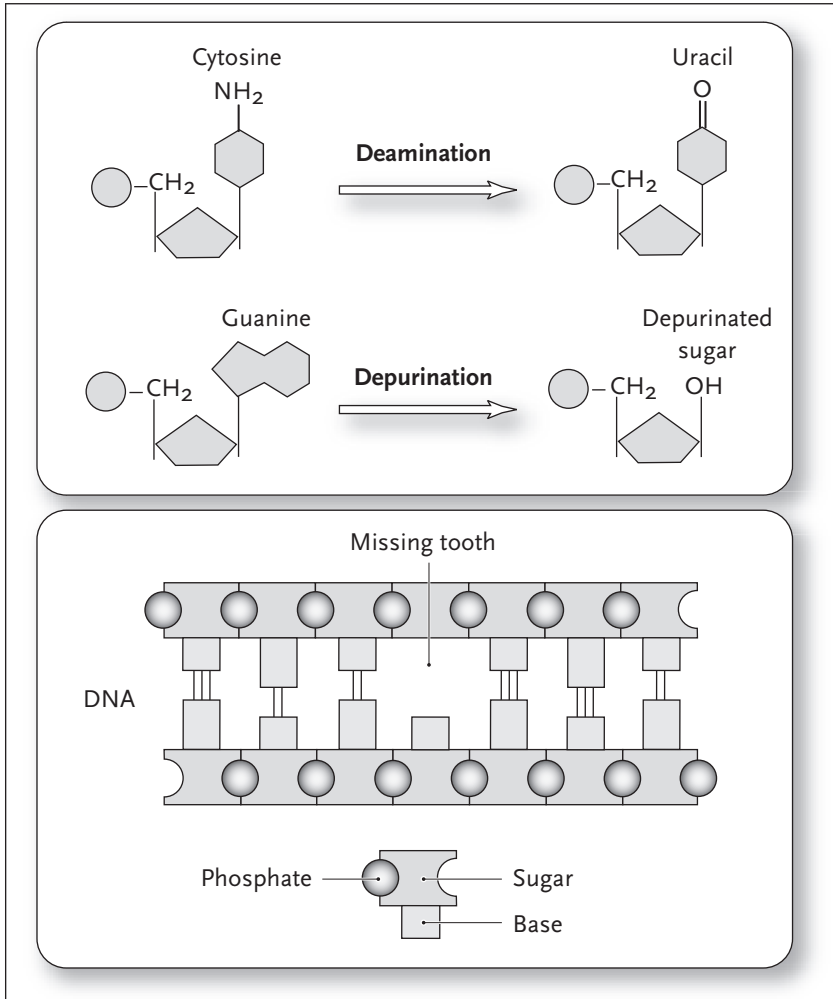
critical for multicellular eukaryotes, all of which are absolutely dependent on careful regulation of gene expression.

Gene Monitors and Repair Crews

DNA is a very stable molecule, but it is not immutable. Every day, in a typical human cell, thousands of nucleotides are being damaged by spontaneous chemical events, environmental pollutants, and radiation. In many cases, it takes only a single defective nucleotide within the coding region of a gene to produce an inactive, mutant protein. The most common forms of DNA damage are depurination and deamination. Depurination is the loss of a purine base (guanine or adenine) resulting in a gap in the DNA sequence, referred to as a “missing tooth.” Deamination converts cytosine to uracil, a base that is normally found only in RNA.

It has been estimated that about 5,000 purines are lost from each human cell every day, and that over the same time period, 100 cytosines are deaminated per cell. Depurination and deamination produce a great deal of damage, and in either case, the daughter strand ends up with a missing nucleotide, and possibly a mutated gene, as the DNA replication machinery simply bypasses the uracil or the missing tooth. If left unrepaired, the mutated genes will be passed on to all daughter cells, with catastrophic consequences for the organism as a whole.

DNA damage caused by depurination is repaired by special nuclear proteins that detect the missing tooth, excise about 10 nucleotides on either side of the damage, and then, using the complementary strand as a guide, reconstruct the strand correctly. Deamination is dealt with by a remarkable group of DNA repair enzymes known as base-flippers. These enzymes monitor the DNA one nucleotide at a time. After binding to a nucleotide, a base-flipper breaks the hydrogen bonds holding the nucleotide to its complementary partner. It then performs the maneuver for which it gets its name. Holding on to the nucleotide, it rotates the base a full 360 degrees, inspects it carefully, and, if it detects any damage, cuts the base out and discards it. In this case, the base-flipper leaves the final repair to the missing-tooth crew that detects and repairs the gap as described previously. If the nucleotide is normal, the base-flipper rotates it back into place and reseals the hydrogen bonds.



DNA damage from deamination and depurination. Deamination, or loss of an amino group (NH_2), converts cytosine to uracil. Depurination results in the loss of a purine nucleotide (top panel), leaving a gap, or “missing tooth,” in the DNA molecule (bottom panel). Both types of damage can lead to a catastrophic buildup of mutations if not repaired.

Base-flippers and the missing-tooth crew work 24 hours a day, seven days a week. Scientists have estimated that these crews inspect and repair the entire genome of a typical human cell in less than 24 hours.

Jumping Genes

When first proposed by the American geneticist Barbara McClintock in 1951, the idea that genes could move, or jump, from one location in the genome to some other location was greeted with disbelief and disdain. For more than 20 years, this idea languished in a kind of scientific limbo until the advent of recombinant technology made it possible to prove the existence of these wandering genes, formally known as transposons or transposable elements. By the 1980s, McClintock's work was finally given the recognition it deserved, and in 1983, at the age of 81, she was awarded the Nobel Prize in medicine. She died on September 2, 1992.

McClintock's work provided several insights into the organization and evolution of the eukaryote genome: First, the position of some genes, within the genome, is flexible; second, the roles of transposable elements, introns, and intervening sequences are interconnected; and third, viruses are direct descendants of jumping genes. With the completion of the genome project, we now know that many of our genes were once transposable elements, including the 220 genes obtained by horizontal transfer from bacteria.

Insertional mutagenesis, discussed earlier in the chapter, is the connection between a jumping gene and an intervening sequence. As mentioned, only 1.0 percent of the human genome contains genes, with the rest of the DNA consisting of intervening sequences and introns. Our genome has evolved into a form that accommodates mobile genes. In such a genome, the odds of a transposable element damaging an existing gene are extremely small. When a jumping gene moves, it will, in all likelihood, reinsert into an intervening sequence, a region that lacks genes, where it may stay for thousands or millions of years. After having been duplicated, most transposons move again, thus producing many copies of the same gene, which are sprinkled around the genome. These copies of the original gene are then free to mutate into other genes that may eventually become useful to the organism. Thus copies of the original transposon are the source of many of the genes now present in the human genome.

The flexibility of a transposable genome is perhaps the single most important characteristic that led to the explosive adaptability of eukaryotes and the many life forms, especially the multicellular creatures, they

produced. Such a genome is also important to modern medical therapies, such as gene therapy, which attempts to cure a disease by introducing a normal gene into the patient's genome. If our genome were organized like that of the prokaryotes, such a therapy would be nearly impossible.

Transposable elements have given us a flexible and extremely powerful genomic organization, but they have also given us the viruses and all the illnesses they produce, such as AIDS, polio, and the common cold. Somehow, millions of years ago, a jumping gene learned how to jump right out of the cell. It acquired this ability in small steps as it moved from one place in the genome to another. A jumping gene that moves usually reinserts into an intervening sequence, devoid of other genes; occasionally, a transposon reinserts next to a gene, possibly one that codes for a protein that could serve as a capsid, a protein that protects a virus's genome, and forms the overall structure of the viral particle. The next time such a transposon moved, it would take a copy of the potential capsid gene with it. Eventually, by moving from place to place, the transposon would have collected a large number of genes that not only made it possible for it to escape from the cell but also gave it the power to reinfect other cells. When that happened, a simple mobile genetic element went from being a molecular curiosity to a living thing, equipped with a life cycle and the power of reproduction.

.6.

FROM CELLS TO BODIES

Eukaryotes seem to have been destined for multicellularity. Not that they fared poorly as single cells; quite to the contrary, the phylogenetic class protista is one of the most successful and diverse groups of organisms around. But becoming multicellular requires a complex communication network backed up by far more genes and proteins than a prokaryote can muster. Prokaryotes did, however, lay the groundwork and formed the first, though temporary, association of cells in the form of fruiting bodies containing the spores of many individual cells.

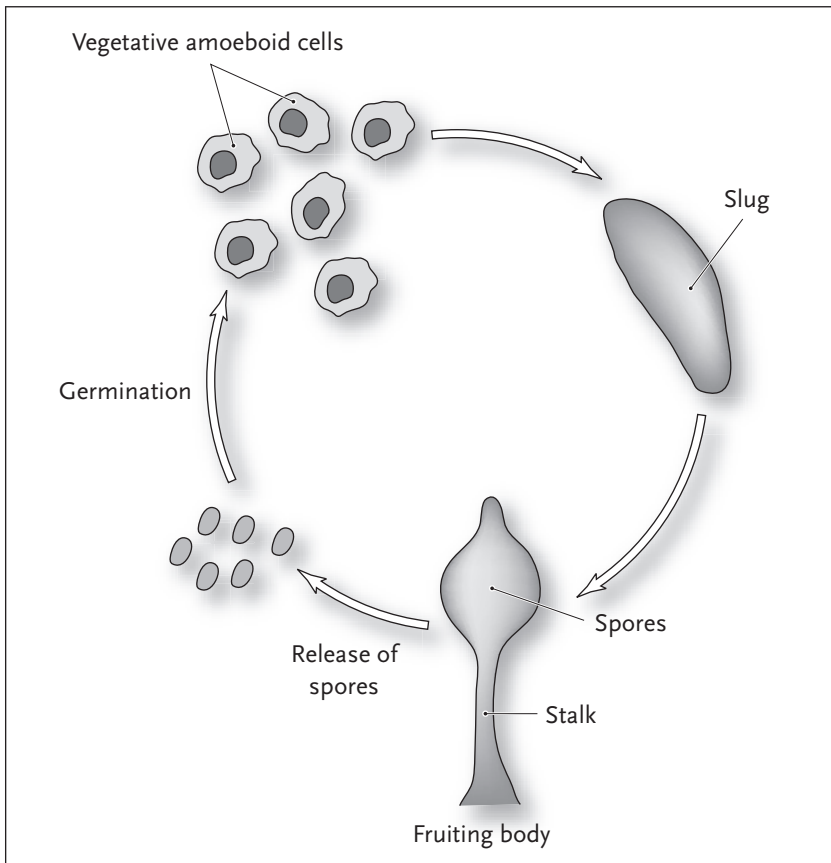
The Road to Multicellular Creatures

Myxobacteria, discussed in chapter 2, were the first cells to form brief multicellular associations for the purpose of reproduction. These prokaryotes spend most of their time as free-living vegetative cells. Starvation is the trigger that leads to the aggregation of these cells into a sporulating fruiting body. As spores, the cells can hibernate until conditions improve, and in addition, wind currents may carry them to other, more favorable locations, where they can germinate with the hope of finding food. This type of multicellular colonial behavior was refined by eukaryotes such as the amoeba *Dictyostelium discoideum* and the green algae *Volvox*. Colonial forms eventually gave rise to true multicellular organisms such as sea sponges and sea cucumbers, followed by higher organisms, both aquatic and terrestrial (see table on page 93).

Dictyostelium, a social amoeba that feeds on bacteria among the leaf litter of forest floors, is the eukaryote version of the myxobacterial life cycle. In this case, the formation of the fruiting body is preceded by the

PROGRESSIVE STAGES OF MULTICELLULARITY	
Stage	Description
Cellular slugs	Temporary associations of identical cells, cooperating for a common purpose. The only known example among eukaryotes is <i>Dictyostelium</i> .
Cellular colonies	Permanent associations among identical cells. The colonies are capable of producing gametes (sperm and eggs). Two examples are <i>Gonium</i> and <i>Volvox</i> .
Sponges	Aquatic creatures constructed from different types of cells, including gametes, that differentiate during development. The cells are not organized into tissues and there is no nervous system.
Cnidarians	Aquatic creatures that include the jellyfish, sea anemones, hydroids, and corals. In these organisms are the first appearance of tissues, a rudimentary nervous system, a mouth, and digestive tract. These animals are constructed from only eight cell types: epithelial, muscle, neural, glandular, interstitial, gonadal, mesenchyme, and the cnidoblast, or stinging cell.
Mollusks	Organisms that include clams, octopoids, and snails, some of which are terrestrial. These animals have a well-developed mouth and digestive tract. In addition, some, like the octopus, have a highly developed central nervous system.
Echinoderms	Aquatic animals that include sea stars, sea urchins, and sea cucumbers. All have well-developed tissues, including a digestive tract and nervous system.
Cartilaginous fish	Fish having a full set of organs and tissues but lacking a true skeleton. This group includes sharks and rays.
Vertebrates	A subphylum of animals found in the water, on land, and in the air. They include bony fishes, amphibians, reptiles, birds, and mammals. Organs and tissue systems are refined to an extraordinary degree, allowing reptiles, birds, and mammals to colonize the land.

formation of a migrating slug, consisting of thousands of individual *Dictyostelium* cells. It is the slug phase that distinguishes this life cycle from that of the myxobacteria. The slug behaves like a single entity with a common purpose. It moves among the leaf litter as though it were a real slug or worm, testing the environment for a suitable place to form a fruiting body. Usually, the slug tries to migrate as far above the leaf litter as possible. In this way, the fruiting body gets maximum exposure to



Life cycle of *Dictyostelium*. Vegetative amoeboid cells feed on bacteria in the soil. When food runs low, the cells aggregate to form a migrating slug eventually transformed into a fruiting body containing spores. Air currents spread the spores to new locations, and if conditions are favorable, the spores germinate into vegetative cells.

wind currents, thus improving the chance that the spores will be carried to a new location.

As with myxobacteria, the main stimulus for slug formation is starvation. When the individual cells run out of food, they release a chemical called cyclic adenosine monophosphate (cAMP) that serves as a call for all the *Dictyostelium* cells in the immediate neighborhood to aggregate. The hungriest cells answer the call first, forming the headend of the slug. The not-so-hungry cells also answer the call, but they take their time in getting there. As a consequence, arriving late, they form the posterior of the slug. When the slug finds a suitable location, the anterior portion anchors to the substratum, and all the cells in this region form the stalk, while the cells at the posterior form the spores. The cells forming the stalk are doomed; they die soon after the fruiting body is formed. There is an interesting logic associated with this arrangement. Stalk cells are the hungry ones, and therefore they do not have the reserves necessary to make viable spores. The hungry cells, in effect, are being sacrificed for the good of the community as a whole. It is also interesting to note that this early form of multicellularity is associated with such complex behavior and the principle of cooperation for the good of the community.

The green algae, represented by *Gonium* and *Volvox*, produced the first permanent cellular colonies. *Gonium* colonies are small, concave discs, consisting of about 32 cells, whereas *Volvox*, resembling a green glass ball, has up to 50,000 cells. *Volvox* colonies have cytoplasmic bridges that connect the cells together, allowing them to communicate with each other and to share nutrients. Each *Volvox* cell has a single flagellum, the beating of which is coordinated to propel the entire colony through the water like a rolling ball. The various cells within the colony seem to have specialized functions, as they cannot survive if the colony is disrupted. A small number of the cells are specialized for reproduction and serve as precursors for new colonies.

Although *Volvox* cells appear to be specialized, they are all essentially the same kind of cell, looking very much like *Chlamydomonas*, the unicellular green algae that is believed to have given rise to the colonial forms. The next stage in the development of multicellularity came with the appearance of the sea sponges, animals that are constructed from several distinct cell types. These simple animals produce gametes that

fuse to form an embryo. Development of the embryo is associated with cellular differentiation by mechanisms that have been retained by higher animals. Sponges, though multicellular, do not have tissues or a nervous system.

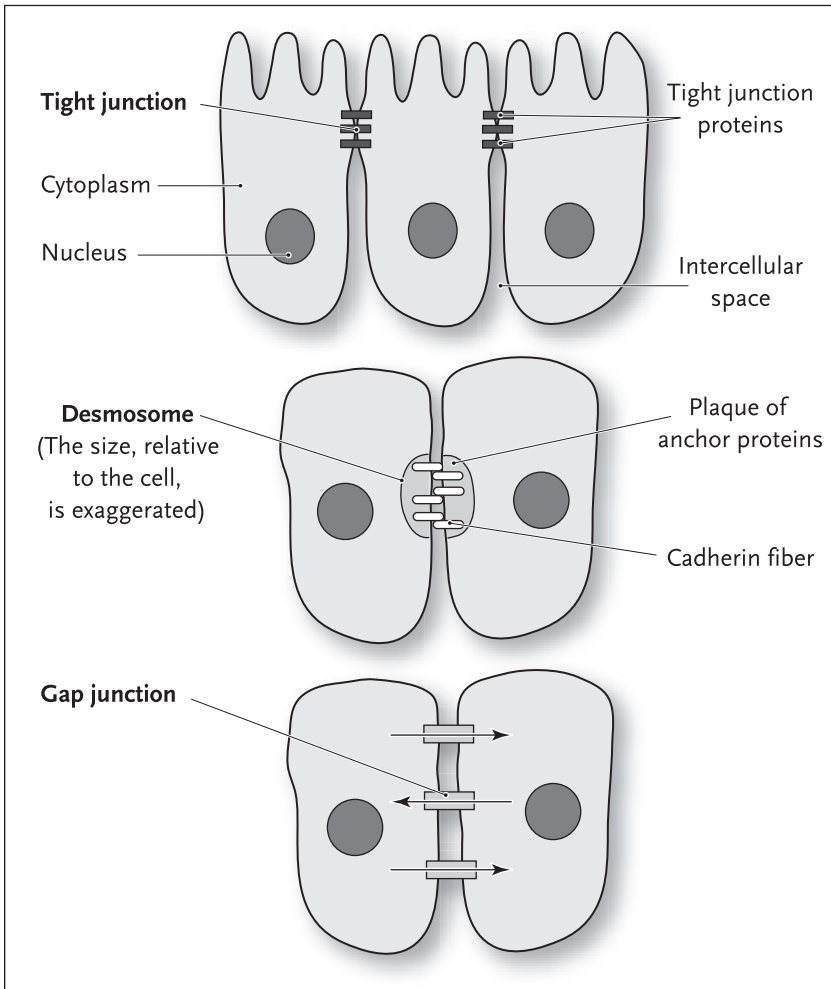
A true nervous system made its first appearance with the evolution of the cnidarians (e.g., jellyfish and corals), animals that are made from only eight cell types but which possess a mouth, digestive tract, and tissues, such as epithelia and muscles. These tissues and organ systems were refined by the mollusks (snails and octopods) and echinoderms (sea stars and cucumbers), reaching an exquisite level of sophistication with the appearance of the vertebrates.

Cell Junctions

A *Volvox* colony is held together with simple cytoplasmic bridges. True multicellular creatures have evolved more sophisticated structures, called cell junctions, for holding the tissues and organs together. Three of the most important of these are tight junctions, desmosomes, and gap junctions.

TIGHT JUNCTIONS These junctions, found primarily in the epithelial lining of an animal's digestive tract, stitch two cells together so tightly that the intercellular space, at the site of the junction, is obliterated. This junction not only keeps the cells together but also performs a very important secondary function of blocking the movement of bacteria and small molecules from the lumen (interior) of the gut into the intercellular space. In other words, tight junctions make the gut leak-proof. Nutrients obtained from the food we eat must pass through the gut's epithelial lining, where they are screened before being allowed to enter general circulation. Without the tight junctions, unwanted chemicals, possibly of a toxic nature, would be absorbed. In addition, the millions of bacteria that inhabit an animal's intestinal tract would quickly spread throughout the body, leading to a fatal infection.

DESMOSOMES Most of the cells in an animal's body are held together by desmosomes, junctions that are like tiny organic rivets. A protein plaque, analogous to a rivet head, is located beneath the



Cell junctions. Tight junctions are constructed from membrane proteins that stitch the two cells together very tightly. Desmosomes, constructed from plaques and protein fibers called cadherins, hold the cells together but allow material to pass through the intercellular space. The gap junction is a hollow tube formed from six identical proteins called connexins (not shown). These junctions hold the cells together while allowing small molecules to pass freely between the cells, as indicated by the arrows. The “gap” refers to the large intercellular space.

membranes of adjacent cells. Another protein, called cadherin, projects from the plaques into the intercellular space where they form chemical bonds with each other, thus holding the two cells together. This junction

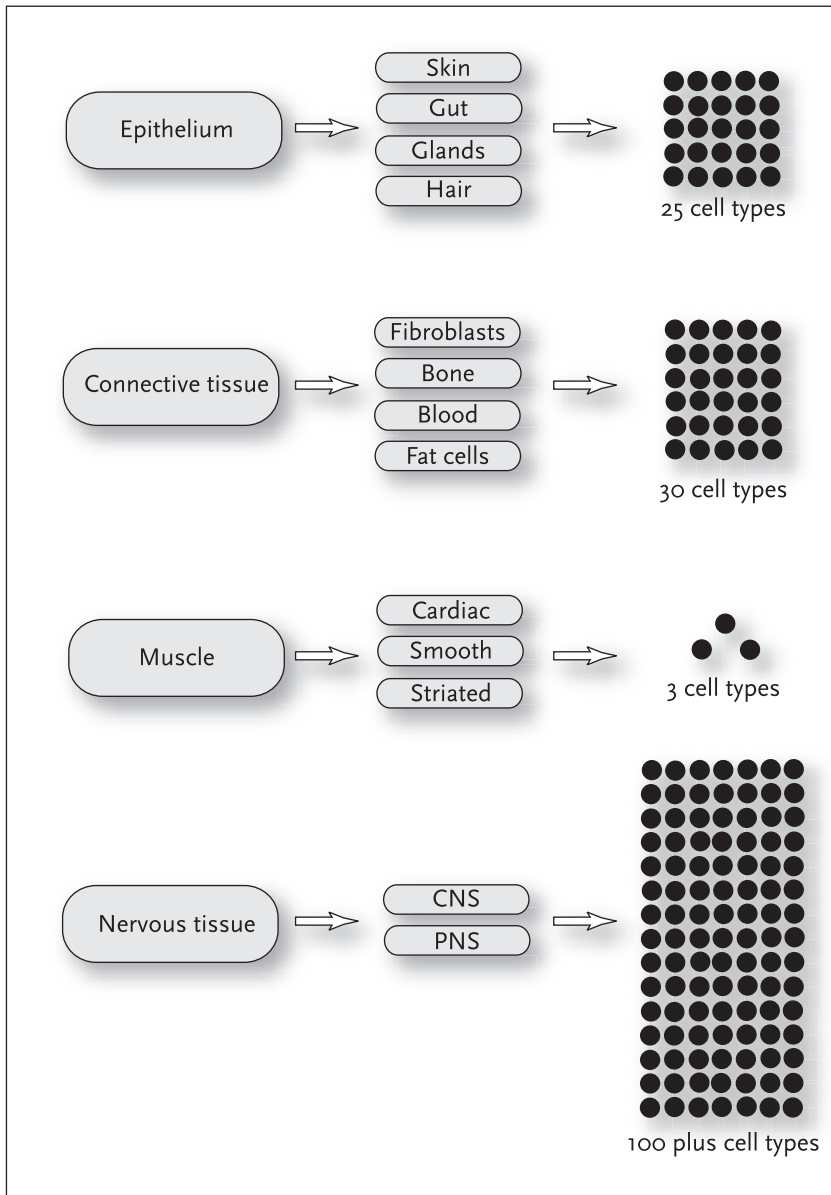
is more relaxed than a tight junction, and molecules are free to diffuse throughout the intercellular space.

GAP JUNCTIONS Cells need to be anchored to each other. In many cases, they also need to exchange fluids. The cytoplasmic bridges found in *Volvox* serve both these functions. Gap junctions, consisting primarily of hollow protein tubes, provide both an anchor and a fluid conduit in higher organisms. The sharing of cytoplasm is especially important in cardiac muscle, where the contraction of each myocyte is coordinated by the movement of ions through gap junctions.

Cells in the Human Body

The evolution of multicellular creatures was accompanied by the diversification of cell types. The simplest multicellular animals, such as the sponges and cnidarians, are made from fewer than a dozen cell types, whereas humans are constructed from more than 200 different kinds of cells. These cells represent four primary tissues: epithelium, connective tissue, muscle, and nervous tissue. The primary tissues give rise to all the organs and fluids of the body. For example, epithelium produces skin, gut, glands, and hair, whereas connective tissue gives rise to fibroblasts, bone, blood, and fat cells. Each of the primary tissues is constructed from many different cell types. Based on cell variety alone, the most complex primary tissue is neural, consisting of more than 100 different kinds of neurons.

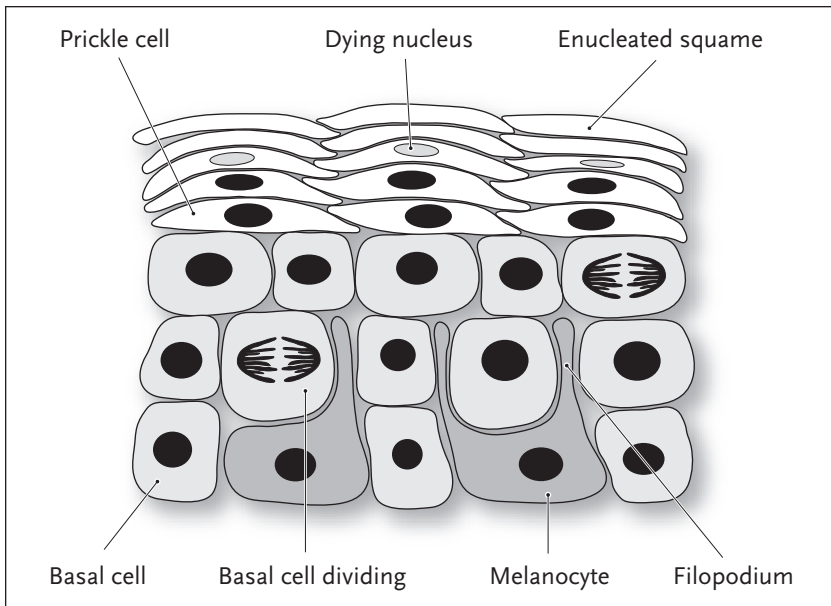
The many kinds of cells that make up an animal's body differ both in structure and function. This is not to say that a muscle cell, for example, contains organelles different from a neuron or liver cell; these cells are all eukaryotes and therefore have the same basic composition as shown previously. However, the external shape, the number of organelles, and even the DNA content vary widely among the differing cell types. To illustrate the extraordinary variety of cells that make up the human body, this section will describe examples of epithelium and muscle. Examples of connective tissue (cells of the immune system) will be described in the next section, while nervous tissue (neurons) is covered in the following chapter.



Cells of the human body. There are four primary tissues in the body: epithelium, connective tissue, muscle tissue, and neural tissue. These tissues, representing more than 200 cell types, give rise to the body's structures, blood, and other fluids. The greatest diversity of cell types is found among the tissues of the central nervous system (CNS) and peripheral nervous system (PNS).

EPITHELIUM

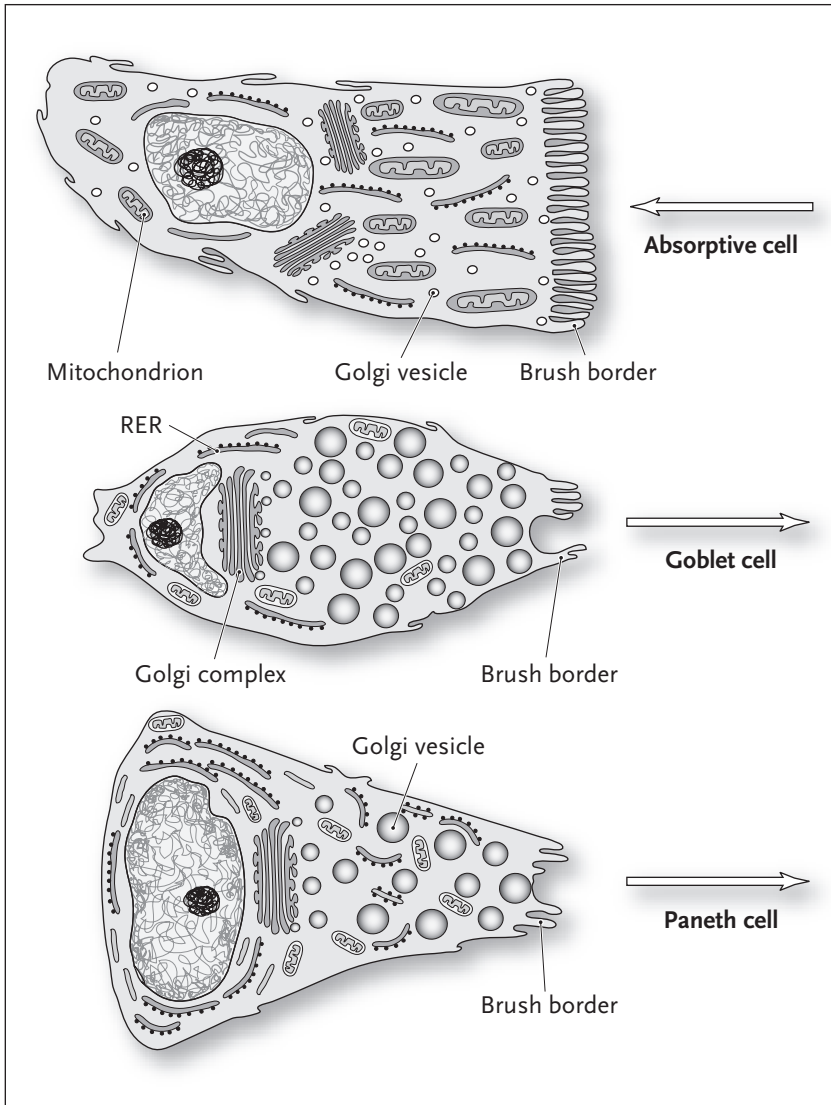
Our skin, made from epithelial cells, protects us from dehydration and microbial infections. It also contains sweat glands to help cool the body, sebaceous glands that oil the skin, and hair follicles. The barrier against microbes is produced by ion pumps, located in the membranes of every epithelial cell, that deposit hydrogen ions on the surface of the skin, producing an acid mantle capable of inhibiting the growth of most bacteria. Human epidermis is a stratified epithelium that consists of two populations of squamous cells: the outermost enucleated squames and



Structure of the epidermis. The epidermis is a stratified epithelium that forms the outer layer of the skin. It consists of three cell populations: squamous cells (the enucleated squames and the prickle cells), the cuboidal basal cells, and at the deepest layers, pigment-containing melanocytes. The basal and squamous cell layers are in a constant state of change. Division of a basal cell is followed by keratinization, a process by which the daughter cells are transformed into prickle cells. Keratin is a tough protein that makes the outer cell layer resistant to abrasion. In the final stage of keratinization, the prickle cell loses its nucleus. The now dead and fully keratinized squame eventually flakes off from the surface. Keratinized squames from the scalp are called dandruff.

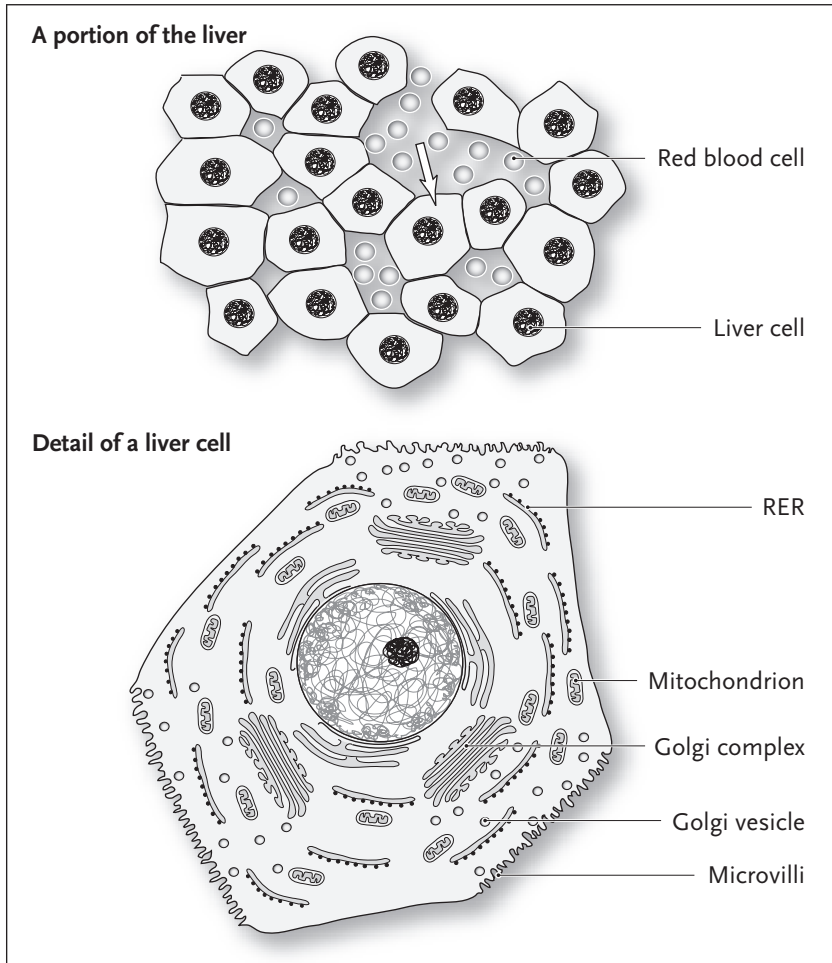
the nucleated prickly cells. Just beneath the squamous cells are the cuboidal basal cells, and at the deepest layers are pigment-containing melanocytes that give the skin color and protect the outer cell layers from the damaging effects of ultraviolet radiation. Melanocytes have long, thin finger-like projections, called filipodia, that work their way in between the basal cells. The basal and squamous cell layers are in a constant state of change. Division of a basal cell is followed by a process called keratinization, by which the daughter cells are transformed into prickly cells. Keratin is a tough protein that makes the outer cell layer resistant to abrasion. In the final stage of keratinization, the prickly cell loses its nucleus and the now dead and fully keratinized squame eventually flakes off from the surface. Keratinized squames from the scalp are called dandruff.

The digestive tract is made from different kinds of epithelial cells that are specialized for absorbing nutrients from the food we eat and for secreting large quantities of proteins, lipoproteins, and glycoproteins needed both for digestion and to help control the enormous population of intestinal bacteria. These cells have evolved a shape that reflects the directionality of their activity. That is, they either absorb material from the intestinal tract or secrete material into the lumen of the tract. Consequently, these cells have an elongated shape, with one surface helping to form the lumen of the gut. This surface is the only one involved in absorption or secretion. The absorptive, goblet, and paneth cells of the small intestine are striking examples of epithelium. The *absorptive cells* (also known as brush-border cells) absorb nutrients from the intestine along one surface, covered in microvilli (the brush border), that projects into the lumen of the tract. The cytoplasm of these cells is highly polar, with its large population of mitochondria and extensive rough endoplasmic reticulum (ER) localized near the absorptive surface. The mitochondria contain the enzymatic machinery needed to process the incoming nutrients for the final production of ATP (for more on this process, see chapters 2 and 3). These cells, being specialized for absorption, do not have an extensive Golgi complex. *Goblet cells* are devoted to the secretion of mucus into the gut, which serves to lubricate the walls of the intestine, improving absorption and the movement of the gut contents. The cytoplasm of these cells has an extensive RER and a single large Golgi complex, aimed directly at the brush border, that releases



Epithelium of the digestive tract. Essential molecules are extracted from the food we eat by absorptive cells, equipped with many mitochondria to convert the molecules to ATP. Goblet cells secrete mucus into the intestinal tract for lubrication, and paneth cells secrete an antibacterial protein. All these cells have brush borders to maximize the surface area of the membrane, either for secretion or absorption (indicated by the arrows). Goblet and paneth cells, both specialized for secretion, have a single large Golgi complex aimed directly at the brush border that releases enormous vesicles.

enormous vesicles full of mucus (a special kind of glycoprotein). *Paneth cells* are part of the innate immune system (discussed later in this chapter) that helps control bacterial populations in the lumen of the gut



Organization of the liver and structure of liver cells. The five- or six-sided cells, shown at the top, are loosely arranged around an intricate system of blood vessels and a system of liver canals, called the portal system, that contain bile secretions. A typical liver cell (bottom) has an extensive RER, a large number of mitochondria, and multiple Golgi complexes, in addition to a large polyploid nucleus. The cell membranes lining blood vessels, or the portal system, have many microvilli to facilitate transport into and out of the cell. The magnified cell is from the image at the top. Its position is marked with an arrow.

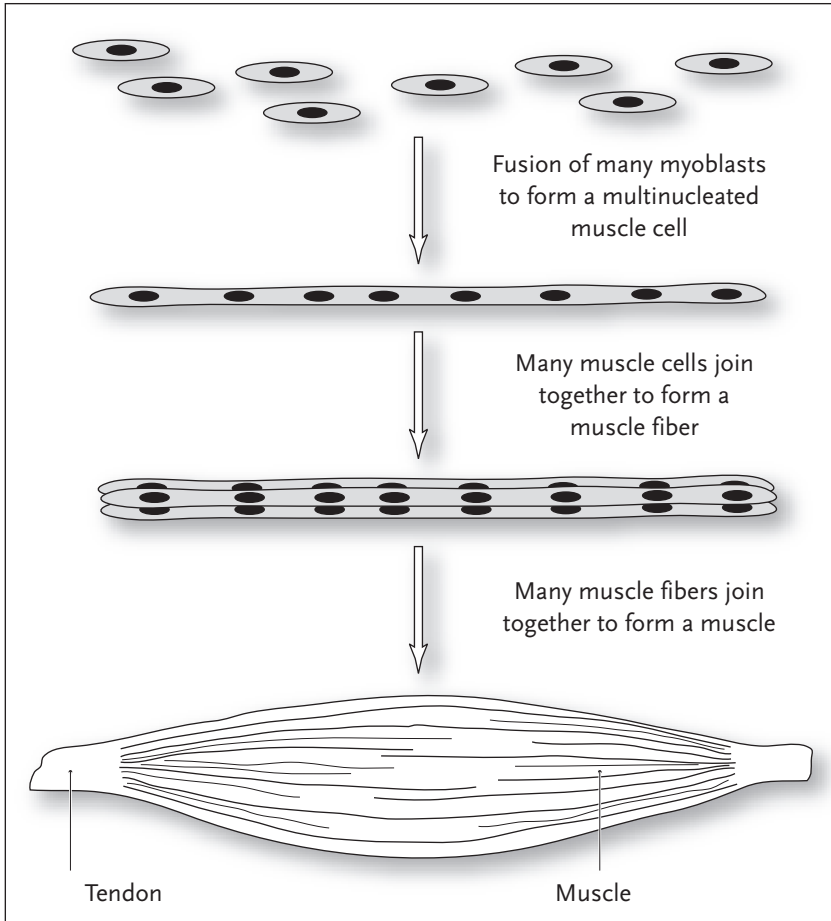
by secreting proteins called cryptdins that kill bacteria. Like the goblet cells, paneth cells have a highly developed rough ER and Golgi complex. Release of the enormous secretory vesicles is polarized, thus directing all vesicles to the secretory surface.

The liver, weighing almost three pounds, is the heaviest gland in the body and is constructed primarily of epithelium (the liver cells). The five- or six-sided cells are loosely arranged around an intricate system of blood vessels and liver canals. The canals, called the portal system, deliver a substance called bile to the intestinal tract for the digestion of fatty compounds (see top of the liver cells figure on page 103). Liver cells are extremely active metabolically, being involved in detoxifying the blood as well as manufacturing bile. Consequently, these cells have a large number of mitochondria, a very extensive endoplasmic reticulum, and several Golgi complexes. The cell membranes lining blood vessels, or the portal system, have many microvilli to facilitate transport into and out of the cell (see bottom of the liver cells figure on page 103).

To maximize their biosynthetic capabilities, liver cells in humans and many other vertebrates have polyploid nuclei. That is, the DNA content has been duplicated fourfold to eightfold, as compared to a haploid nucleus (i.e., a sperm or an egg nucleus). Consequently, liver cells may be found that are diploid, tetraploid, or octoploid (abbreviated as $2n$, $4n$, and $8n$). Duplicating the genome provides additional copies of crucial genes, all of which may be transcribed simultaneously in order to increase the amount of protein being produced over a given period of time. This is a very clever strategy in that a single octoploid cell is the functional equivalent of four diploid cells. Thus a liver consisting of octoploid cells can be smaller than a diploid liver and still achieve the same output. The practice of maximizing cellular output by duplicating the entire genome is ancient, and this strategy is commonly employed by insects. Insect cells have been found containing a quantity of DNA that is 100,000 times greater than is present in the animal's sperm or eggs.

MUSCLE

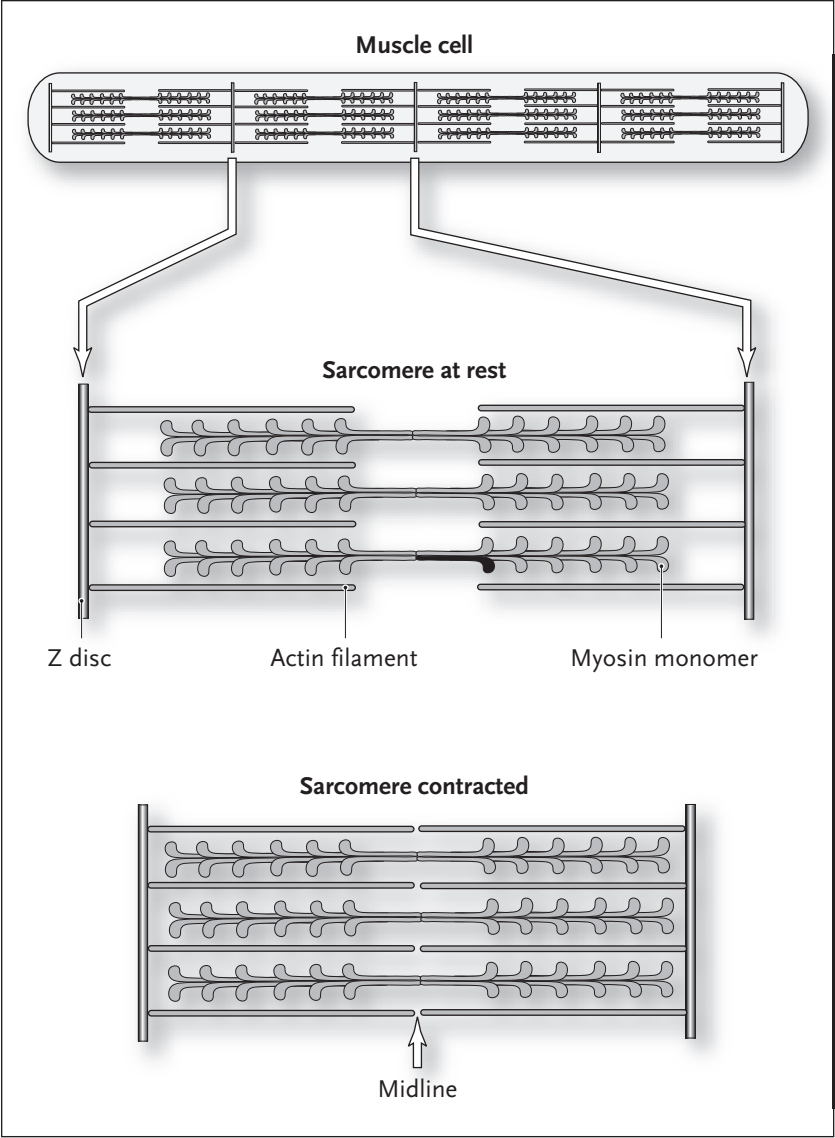
As one of the four primary tissues, muscle gives rise to only three cell types: skeletal, cardiac, and smooth muscle. Cardiac muscle forms the



Skeletal muscle. During embryonic development, myoblasts fuse to form long, multinucleated muscle cells. The filamentous muscle cells, also called myofibrils, bundle together to form muscle fibers. Finally, many muscle fibers are joined together to form a muscle, anchored at both ends by tendons.

heart; smooth muscle is found in the digestive tract, blood vessels, and the uterine lining; and skeletal muscles move the limbs, tongue, and eyes.

Skeletal muscle is formed during embryonic development, when myoblasts fuse to form a multinucleated muscle cell (also known as a myofibril). Dozens of muscle cells join together to form a muscle fiber, after which the fibers are joined together to make a muscle. The con-

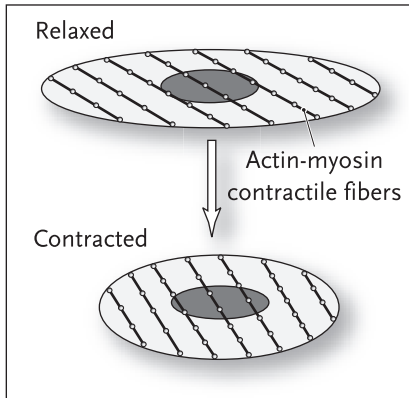


The sarcomere. A skeletal muscle cell (top) contains many repeats of the basic contraction unit, called a sarcomere. Each sarcomere (middle) is constructed from actin filaments, which are attached to a Z disc, and myosin monomers, the shape of a golf club (black). The Z discs are constructed from the proteins capZ and alpha-actinin. The sarcomere contracts (bottom) when the club, or headend, of the myosin monomers binds to the actin and pulls the Z discs toward the midline (arrow).

tractile unit of a muscle cell is called a sarcomere. These units are kept perfectly aligned during the formation of the muscle fiber and, finally, the muscle itself. In this way, the muscle, composed of thousands of individual sarcomeres all aligned in register, contracts as though it were a single unit. Sarcomeres are constructed from four kinds of protein: actin, myosin, CapZ, and alpha-actinin. Additional, accessory proteins are present, which stabilize muscle structure and help activate contraction. The precise and regularly repeating arrangement of the sarcomeres is visible in microscopic images of skeletal muscle as cross-striations. The sarcomere contracts when the myosin heads bind to the actin and pull the Z discs toward the center of the sarcomere, which is called the midline. This description of muscle contraction, called the sliding-filament model (so named because the actin “filament” slides along the myosin), was first proposed almost 50 years ago by the British physiologist Sir Andrew Huxley (half brother of Aldous Huxley, the author of *Brave New World*). Huxley based his model on the biochemical analysis of contracting muscles, as well as microscopic images of relaxed and contracting skeletal muscle.

Muscle, like neurons, is an excitable tissue, meaning that it responds to electrochemical signals. Neurons pass the signal along to other neurons in a complex communication network, whereas muscles use the signal to activate contraction. When we move an arm, a leg, or a finger, it is because our brain sends a signal to a neuromuscular junction; a place where the nerve fiber contacts the muscle. The neural stimulus initiates the release of calcium from an internal storage depot, which activates simultaneous contraction of every sarcomere in the muscle. The length of time the calcium remains free depends on the length of the stimulus. If we pick up a heavy object and hold it in the air for a while, the muscle receives a steady stream of neural stimulation, thus keeping the muscle’s concentration of free calcium at high levels. When we set the object down, the neural stimulation stops, and the calcium is immediately returned to its storage depot so the muscle can relax.

Cardiac muscle has the same sarcomere-based contraction machinery as that found in skeletal muscle, and thus is another example of striated muscle. The main difference between cardiac muscle and skeletal muscle is that the cardiac muscle has large numbers of



Smooth muscle. Fibers containing actin and myosin span the cell diagonally with respect to the long axis. Contraction of these fibers pulls the ends of the cell toward the center, greatly reducing its overall length.

gap junctions between the muscle cells, permitting a neural signal to pass freely through every cell in the organ. Consequently, a single impulse will stimulate all the cells simultaneously, allowing the heart to beat as a single unit.

Smooth muscle gets its name from the fact that the contraction machinery is not organized into a sarcomere, and therefore these muscles have a smooth, rather than striated, appearance. Contraction is based on a simple and probably ancient arrangement of actin-myosin filaments. These filaments are arranged in a diagonal pattern from one end

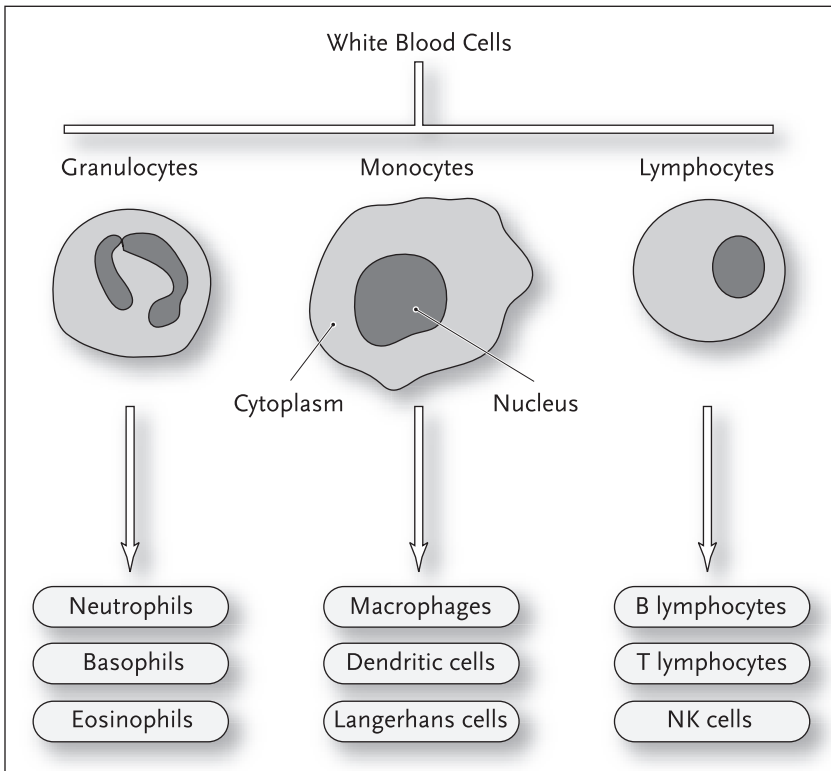
of the cell to the other, and as they contract, they pull the ends of the cell toward the center, greatly reducing its overall length. Smooth muscle is not capable of the speed or force of contraction that is typical for striated muscles, but they are ideal for tissues that require slow, rhythmic contractions, such as intestine or uterine lining.

Cells of the Immune System

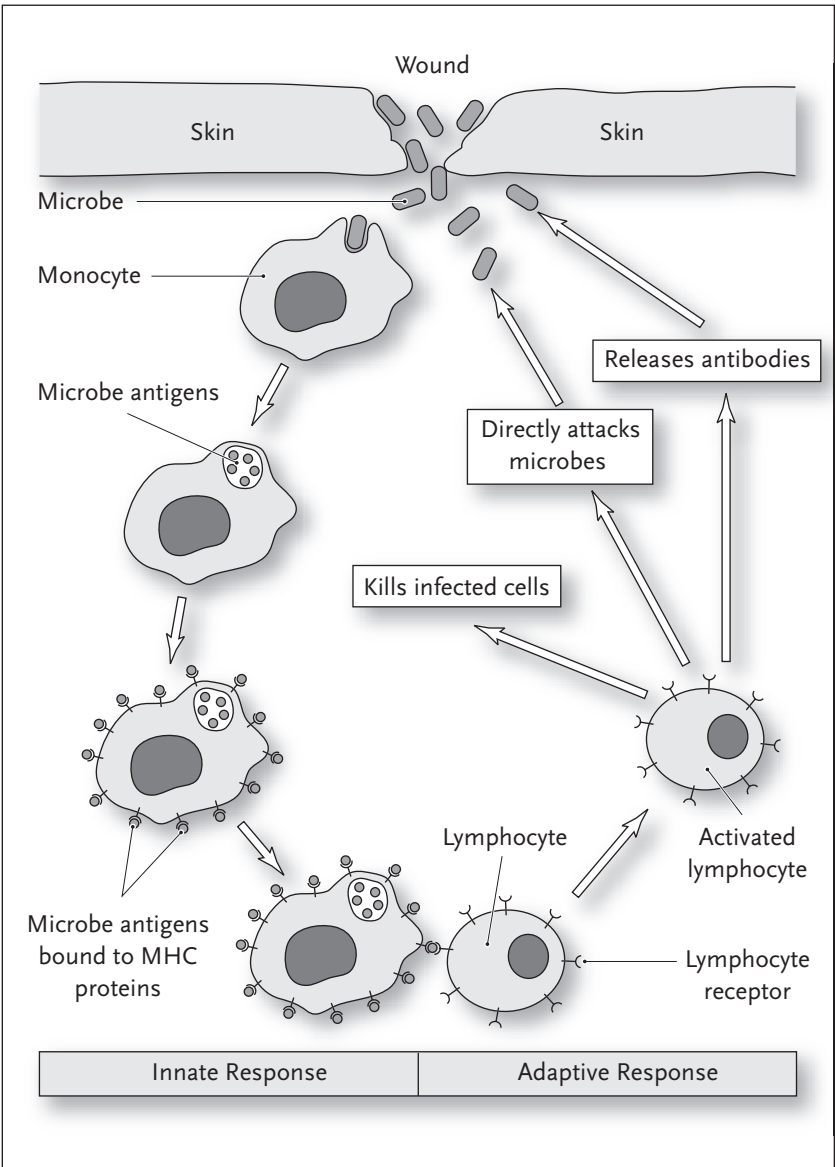
The human immune system is composed of a diverse group of white blood cells that are divided into three major categories, all of which are derived from connective tissue: granulocytes, monocytes, and lymphocytes (see figure page 109). Granulocytes have a distinctive, lobular nucleus and are phagocytic (can eat cells and viruses). Monocytes are large phagocytic cells with an irregularly shaped nucleus. The largest monocytes, the macrophages, can engulf whole bacteria as well as damaged and senescent body cells. Lymphocytes have a smooth morphology and a large round nucleus. T-lymphocytes (also called T cells) and natural killer (NK) cells deal primarily with coordinating

the immune response and with killing already infected body cells. B-lymphocytes (B cells) are nonphagocytic; they deal with an invading microbe by releasing antibodies.

Phagocytosis of an invading microbe by granulocytes and monocytes represents a first-line defense called the innate response. All animals are capable of mounting this kind of defense. Activation of the



White blood cells. These cells are divided into three major categories: granulocytes, monocytes, and lymphocytes. Granulocytes have a distinctive, lobular nucleus and are phagocytic (eat cells, viruses, and debris). Monocytes are large cells with an irregularly shaped nucleus. All monocytes are phagocytic; the largest members, the macrophages, can engulf whole bacteria and damaged or senescent body cells. Lymphocytes have a smooth morphology with a large round nucleus. B-lymphocytes are nonphagocytic but produce antibodies. T-lymphocytes and natural killer (NK) cells coordinate the immune response and can force infected cells to commit suicide.



Innate and adaptive immune response. Phagocytosis of invading microbes is called the innate response. In higher vertebrates, microbe antigens, bound to special monocyte surface proteins called the major histocompatibility complex (MHC), are then presented to lymphocytes. Contact between the lymphocyte receptor and the antigen activates the lymphocyte and the adaptive response, consisting of a three-prong attack on the microbe and microbe-infected cells.

lymphocytes leads to a more powerful, second line of defense called the adaptive response, which is found only in higher vertebrates. The adaptive response is initiated by monocytes, specifically, dendritic and Langerhans cells. These cells, after engulfing a virus or bacteria, literally tear the microbe apart and then embed the pieces, now called antigens, in their membrane. The antigens are presented to lymphocytes, which become activated when their receptors bind to the microbial antigens. Activated B-lymphocytes secrete antibodies specifically designed for that particular microbe. Activated T-lymphocytes and NK cells attack the microbe directly but are primarily concerned with locating and killing infected body cells.

The adaptive system can remember a pathogen long after it has been removed from the body. This is why a specific bacteria or virus cannot make us sick twice. Once infected, we develop a natural, lifelong immunity. We can also immunize ourselves against many diseases by injecting a crippled version of the pathogen or specific antigens from a pathogen into our bloodstream. This concoction of bits and pieces from a pathogen, called an immunizing serum, activates the adaptive response, leading to a lasting (though not always lifelong) immunity against the disease.

.7.

NEURONS

Pushing Back the Night

A world without neurons is a dark world, populated by cells, plants, and simple creatures that never see the light of day. That world is a place devoid of vision, where there are no fish in the sea and no animals on the land; where there is no music, no laughter, no emotion, and no intellect. The first cells appeared on Earth 3.5 billion years ago, but life as we know it really began with the appearance of the neuron, about 2 billion years ago.

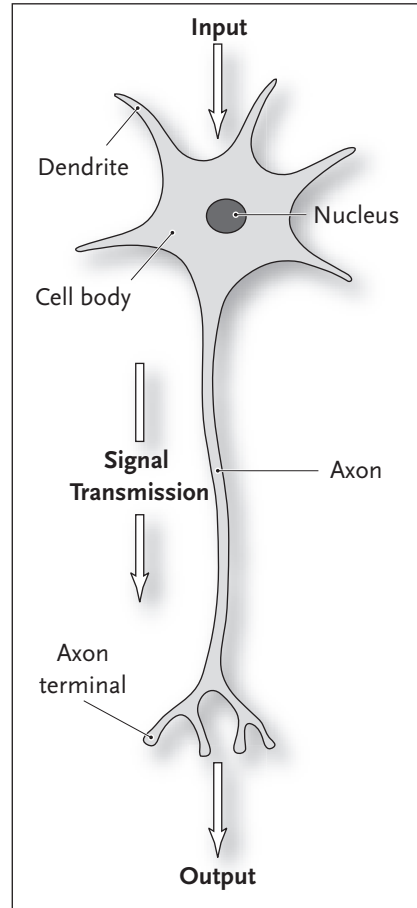
Neurons, and the things they do, defy the imagination. They make it possible for us to instantly recognize friends that walk into a room. We can tell at a glance if they are happy or sad, and whether they are wearing the same clothes they wore the day before. We can talk to them, for hours on end sometimes, or we can get up and dance around the room. Neurons make all this possible. They do it by forming intricate communication networks, usually consisting of billions of cells. By sending signals, fast and furious, through that network, neurons let us see the world, shape our intellect, and give us our personality.

We have only a vague notion of how neural networks accomplish so much. If we dwell on our personality and the thing psychiatrists call a persona, or self-image, it is difficult to see how these intangible, possibly spiritual, aspects of our brain are generated by neurons simply talking to one another. Equally difficult to understand is the manner by which these networks store and access memories. We know exactly how much memory our personal computers have; it is

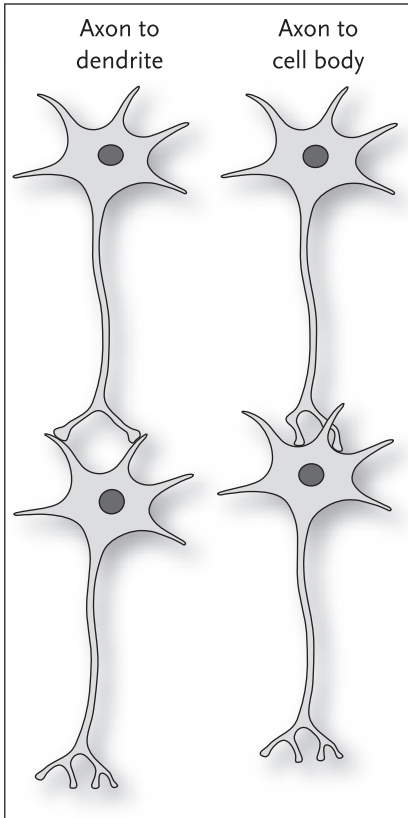
advertised with every machine. Yet the memory capacity of our own brain is still a mystery. Research in this area is still highly speculative, but many scientists believe the memory capacity of the human brain is infinite. This estimate is based on the observation that on any given day, we always have room for new memories. We never seem to reach that point in time when we have to stop and say, “Sorry, teacher, I can’t learn any more, my brain is full.” Skeptics would argue that our brains simply erase old memories to make room for the new. We do this sort of thing all the time with our computers. And yet, psychologists insist that under hypnosis, people are able to remember things they once thought were lost. The memories are there, but sometimes we have trouble retrieving them.

Neural Anatomy

Neurons are cells that are specially designed for communication and, like computers or other communication hardware, have a polarized anatomy that contains structures for receiving input, and other structures for processing the output. A signal, in the form of an electrochemical current, enters a neuron at finger-like projections, called dendrites, and is passed along



Neural anatomy. A neuron receives signals at finger-like projections from the cell body, called dendrites, and passes them on to other neurons through an elongated process called an axon. The tip of an axon, which often branches into several terminals, makes contact with the dendrites of other neurons.



Synaptic junctions. The synapse is usually formed between the axon, and one or more dendrites, but axons may also form a synapse with the cell body.

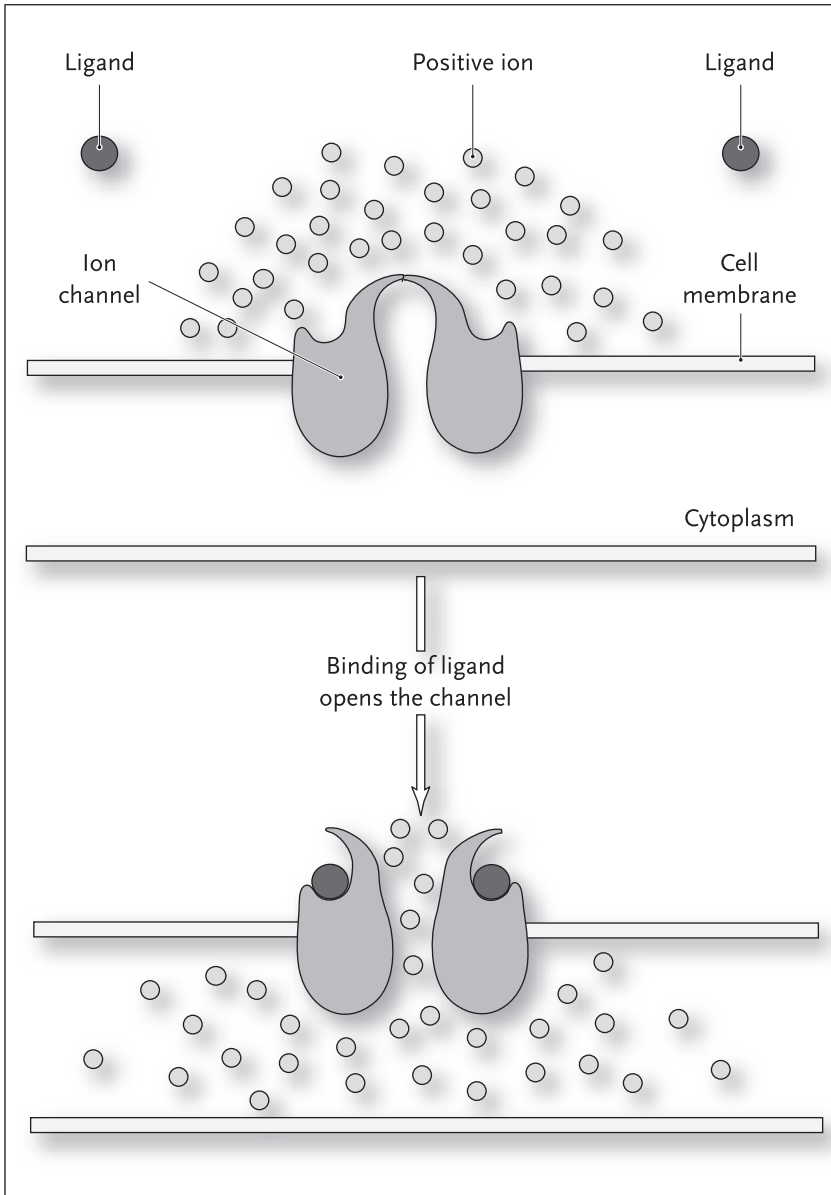
to another neuron through the axon, a single projection, much longer than a dendrite. Signal reception, propagation through the cell, and transmission to another neuron takes less than a millisecond and travels at a rate of 120 meters/sec (270 miles/hr). Every axon forms a communication junction, called a synapse, with a dendrite or the cell body (soma) of another neuron. This simple scheme, repeated millions of times, makes possible the immensely complex neural circuitry of an animal's brain.

Signal Transduction

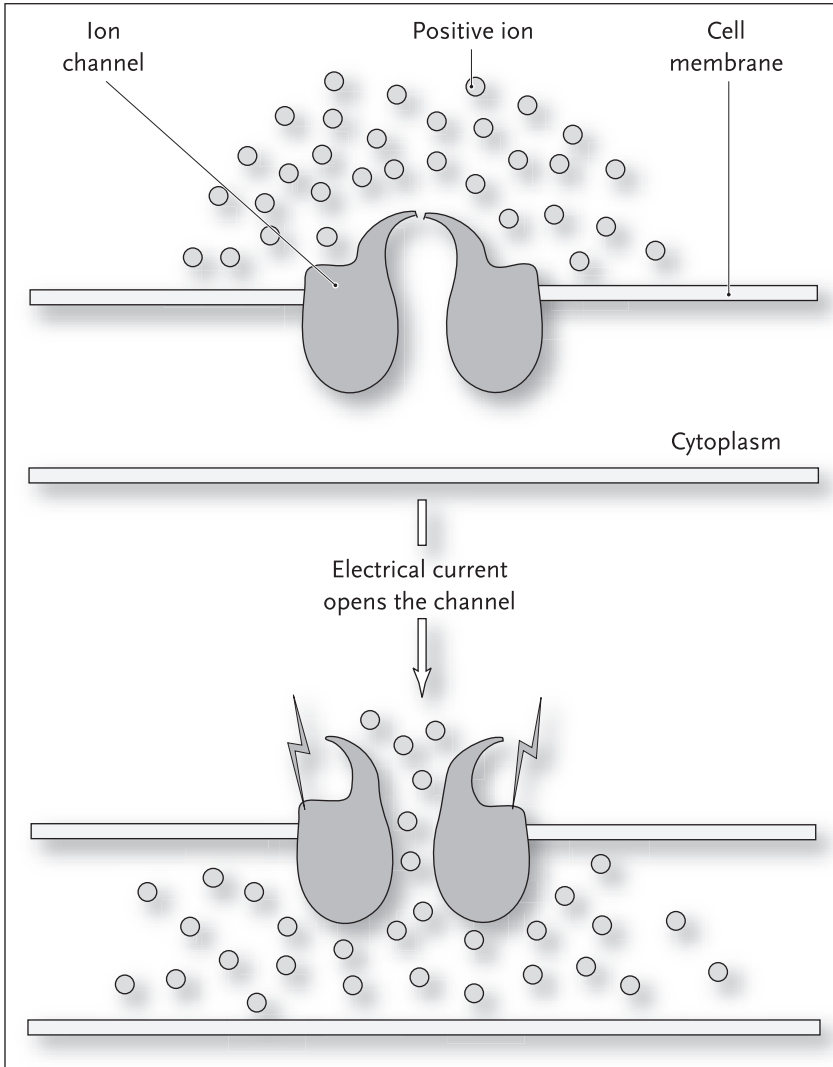
Propagation of a signal from one neuron to another depends on the coordinated operation of ion channels, located in the cell's membrane. Ion channels vary according to the type of ion that can pass through them, most commonly sodium (Na^+) and calcium (Ca^{++}), and the

mechanism by which they are activated. Ligand-gated (Lg) ion channels are activated, or opened, when a signaling molecule, called a ligand, binds to the channel. Lg Na^+ channels are the most common type and serve to initiate neural signaling. Voltage-gated (Vg) channels are opened by an electrical stimulus. The most common channels of this kind permit the entry of Na^+ or Ca^{++} ions.

When Na^+ channels are closed, the ion concentration is greater on the exterior surface of the neuronal membrane than it is on the inside of the cell. Thus the exterior is positively charged relative to the interior.



Ligand-gated (Lg) ion channel. The closed channel keeps positively charged ions outside the cell, and in this state, the membrane is said to be polarized (i.e., positive on the outside, relative to the inside). The channel opens when bound to a signaling (ligand) molecule, allowing the ions to rush inside, thus de-polarizing the membrane and initiating an electric current.



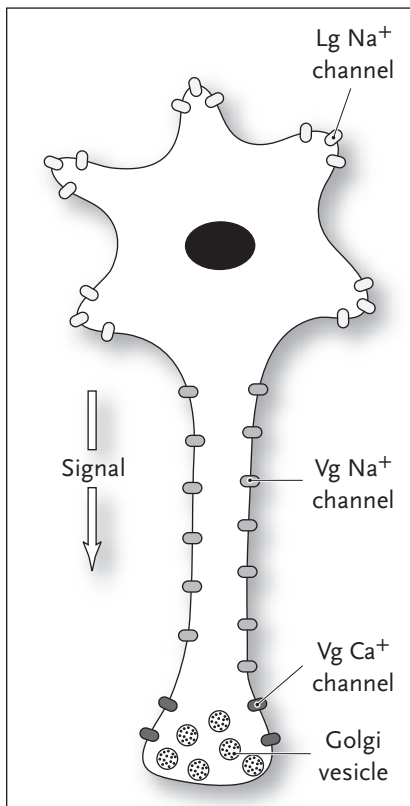
Voltage-gated (Vg) ion channel. The electric current that opens these channels originates with the opening of ligand-gated channels, after which Vg channels are opened by the current generated by other Vg channels.

In this resting state, the membrane is said to be polarized, and the potential difference across the membrane, known as the voltage, is relatively high. When Lg ion channels open, the Na^+ rushes inside the

cell, and in so doing, establishes an electrical current. The same applies to a $Vg Na^+$ channel or a $Vg Ca^{++}$ channel.

Neurons receive, propagate, and transmit signals by coordinating the activation of $Lg Na^+$ channels, $Vg Na^+$ channels, and finally $Vg Ca^{++}$ channels. $Lg Na^+$ channels are always located in dendritic membranes (the input end of the cell). A special ligand, known as a neurotransmitter, sent by a neuron, binds to and activates $Lg Na^+$ channels. The opening of these channels initiates an electrical stimulus through membrane depolarization. This signal spreads electrotonically, or passively, through the cytoplasm, and decays rapidly with distance, but usually has enough strength to activate the first $Vg Na^+$ channels, located in the axonal membrane close to the cell body. The signal propagates along the axon, toward the axonal terminus, by sequential activation of $Vg Na^+$ channels. The strength of this signal is boosted each time a Vg channel

opens, so it does not decay over the full length of the axon. The signal through the axon is called a spike potential, because the opening of each channel results in a renewed burst (spike) of the electrical current. Finally, the $Vg Na^+$ channels activate $Vg Ca^{++}$



Signal transmission. A ligand (neurotransmitter) binds to the Lg sodium (Na^+) channels, depolarizing the membrane and initiating an electric current that opens the first $Vg Na^+$ channels in the axon. The signal propagates along the axon by sequential activation of the $Vg Na^+$ channels. Finally, the axonal current activates the Vg calcium (Ca^{++}) channels. Influx of calcium stimulates exocytosis of the Golgi vesicles, containing neurotransmitters that send the signal on to another neuron.

channels, located at the axonal terminus. The influx of calcium stimulates exocytosis of Golgi vesicles (fusion of the vesicles with the cell membrane), containing neurotransmitters that send the signal on to other neurons.

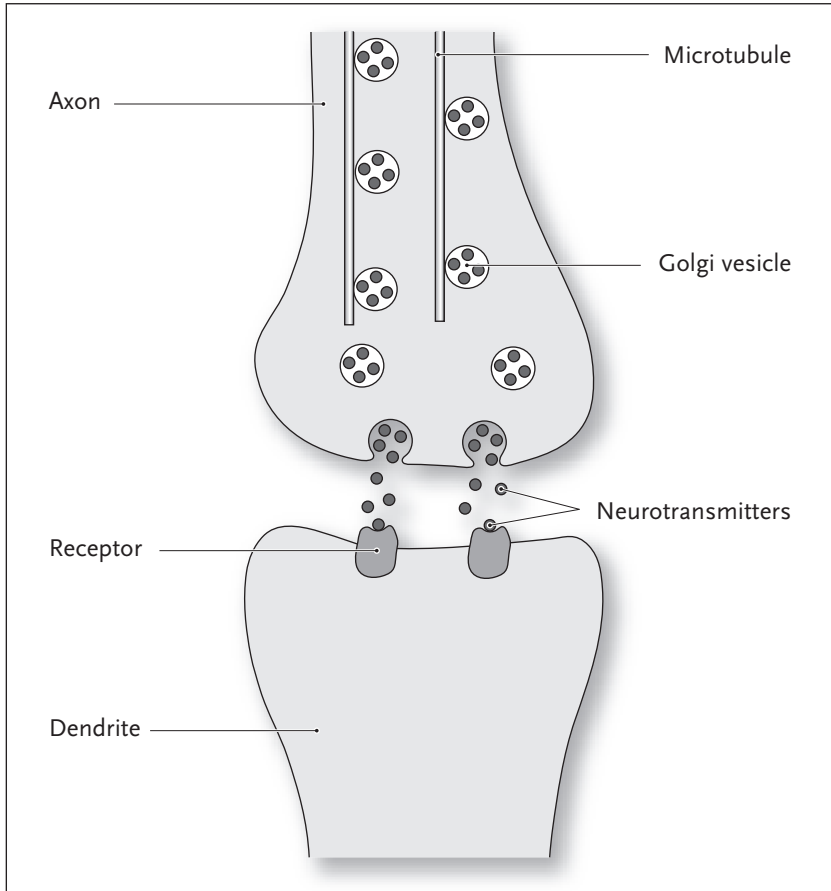
Signal propagation along the axon is unidirectional because the Na^+ channels, just behind the advancing signal, are deactivated. Special Na^+ pumps, driven by the hydrolysis of ATP, return the ion to the extra cellular space, thus repolarizing the membrane. This also serves to reactivate the channels, in preparation for the next signal. The Vg Na^+ channels are capable of processing 500 signals per second, and depolarization of the membrane involves the passage of up to 10 million Na^+ per second.

The Synaptic Junction

The connection between an axon and a dendrite is called a synapse. Although neurons communicate through the synapse, they do not actually touch one another. Close inspection of a synapse shows a small gap separating the axon from the dendrite. A signal is transmitted across the gap by the release of neurotransmitters that are stored at the axon terminus in Golgi-derived vesicles. The vesicles travel to the axon terminus on a railroad-like transport network constructed of microtubules. When a neuron receives a signal, the Golgi vesicles, at the terminus, are released from the microtubules and fuse with the axonal membrane, dumping their cargo into the synaptic gap. The neurotransmitters quickly diffuse across the gap and bind to receptors on the dendritic membrane, triggering an electrochemical impulse in the target neuron, thus completing transmission of the signal. This may seem like an awkward way for neurons to signal each other, but the synaptic gap, and the use of neurotransmitters, is crucial for maintaining the strength of the signal over a network that consists of 100 billion cells. Each neuron in such a circuit boosts the signal to its original strength, in a manner analogous to the signal boost that each Vg Na^+ channel provides as the signal travels along each axon.

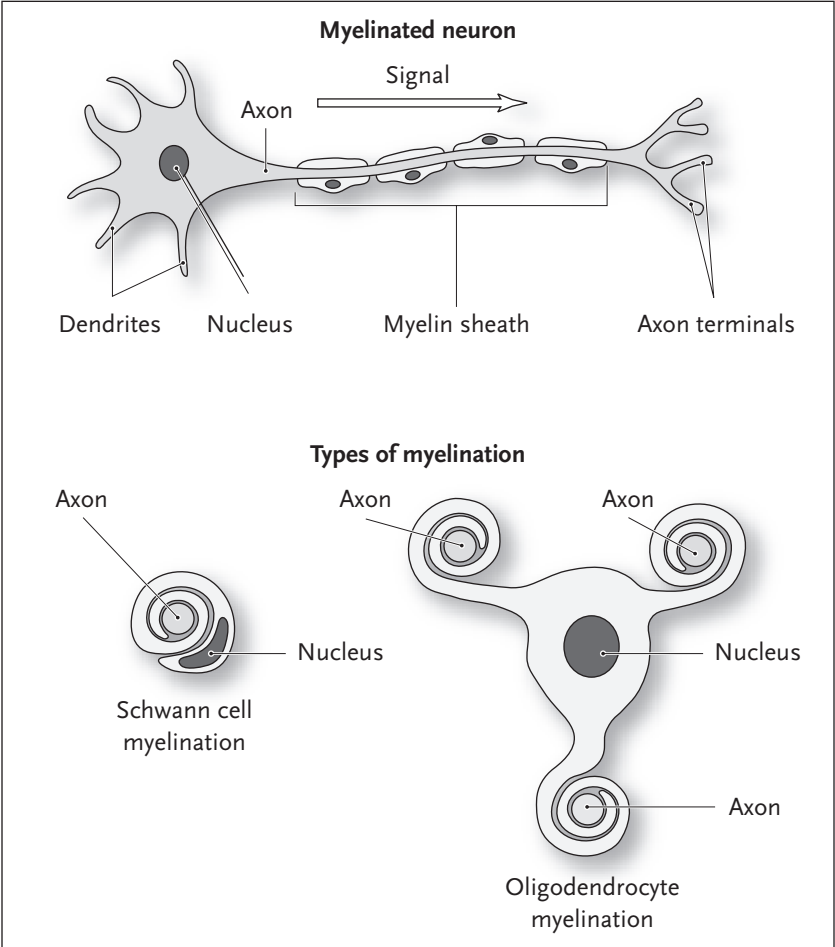
Myelination

A neural circuit, much like an ordinary household electrical circuit, works better if the axons are insulated. Electrical wire is insulated with a plastic



Synaptic junction. Axons and dendrites do not touch each other but are separated by a small gap called the synapse or synaptic junction. A signal is transmitted by the release of small molecules called neurotransmitters that are stored at the axon terminus in Golgi vesicles. Binding of the neurotransmitter to the receptor on the dendrite membrane completes the transmission. The Golgi vesicles travel to the axon terminus on a transportation network constructed from microtubules.

coating, but neurons are insulated by special cells that form a coating, called myelin, around each axon. Myelin is constructed from cells called oligodendrocytes and Schwann cells that wrap around axons to form a protective multilayered sheath. Oligodendrocytes, located in the central nervous system (CNS, consisting of the brain and spinal cord), can myeli-



Neural signaling and myelination. Communication between neurons is much more efficient when the axons are insulated with a myelin sheath. Myelin is made by Schwann cells or oligodendrocytes wrapping around the axon. Oligodendrocytes can insulate more than one axon at a time.

nate more than one axon at a time. Schwann cells, located in the peripheral nervous system (i.e., any nerves outside the CNS, such as those in the toes and fingers), wrap around a single axon. The existence of the myelin sheath greatly complicates medical procedures aimed at repairing a severed spinal cord or neural damage caused by Alzheimer's or Parkinson's disease.

Neurotransmitters

Neurons synthesize the ligand, called a neurotransmitter, that is released at the axonal terminus to relay a signal to another neuron. Neurotransmitters are divided into two major groups. The first group is composed of fast, direct neurotransmitters that include acetylcholine (synthesized from acetyl-CoA and choline) and several amino acids (glutamate, aspartate, gamma-amino butyric acid, and glycine). The second group is composed of slow, indirect neurotransmitters that include the monoamines (derived from various amino acids), such as epinephrine, dopamine, and serotonin. This group also includes a large number of neuropeptides. Fast, direct neurotransmitters bind to receptors that are ion channels. By contrast, the slow, indirect neurotransmitters bind to receptors that are linked to a signaling pathway and thus modulate the behavior of the target neuron through a series of biochemical steps, rather than generating an electrical current.

Acetylcholine (ACh) is one of the most ubiquitous neurotransmitters, being used by most vertebrate species in motor neurons (neurons that directly stimulate muscle) and various neurons in the CNS. Neurons that use ACh are referred to as being cholinergic. GABA and glycine are both inhibitory neurotransmitters released by neurons in the CNS and peripheral nervous system (PNS) in vertebrates, crustaceans, and worms. Glutamine is an excitatory neurotransmitter found in the CNS of vertebrates, insects, and crustaceans. Glutamate receptors participate in modifying synaptic behavior and organization during long-term potentiation (learning and memory).

Norepinephrine (NE), synthesized from the amino acid phenylalanine, is released by various neurons in the CNS, as well as cells in the adrenal gland. Neurons releasing NE are said to be adrenergic. Psychoactive drugs, such as mescaline (from the peyote cactus), interfere with NE signal transmission in the CNS. Amphetamines mimic the function of NE, and cocaine interferes with NE inactivation, thus prolonging the stimulation of an adrenergic network. Dopamine, synthesized from the amino acid tyrosine, occurs throughout the CNS and acts through a signaling pathway. Dopamine is primarily a neuromodulator and is able to dampen the excitability of the target neuron. Parkinson's disease is a neurological disorder characterized by the degeneration and

death of dopaminergic neurons in an area of the brain known as the substantia nigra. Serotonin, synthesized from the amino acid tryptophan, also occurs throughout the CNS. Some of the receptors are ion channels, while others are linked to signaling pathways.

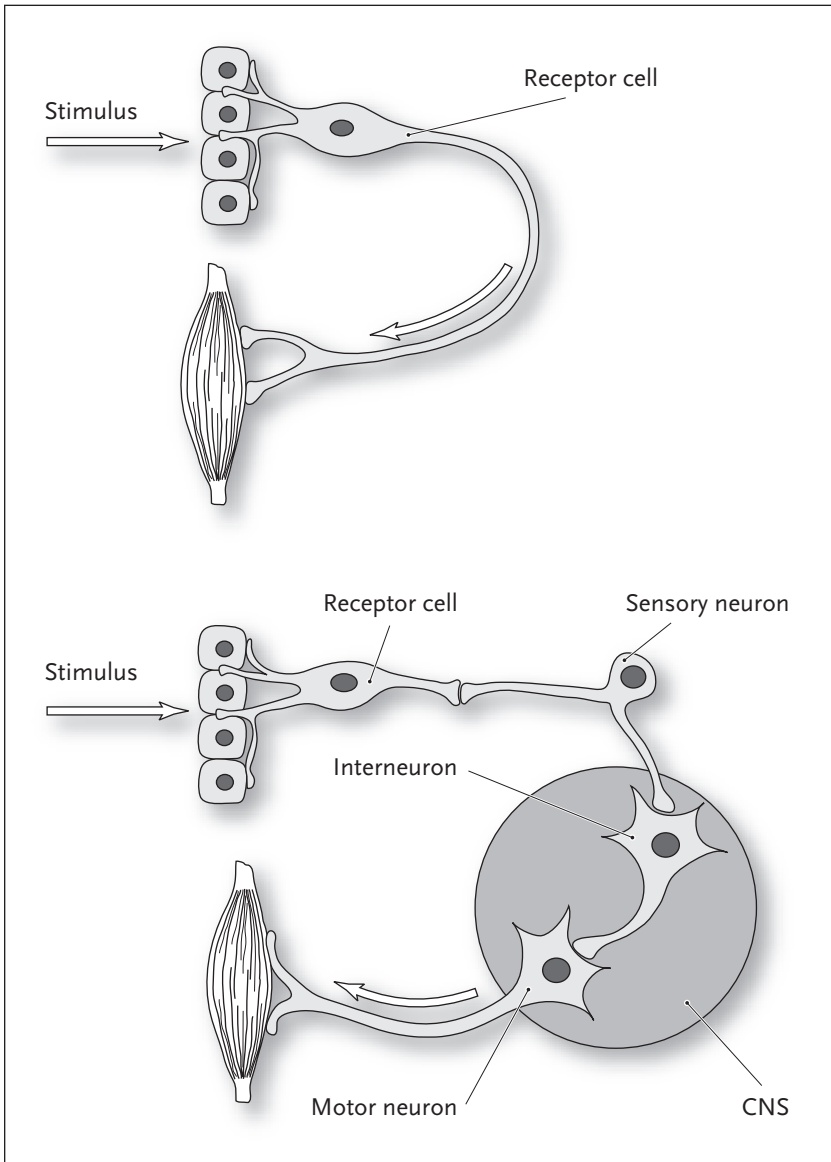
Neuropeptides, consisting of about 30 amino acids, are widely distributed throughout the CNS and, like serotonin, bind to receptors that are ion channels or are linked to signaling pathways. One group of neuropeptides, called the endogenous opioids, are involved in regulating neural circuits controlling many physiological processes, including behavior, appetite, pain tolerance, stress, and shock. They are also involved in regulating the neuroendocrine system, primarily through their ability to reduce the excitability of neurons and neuroendocrine cells.

Neural Circuits

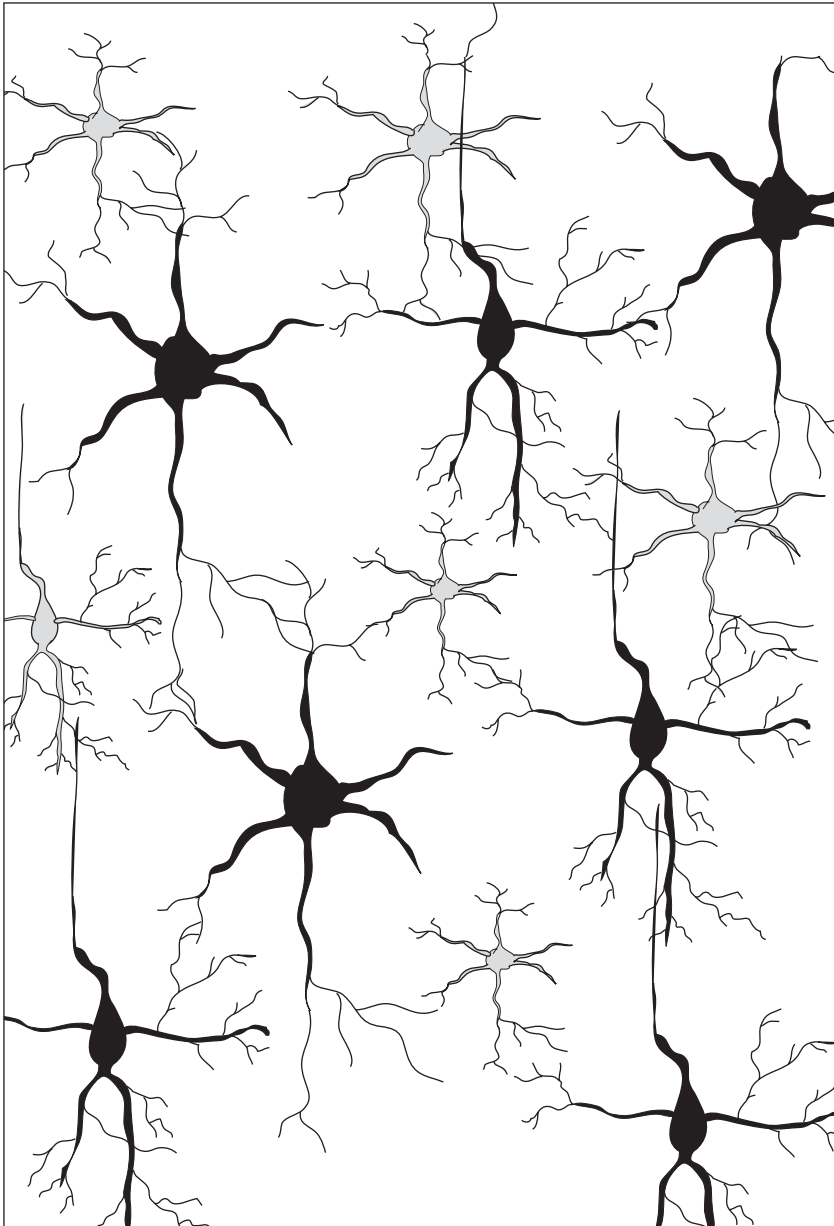
Neural circuits probably began with a single neuron acting as both a receptor cell and as a motor neuron capable of stimulating muscle or other tissue. This simple scheme evolved into a more elaborate system with the addition of specialized neurons, situated between the receptor and the target tissue. The additional circuitry became the central nervous system, capable of processing and evaluating the original stimulus before any action is taken.

Neural circuits in the human brain are enormously complex, consisting of more than 10 billion neurons, each forming thousands of synaptic junctions with other neurons. In total, it has been estimated that the human brain has more than a trillion synaptic junctions. One type of neuron in the cerebellum, called the Purkinje cell, has thousands of dendrites, receiving inputs from more than 200,000 other neurons. Circuits involving these neurons are part of the brain's machinery for controlling complex movements.

The complexity of the brain circuitry increases in humans, and other mammals, soon after birth. The increased complexity is not due to an increase in the number of neurons but rather is due to an increase in the number of synaptic junctions. The addition of new synapses increases the integrative and analytic capabilities of the circuit and may be necessary for the storage of new memories. Experiments with rats



Evolution of neural circuits. Circuits evolved from single-cell circuits consisting of a neuron that had the dual function of a receptor cell and a motor neuron capable of stimulating a muscle or other tissue. More advanced circuits include sensory neurons and a central nervous system (CNS), consisting of many different kinds of neurons that process and evaluate the stimulus before activating a motor neuron or other output neurons.



A neural circuit in the mammalian brain. Circuits in the brain consist of billions of neurons, each forming thousands of synaptic junctions with other neurons. These complex, highly integrated circuits give us our intellect, our emotions, our ability to see the world, and much more.

have shown that animals reared in a boring environment, with minimal visual stimulation, have fewer synapses than rats reared in stimulating environments or subjected to long-term training exercises. Extensive training and learning exercises also modify pre-existing synapses in several ways: synaptic contacts increase in size, the number of Golgi vesicles stored at the axonal terminus increases, and there is an increase in the number of Ca^{++} channels in the pre-synaptic membrane. All these changes are part of a complex process, known as long-term potentiation, that remodels the circuitry for enhanced functionality and memory storage.

Summary

If neurons had never appeared on our planet, there would still be luxurious plant communities covering the surface of the earth, microbes would be everywhere, and the seas would contain simple creatures, like coral and sponges. All these organisms are constructed from cells that communicate with each other, but they do so through a slow process involving the exchange of molecules that bind to cell-surface receptors. Neurons, by exploiting the properties of ion channels, introduced a lightning-fast form of communication, without which the progression from single cells to complex multicellular bodies would not have been possible.

.8.

RESOURCE CENTER

Studying the cell requires a great deal of hardware, technology, and experimental procedures. This chapter provides additional information that covers light microscopy, histology and histochemistry, and recombinant DNA technology.

Light Microscopy

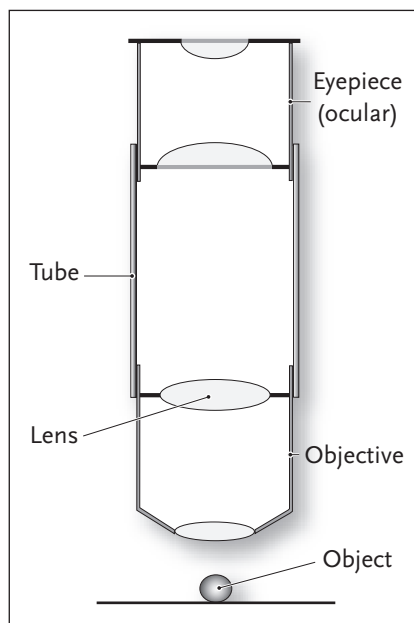
Microscopes are among the oldest of scientific instruments, and yet they are just as important today as they were 120 years ago when microbiologists used them to study bacteria and the many diseases they cause. The earliest microscopes, invented in the 1600s by Antonie van Leeuwenhoek were single lenses mounted in a brass frame that were held in the hand or attached to a table. These simple magnifying glasses could resolve bacteria and other microbes but were extremely difficult, and tiring, to use.

By the 1700s, compound microscopes, using two or more lenses, were being made that offered higher magnifications than Leeuwenhoek's microscope and were much easier to use, but the image quality was not as good. These microscopes consisted essentially of an eyepiece, a metal tube, and an objective that formed the initial magnified image of the object. Adding two or more lenses together increased the overall power of a microscope but, at the same time, introduced optical artifacts that seriously degraded the image. The two most important of these artifacts are spherical and chromatic aberration.

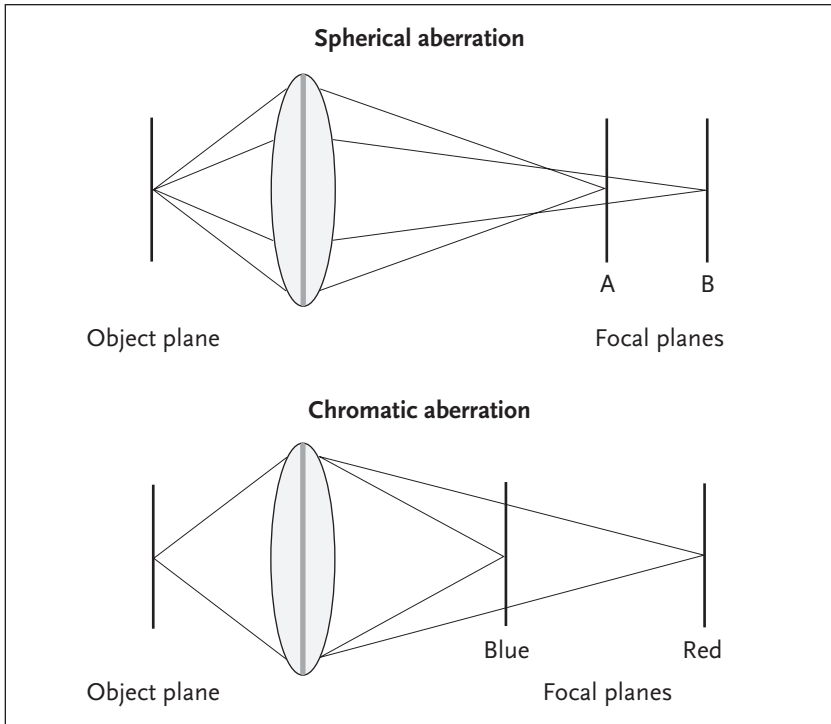
Spherical and chromatic aberrations arise as a natural consequence of the way light behaves when it passes through a glass lens, which has

a higher density than the air around it. There are two simple rules of geometric optics that govern this behavior. First, light travels in a straight path, and second, the path bends (refracts) at an interface between two transparent media (i.e., the air and the lens). Lenses are designed with curved surfaces to magnify an object, and to bring the image of that object to a single focal point. Spherical aberration occurs when light rays from an object are refracted by different parts of the lens and, therefore, do not come to a single focal point, producing an image that is fuzzy and out of focus. Chromatic aberration is a consequence of the multiwavelength composition of white light, ranging from red (long wavelength) to blue (short wavelength). The extent to which light is refracted at an interface depends on its wavelength, being high for blue light and low for red light. Therefore, the blue and red components of white light are refracted by a lens to separate focal points, producing a fuzzy image that has a red and blue halo.

In 1868, the German physicist, Ernst Abbe, discovered a way to construct objectives containing lens pairs and triplets that corrected chromatic and spherical aberration simultaneously. These lens systems, called apochromatic objectives, provide clear and undistorted images at the highest magnification possible (1,000 $\times$) while retaining a high resolution. The measure for resolution is the smallest distance at which two points can still be discerned. A resolution of one micrometer



An 18th-century light microscope. The image is magnified by the objective and brought to focus at the eye by the eyepiece. The eyepiece and objective are compound-lens systems that give high magnification but poor resolution and image quality.



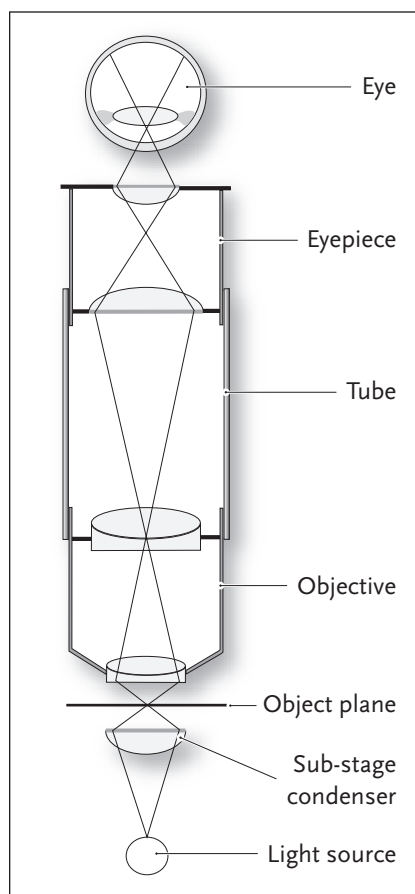
Spherical and chromatic lens aberrations. Spherical aberration is caused by variations in the amount of refraction that occurs over the surface of the lens. Consequently, light passing through peripheral areas of the lens come to a different focal point (A) than light passing through central regions (B). Chromatic aberration occurs because the amount of refraction is greater for blue light than it is for red light. Thus these two wavelengths, and all wavelengths in between, come to different focal points.

(μm), for instance, indicates that two objects one μm apart can still be seen as separate objects, whereas if the objects were $0.5 \mu\text{m}$ apart, they would appear to be a single object. Apochromatic objectives provide a resolution of $0.4 \mu\text{m}$, very close to the theoretical limit for light microscopes of $0.2 \mu\text{m}$ (a distance of $0.2 \mu\text{m}$ is about $1/500\text{th}$ the diameter of a period on this page). The high resolution, and clarity of the image, provided by apochromatic objectives made it possible for scientists to study the structure and function of individual cells in a way that was never before possible. The quality of Abbe's lenses has never been

surpassed, and apochromatic objectives are still standard on light microscopes today.

With the lens system perfected, microscopes quickly evolved from a simple monocular construction to binocular compound microscopes that are equipped with rotating turrets, holding a variety of objectives, camera attachments, video equipment, and special filtering systems for observing fluorescent images. In addition, microscopes with special optical systems, known as phase contrast and Nomarski's interference, were invented in the 1950s that made possible the observation of living cells without having to stain them with dyes. An important advance in the field of microscopy came with the introduction of the fluorescent microscope, an instrument that is used to observe cells and tissue sections that

have been stained with fluorescent dyes. This microscope has standard objectives, but it includes special optical filters, and a beam splitter, for observing cells and tissues emitting fluorescent light. A fluorescent dye is a molecule that absorbs light of one wavelength, emitting it at another, longer wavelength.



A light microscope corrected for chromatic and spherical aberration. Light is focused on the object by a condenser to enhance image quality and contrast. Chromatic and spherical aberrations are corrected at the objective with precisely ground lens-pairs or triplets (two or three lenses fitted together). High power objectives (100x) may have up to 10 separate lenses to correct for aberrations. The example shown at left is a 10x objective, consisting of four lenses.



A 19th-century monocular microscope. These instruments were capable of high resolution but were difficult to use for extended periods of time. (Courtesy of Dr. Joseph Panno)

An example is the molecule fluorescein, which emits green light, at 520 nanometers (nm), when stimulated with blue light at 450 nm.

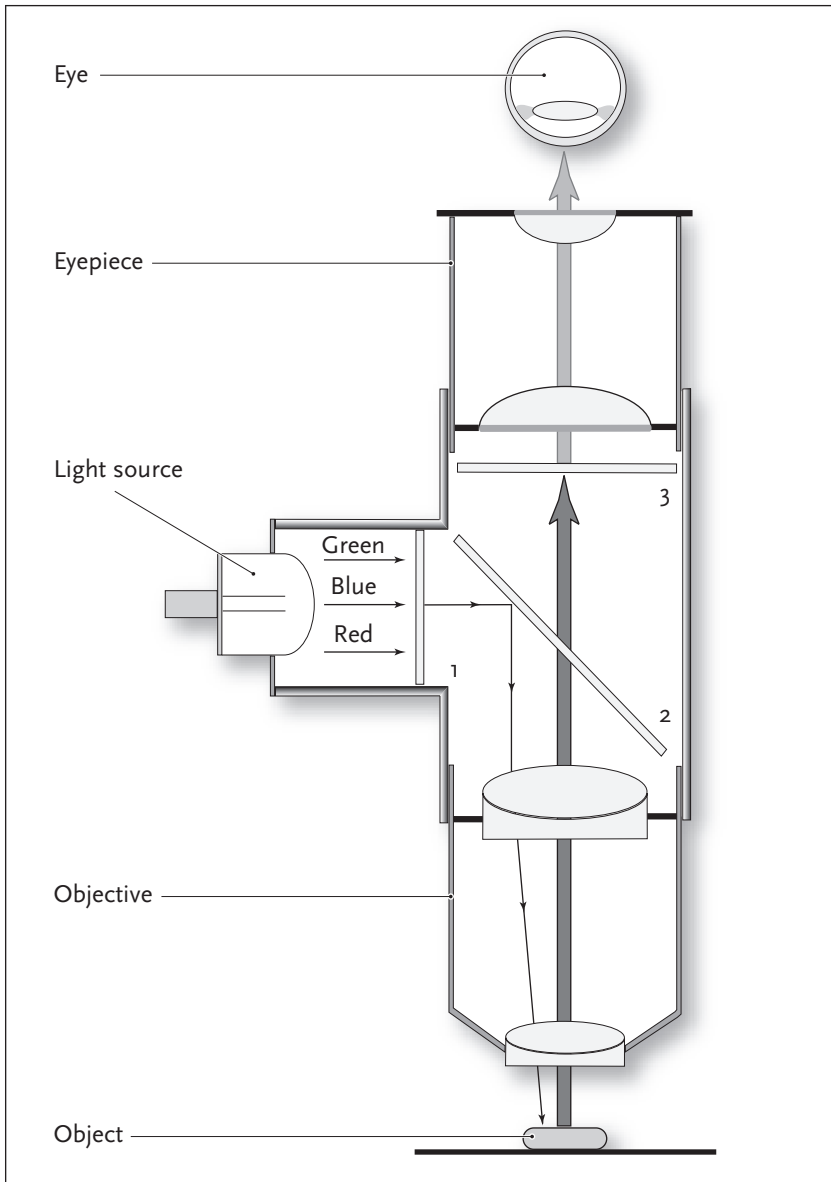
Monocular microscopes used at the turn of the 20th century could be carried around in a container not much bigger than a shoebox. Modern, fully equipped research microscopes are not only more powerful, they are also much bigger, weighing more than 100 pounds, and easily filling the top of a lab bench. Indeed, some of these microscopes are so large, and so elaborate, they often need their own room.

Histology and Histochemistry

Histology and histochemistry represent an overlapping collection of techniques that are used to study the structure and chemical properties of intact

tissues. Histology means literally the study of tissue, whereas, histochemistry refers to the use of specific stains to enhance the detection of microscopic detail, based on known chemical or physical properties of the tissue. Thus progress in histochemistry improved histology.

The microscopic characterization of human tissues, organs, and more than 200 human cell types, some of which are described in chapters 6 and 7, were originally obtained with a combination of histological and histochemical techniques. Studies involving these procedures are generally conducted on tissue sections or tissue smears that have been



A fluorescence microscope. A light source is filtered by a barrier filter (1) that blocks passage of all wavelengths, except for blue light, which is deflected by a beam-splitting mirror (2), directing it through the objective, where it stimulates fluorescent molecules in the object. Fluorescent light from the object passes freely through the beam splitter and is filtered of spurious fluorescence by a second barrier filter (3) before reaching the eye.

preserved with a chemical fixative and then stained with colored dyes. Histological preparations of tissue sections form the basis for our understanding of organ structure at the cellular level and is indispensable for the study of physiology. Histochemical analysis is usually focused more at the cellular level and was crucial for localizing DNA to the cell nucleus and for tracking the movement of molecules between the various sub-cellular compartments. Prominent examples included the movement of RNA from the nucleus to the cytoplasm, the movement of certain proteins from the cytoplasm back to the nucleus, and the transport of other proteins through the endoplasmic reticulum and Golgi complex.

HISTOLOGY

Tissues are usually fixed, or preserved, before they are stained with a dye. Two fixatives that are commonly used are formalin (10 percent formaldehyde solution) and a mixture of alcohol and glacial acetic acid (3:1), called Clark's solution. Formalin is preferred for most applica-



A modern binocular microscope that is used in cancer research. This particular instrument is connected to a computer, which displays the image field on the monitor and is equipped with software for analyzing and enhancing the image. (Courtesy of Jean-Paul Chassenet/Photo Researchers, Inc.)

tions, but Clark's solution is ideal for blood or chromosome smears. Staining tissues with various dyes to enhance the contrast of their microscopic image is a standard histological procedure. This approach takes advantage of biochemical differences between the various cellular compartments. The nucleus, for example, consists primarily of the negatively charged DNA, whereas the cytoplasm carries a net positive charge. Thus a positively charged dye, such as hemotoxylin, can be used to stain the nucleus, while a negatively charged dye, such as eosin, will stain the cytoplasm. These two dyes are normally used together; hemotoxylin stains the nucleus blue, while eosin counterstains the cytoplasm pink.

HISTOCHEMISTRY

Fixed tissues can be stained for specific cellular compounds such as DNA, RNA, lipids, and specific proteins. Of these procedures, the most important is the Feulgen reaction, which is used for detecting DNA. The Feulgen reaction involves hydrolyzing DNA nucleotides in a strong acid solution to expose aldehyde groups. The aldehydes, in turn, react with a special dye solution, called Schiff's reagent, resulting in a deep purple color. The Feulgen reaction is specific for DNA and does not stain other molecules in the nucleus, or the cytoplasm. This method has been used to determine the amount of DNA in individual cell nuclei and to assess the degree of chromatin condensation.

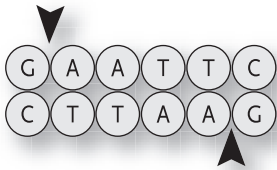
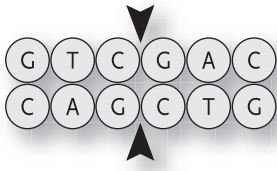
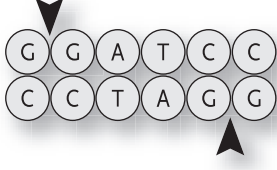
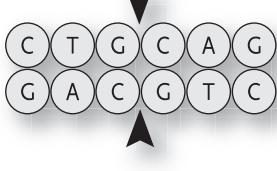
Recombinant DNA Technology

Recombinant technology is a collection of procedures that make it possible to isolate a gene and produce enough of it for a detailed study of its structure and function. Central to this technology is the ability to construct libraries of DNA fragments that represent the genetic repertoire of an entire organism, or of a specific cell type. Constructing these libraries involves splicing different pieces of DNA together to form a novel or recombinant genetic entity, from which the procedure derives its name. DNA cloning and library construction were made possible by the discovery of DNA modifying enzymes that can seal two pieces of DNA together, or can cut DNA at sequence-specific sites. Many of the procedures that are part of recombinant technology, such as DNA sequencing or filter hybridization, were developed in order to characterize DNA

fragments that were isolated from cells or gene libraries. Obtaining the sequence of a gene has made it possible to study the organization of the genome, but more important, it has provided a simple way of determining the protein sequence, and the expression profile for any gene.

DNA-MODIFYING ENZYMES

Two of the most important enzymes used in recombinant technology are those that can modify DNA by sealing two fragments together, and others that can cut DNA at specific sites. The first modifying enzyme to be discovered was DNA ligase, an enzyme that can join two pieces of

RESTRICTION ENZYMES		
Enzyme	Source	Sequence
<i>EcoRI</i>	<i>Escherichia coli</i>	
<i>HindII</i>	<i>Haemophilus influenzae</i>	
<i>BamHI</i>	<i>Bacillus amyloliquefaciens</i>	
<i>PstI</i>	<i>Providencia situartii</i>	

Arrows indicate the sites at which the enzyme cuts the DNA.

DNA together and is an important component of the cell's DNA replication and repair machinery. Other DNA modifying enzymes, called restriction enzymes, cut DNA at sequence-specific sites, with different members of the family cutting at different sites. Restriction enzymes are isolated from bacteria, and since their discovery in 1970, more than 90 such enzymes have been isolated from more than 230 bacterial strains; a few examples are shown in the figure on page 135.

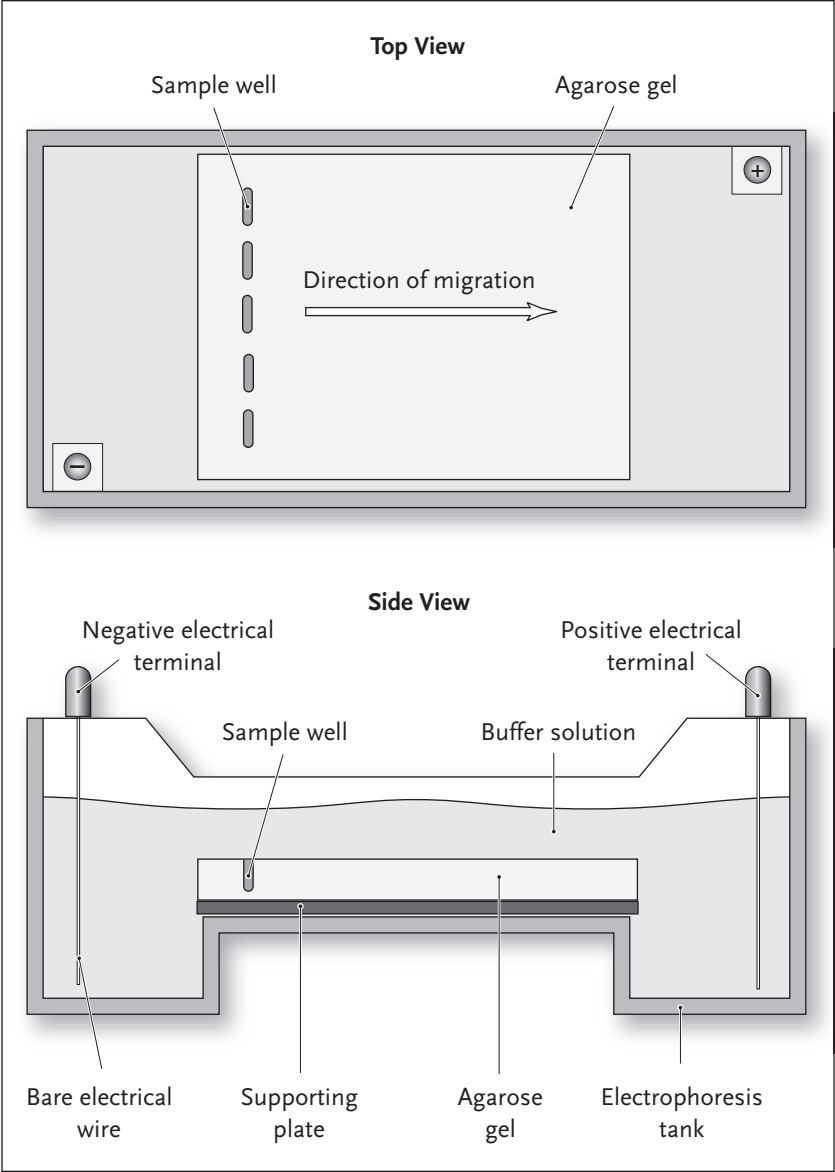
The name "restriction enzyme" is cryptic and calls for an explanation. During the period when prokaryotes began to appear on earth, their environment contained a wide assortment of molecules that were released into the soil or water by other cells, either deliberately or when the cells died. DNA of varying lengths was among these molecules and was readily taken up by living cells. If the foreign DNA contained complete genes from a competing bacterial species, there was the real possibility that those genes could have been transcribed and translated by the host cell with potentially fatal results. As a precaution, prokaryotes evolved a set of enzymes that would *restrict* the foreign DNA population by cutting it up into smaller pieces before being broken down completely to individual nucleotides.

GEL ELECTROPHORESIS

This procedure is used to separate different DNA and RNA fragments in a slab of agar or polyacrylamide subjected to an electric field. Nucleic acids carry a negative charge and thus will migrate toward a positively charged electrode. The gel acts as a sieving medium that impedes the movement of the molecules. Thus the rate at which the fragments migrate is a function of their size; small fragments migrate more rapidly than large fragments. The gel, containing the samples, is run submerged in a special pH-regulated solution, or buffer, containing a nucleic acid-specific dye, ethidium bromide. This dye produces a strong reddish-yellow fluorescence when exposed to ultraviolet (UV) radiation. Consequently, after electrophoresis, the nucleic acid can be detected by photographing the gel under UV illumination.

DNA CLONING

In 1973, scientists discovered that restriction enzymes, DNA ligase, and bacterial plasmids could be used to clone DNA molecules. Plasmids are



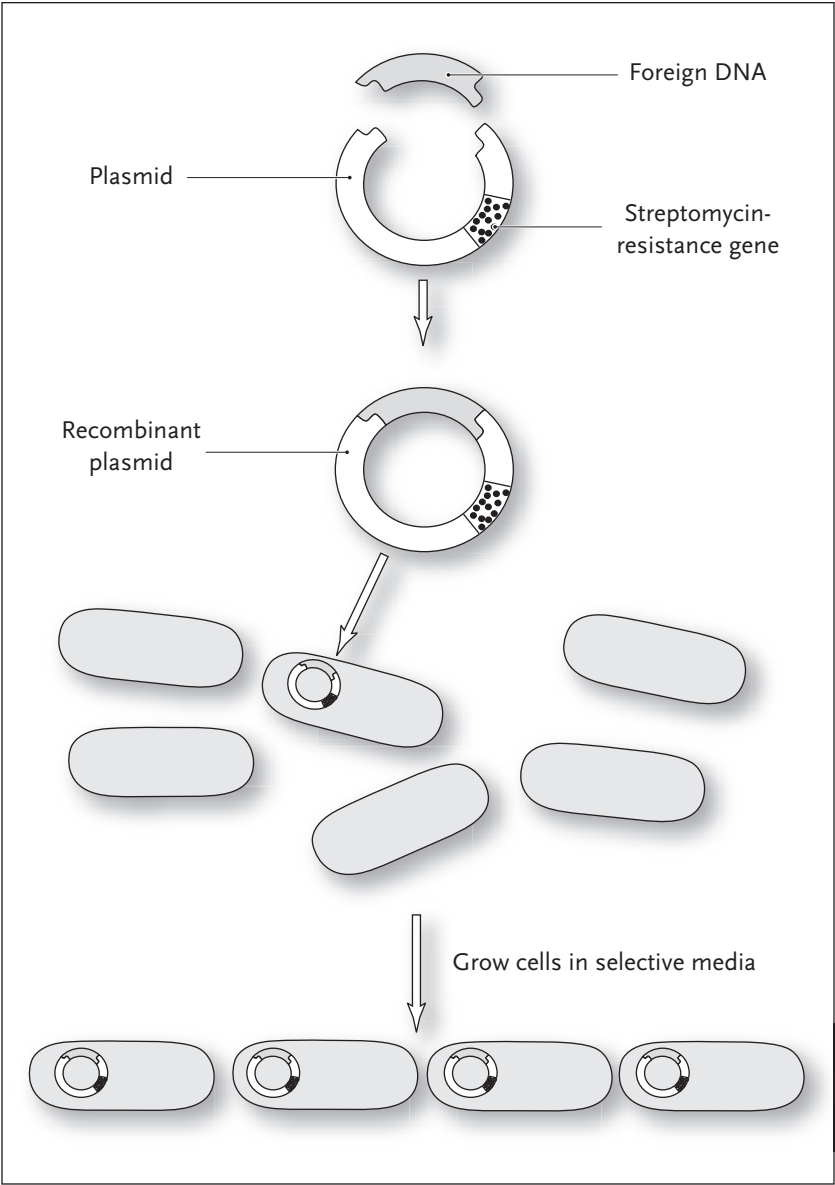
Agarose gel electrophoresis. An agarose gel is placed in an electrophoresis tank and submerged in a buffer solution. The electric terminals are connected to a power source, with the sample wells positioned near the negative terminal. When the current is turned on, the negatively charged nucleic acids migrate toward the positive terminal. The migration rate is an inverse function of molecular size (large molecules travel slower than small ones).

small (3–4 kb) circular minichromosomes that occur naturally in bacteria and are often exchanged between cells by passive diffusion. When a bacterium acquires a new plasmid it is said to have been transfected. For bacteria, the main advantage to swapping plasmids is that they often carry antibiotic resistance genes so that a cell sensitive to ampicillin can become resistant simply by acquiring the right plasmid.

The first cloning experiment used a plasmid from *Escherichia coli* that was cut with the restriction enzyme *EcoRI*. The plasmid had a single *EcoRI* site so the restriction enzyme simply opened the circular molecule, rather than cutting it up into many useless pieces. Foreign DNA, cut with the same restriction enzyme, was incubated with the plasmid. Because the plasmid and foreign DNA were both cut with *EcoRI*, the DNA could insert itself into the plasmid to form a hybrid, or recombinant plasmid, after which DNA ligase sealed the two together. The reaction mixture was added to a small volume of *E. coli* so that some of the cells could take up the recombinant plasmid before being transferred to a nutrient broth containing streptomycin. Only those cells carrying the recombinant plasmid, which contained an anti-streptomycin gene, could grow in the presence of this antibiotic. Each time the cells divided, the plasmid DNA was duplicated along with the main chromosome. After the cells had grown overnight, the foreign DNA had been amplified, or cloned, billions of times and was easily isolated for sequencing or expression studies.

GENOMIC AND cDNA LIBRARIES

The basic cloning procedure described above not only provides a way to amplify a specific piece of DNA, but it can also be used to construct gene libraries. In this case, however, the cloning vector is a bacteriophage, called lambda. The lambda genome is double-stranded linear DNA of about 40,000 base pairs, much of which can be replaced by foreign DNA without sacrificing the ability of the virus to infect bacteria. This is the great advantage of lambda over a plasmid. Lambda can accommodate very long pieces of DNA, often long enough to contain an entire gene, whereas a plasmid cannot accommodate foreign DNA that is larger than 2,000 base pairs. Moreover, bacteriophage has the natural ability to infect bacteria, so that the efficiency of transfection is 100x greater than it is for plasmids.



Cloning DNA in a plasmid. The foreign DNA and the plasmid are cut with the same restriction enzyme, allowed to fuse, and then sealed with DNA ligase. The recombinant plasmid is mixed with bacterial cells, some of which pick up the plasmid, allowing them to grow in a culture medium containing streptomycin. The bacteria's main chromosome is not shown.

The construction of a gene library begins by isolating genomic DNA and digesting it with a restriction enzyme to produce fragments of 1,000 to 10,000 base pairs. These fragments are ligated into lambda genomes, which are subjected to a packaging reaction to produce mature viral particles, most of which carry a different piece of the genomic DNA. This collection of viruses is called a genomic library and is used to study the structure and organization of specific genes. Clones from a library such as this contain the coding sequences, in addition to introns, intervening sequences, promoters, and enhancers. An alternative form of a gene library can be constructed by isolating mRNA from a specific cell type. This RNA is converted to the complementary DNA (cDNA) using an RNA-dependent DNA polymerase called reverse transcriptase. The cDNA is ligated to lambda genomes and packaged as for the genomic library. This collection of recombinant viruses is a cDNA library and only contains genes that were being expressed by the cells when the RNA was extracted. It does not include introns or controlling elements because these are lost during transcription and the processing that occurs in the cell to make mature mRNA. Thus a cDNA library is intended for the purpose of studying gene expression and the structure of the coding region only.

LABELING CLONED DNA

Many of the procedures used in the area of recombinant technology were inspired by the events that occur during DNA replication. This includes the labeling of cloned DNA for use as probes in expression studies, DNA sequencing, and PCR (described below). DNA replication involves duplicating one of the strands (the parent, or template strand) by linking nucleotides in an order specified by the template and depends on a large number of enzymes, the most important of which is DNA polymerase. This enzyme, guided by the template strand, constructs a daughter strand by linking nucleotides together. One such nucleotide is deoxyadenine triphosphate (dATP). Deoxyribonucleotides have a single hydroxyl group located at the 3' carbon of the sugar group, while the triphosphate is attached to the 5' carbon. The procedure for labeling DNA probes, developed in 1983, introduces radioactive nucleotides into a DNA molecule. This method supplies DNA polymerase with a single stranded DNA template, a primer, and the

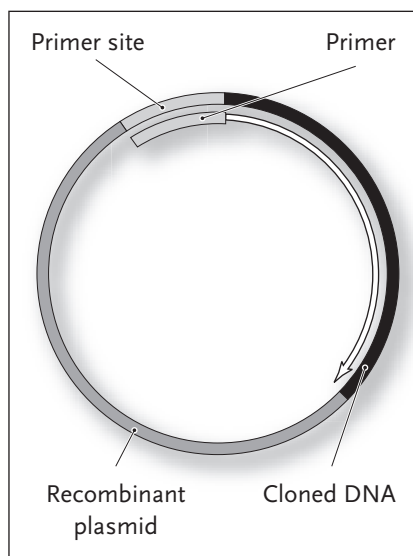
four nucleotides, in a buffered solution to induce *in vitro* replication. The daughter strand, which becomes the labeled probe, is made radioactive by including a ^{32}P -labeled nucleotide in the reaction mix. The radioactive nucleotide is usually deoxy-cytosine triphosphate (dCTP), or dATP. The ^{32}P is always part of the α (alpha) phosphate (the phosphate closest to the 5' carbon), as this is the one used by the polymerase to form the phosphodiester bond between nucleotides.

Single-stranded DNA hexamers (six bases long) are used as primers, and these are produced in such a way that they contain all possible permutations of four bases taken six at a time. Randomizing the base sequence for the primers ensures that there will be at least one primer site in a template that is only 50 bp long. Templates used in labeling reactions such as this are generally 100 to 800 bp long. This strategy of labeling DNA, known as random primer or oligo labeling, is widely used in cloning and in DNA and RNA filter hybridizations (described below).

DNA SEQUENCING

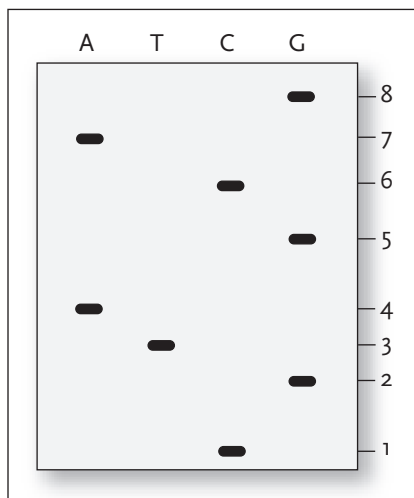
A sequencing reaction developed by the British biochemist Dr. Fred Sanger in 1976 is another technique that takes its inspiration from the natural process of DNA replication. DNA polymerase requires a primer with a free 3' hydroxyl group. The polymerase adds the first nucleotide to this group, and all subsequent bases are added to the 3' hydroxyl of the previous base. Sequencing by the Sanger method is usually

Plasmid primer site for DNA sequencing. The cloned DNA is inserted into the plasmid near an engineered primer site. Once the primer binds to the primer site, the cloned DNA may be replicated, as part of a sequencing reaction, in the direction indicated by the arrow. Only one strand of the double-stranded plasmid and cloned DNA is shown.



performed with the DNA cloned into a plasmid. This simplifies the choice of the initial primers because their sequence can be derived from the known plasmid sequence. An engineered plasmid primer site adjacent to a cloned DNA fragment is shown in the figure on page 140. Once the primer binds to the primer site, the cloned DNA may be replicated. Sanger’s innovation involved the synthesis of artificial nucleotides lacking the 3' hydroxyl group, thus producing di-deoxynucleotides (ddATP, ddCTP, ddGTP, and ddTTP). Incorporation of a di-deoxynucleotide terminates the growth of the daughter strand at that point, and this can be used to determine the size of each daughter strand. The shortest daughter strand represents the complementary nucleotide at the beginning of the template, whereas the longest strand represents the complementary nucleotide at the end of the template (see table below). The reaction products, consisting of all the daughter strands, are fractionated on a polyacrylamide gel. Polyacrylamide serves

EXAMPLE OF A SEQUENCING REACTION		
Tube	Reaction Products	
A	G-C-A-T-C-G-T-C	G-C-A-T-C-G-T-C
	C-G-T- A	C-G-T-A-G-C- A
T	G-C-A-T-C-G-T-C	
	C-G- T	
C	G-C-A-T-C-G-T-C	G-C-A-T-C-G-T-C
	C	C-G-T-A-G- C
	G-C-A-T-C-G-T-C	G-C-A-T-C-G-T-C
G	C- G	C-G-T-A- G
	G-C-A-T-C-G-T-C	
	C-G-T-A-G-C-A- G	
The Sanger sequencing reaction is set up in four separate tubes, each containing a different di-deoxynucleotide (ddATP, ddTTP, ddCTP, and ddGTP). The reaction products are shown for each of the tubes: A (ddATP), T (ddTTP), C (ddCTP), and G (ddGTP). The template strand is GCATCGTC. Replication of the template begins after the primer binds to the primer site on the sequencing plasmid. The di-deoxynucleotide terminating the reaction is shown in bold. The daughter strands, all of different lengths, are fractionated on a polyacrylamide gel.		



A representation of a sequencing gel. The reaction products (shown in the table on page 141) run from the top to the bottom, with the smallest fragment migrating at the highest rate. The sequence is read beginning with the smallest fragment on the gel (band # 1, in the "C" lane) and ending with the largest fragment at the top (band # 8, in the "G" lane). The sequence is CGTAGCAG. The complementary sequence is GCATCGTC. This is the template strand indicated in the table on page 141.

the same function as agarose. It has the advantage of being a tougher material, essential for the large size of a typical sequencing gel. Some of the nucleotides included in the Sanger reaction are labeled with a radioactive isotope so the fractionated daughter strands can be visualized by drying the gel and then exposing it to X-ray film. Thus the Sanger method uses the natural process of replication to mark the position of each nucleotide in the DNA fragment so the sequence of the cloned DNA can be determined.

The fragments are separated by size using polyacrylamide gel electrophoresis and visualized by exposing the dried gel to X-ray film. A representation of such a gel is shown in the figure at left. The sequence of the daughter strand is read beginning with the smallest fragment at the bottom of the gel and ending with the largest fragment at the top. The

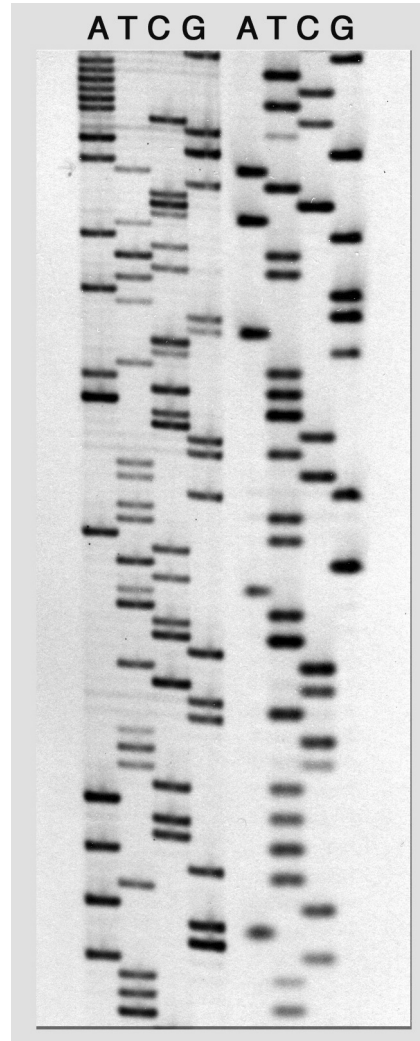
sequence of the template strand is obtained simply by taking the complement of the sequence obtained from the gel (the daughter strand).

SOUTHERN AND NORTHERN BLOTTING

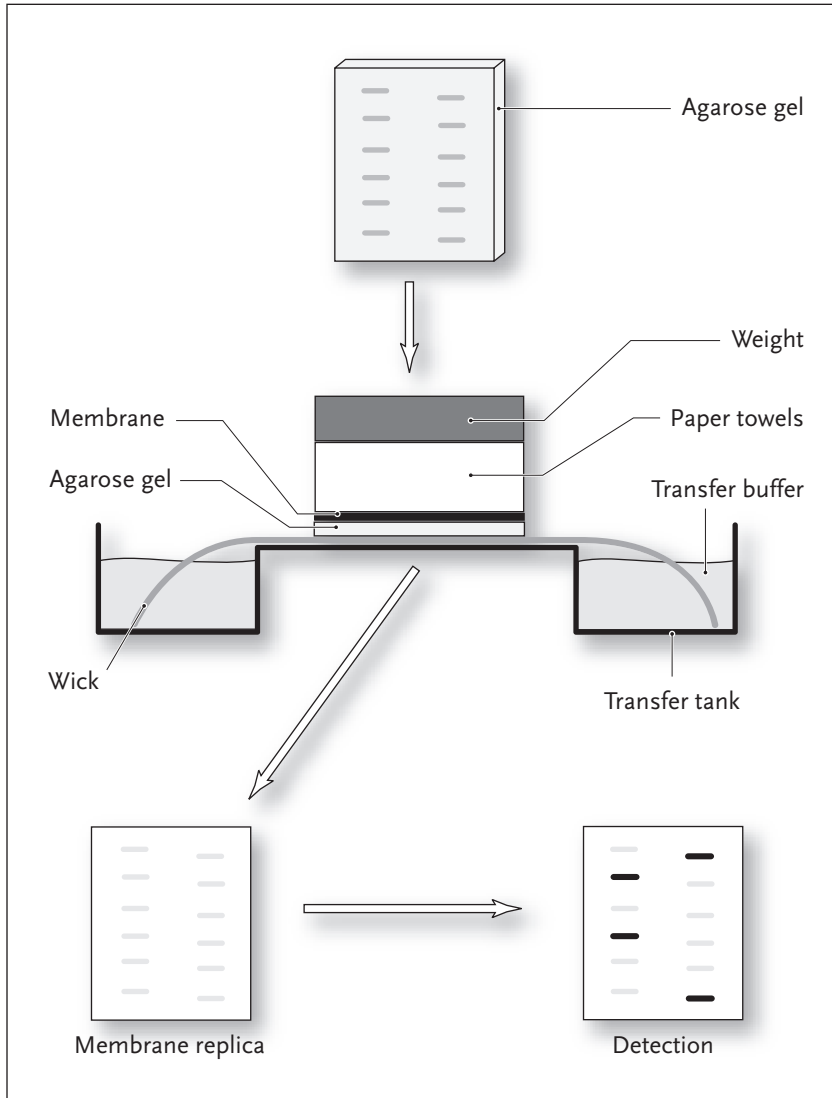
One of the most important techniques to be developed, as part of recombinant technology, is the transfer of nucleic acids from an agarose gel to nylon filter paper that can be hybridized to a labeled probe to detect specific genes. This procedure was introduced in 1975 by the Scottish scientist E. M. Southern for transferring DNA and is now

known as Southern blotting. Since the DNA is transferred to filter paper, the detection stage is known as filter hybridization. In 1980, the procedure was modified to transfer RNA to nylon membranes for the study of gene expression and, in reference to the original, is called northern blotting.

Northern blotting is used to study the expression of specific genes, and is usually performed on messenger RNA (mRNA). Typical experiments may wish to determine the expression of specific genes in normal, versus cancerous tissue, or tissues obtained from groups of different ages. The RNA is fractionated on an agarose gel and then transferred to a nylon membrane. The paper towels placed on top of the assembly pull the transfer buffer through the gel, eluting the RNA from the gel and trapping it on the membrane. The location of specific mRNA can be determined by hybridizing the membrane to a radiolabeled cDNA or genomic clone. The hybridization procedure involves placing the filter in a buffer solution containing a labeled probe. During a long incubation period, the probe binds to the target sequence immobilized on the membrane. A-T and G-C base pairing mediate the binding between the probe and target.



An autoradiogram of a portion of a DNA sequencing gel. A partial sequence (the first 20 bases) of the left set, beginning at the bottom of the "T" lane, is TTTAGGATGACCACTTTGGC. (Courtesy of Dr. Joseph Panno)



Northern transfer and membrane hybridization. RNA is fractionated on an agarose gel and then placed face down on a paper wick in a transfer tank. The gel is overlaid with a piece of nylon membrane, paper towels, and weight. The paper towels draw the buffer through the gel and the membrane. The flow of buffer elutes the RNA from the gel transferring it to the membrane. A radiolabeled cDNA probe is hybridized to the membrane to detect specific mRNA transcripts. Note that the thickness of the membrane is exaggerated for clarity.

The double-stranded molecule that is formed is a hybrid, being formed between the RNA target, on the membrane, and the DNA probe.

FLUORESCENT IN SITU HYBRIDIZATION (FISH)

Studying gene expression does not always depend on northern blots and filter hybridization. In the 1980s, scientists found that cDNA probes could be hybridized to DNA or mRNA *in situ*, that is, while located within cells or tissue sections fixed on a microscope slide. In this case, the probe is labeled with a fluorescent dye molecule, rather than a radioactive isotope. The samples are then examined and photographed under a fluorescent microscope. FISH is an extremely powerful variation on Southern and northern blots. This procedure gives precise information regarding the identity of a cell that expresses a specific gene, information that usually cannot be obtained with filter hybridization. Organs and tissues are generally composed of many different kinds of cells, which cannot be separated from each other using standard biochemical extraction procedures. Histological sections, however, show clearly the various cell types and, when subjected to FISH analysis, provide clear results as to which cells express specific genes. FISH is also used in clinical laboratories for the diagnosis of genetic abnormalities.

POLYMERASE CHAIN REACTION (PCR)

PCR is simply repetitive DNA replication over a limited, primer-defined region of a suitable template. The region defined by the primers is amplified to such an extent that it can be easily isolated for further study. The reaction exploits the fact that a DNA duplex, in a low-salt buffer, will melt (i.e., separate into two single strands) at 75° C but will reanneal (rehybridize) at 37° C. The reaction is initiated by melting the template in the presence of primers and polymerase in a suitable buffer, cooling quickly to 37° C, and allowing sufficient time for the polymerase to replicate both strands of the template. The temperature is then increased to 75° C to melt the newly formed duplexes and then cooled to 37° C. At the lower temperature, more primer will anneal to initiate another round of replication. The heating-cooling cycle is repeated 20 to 30 times, after which the reaction products are fractionated on an agarose gel and photographed. The band containing the amplified fragment may be cut out of the gel and purified for further study. The DNA

polymerase used in these reactions is isolated from thermophilic bacteria that can withstand temperatures of 70° C to 80° C. PCR applications are nearly limitless. It is used to amplify DNA from samples containing, at times, no more than a few cells. It can be used to screen libraries and to identify genes that are turned on or off during embryonic development or during cellular transformation.

GLOSSARY



acetyl A chemical group derived from acetic acid. Important in energy metabolism and for the modification of proteins.

acetylcholine A neurotransmitter released at axonal terminals by cholinergic neurons. Found in the central and peripheral nervous system and released at the vertebrate neuromuscular junction.

acetyl-CoA A water-soluble molecule, coenzyme A (CoA), that carries acetyl groups in cells.

acid A substance that releases protons when dissolved in water. Carries a net negative charge.

actin filament A protein filament formed by the polymerization of globular actin molecules. Forms the cytoskeleton of all eukaryotes and part of the contractile apparatus of skeletal muscle.

action potential A self-propagating electrical impulse that occurs in the membranes of neurons, muscles, photoreceptors, and hair cells of the inner ear.

active transport Movement of molecules across the cell membrane, utilizing the energy stored in ATP.

adenylate cyclase A membrane-bound enzyme that catalyzes the conversion of ATP to cyclic AMP. An important component of cell-signaling pathways.

adherens junction A cell junction in which the cytoplasmic face of the membrane is attached to actin filaments.

adipocyte A fat cell.

adrenaline (epinephrine) A hormone released by chromaffin cells in the adrenal gland. Prepares an animal for extreme activity, increases the heart rate and blood-sugar levels.

adult stem cells Stem cells isolated from adult tissues, such as bone marrow or epithelium.

aerobic Refers to a process that either requires oxygen or occurs in its presence.

allele An alternate form of a gene. Diploid organisms have two alleles for each gene, located at the same locus (position) on homologous chromosomes.

allogeneic transplant A patient receives a tissue or organ transplant from an unrelated individual.

alpha helix A common folding pattern of proteins in which a linear sequence of amino acids twists into a right-handed helix stabilized by hydrogen bonds.

amino acid An organic molecule containing amino and carboxyl groups that is a building block of protein.

aminoacyl-tRNA An amino acid linked by its carboxyl group to a hydroxyl group on tRNA.

aminoacyl-tRNA synthetase An enzyme that attaches the correct amino acid to a tRNA.

amino terminus The end of a protein or polypeptide chain that carries a free amino group.

amphipathic Having both hydrophilic and hydrophobic regions, as in a phospholipid.

anabolism A collection of metabolic reactions in a cell whereby large molecules are made from smaller ones.

anaerobic A cellular metabolism that does not depend on molecular oxygen.

anaphase A mitotic stage in which the two sets of chromosomes move away from each other toward opposite and spindle poles.

anchoring junction A cell junction that attaches cells to each other.

angiogenesis Sprouting of new blood vessels from preexisting ones.

angstrom A unit of length, equal to 10^{-10} meter or 0.1 nanometer (nm), that is used to measure molecules and atoms.

anterior A position close to or at the head end of the body.

antibiotic A substance made by bacteria, fungi, and plants that is toxic to microorganisms. Common examples are penicillin and streptomycin.

antibody A protein made by B cells of the immune system in response to invading microbes.

- anticodon** A sequence of three nucleotides in tRNA that is complementary to a messenger RNA codon.
- antigen** A molecule that stimulates an immune response, leading to the formation of antibodies.
- antigen-presenting cell** A cell of the immune system, such as a monocyte, that presents pieces of an invading microbe (the antigen) to lymphocytes.
- antiparallel** The relative orientation of the two strands in a DNA double helix; the polarity of one strand is oriented in the opposite direction to the other.
- antiporter** A membrane carrier protein that transports two different molecules across a membrane in opposite directions.
- apoptosis** Regulated or programmed form of cell death that may be activated by the cell itself or by the immune system to force cells to commit suicide when they become infected with a virus.
- asexual reproduction** The process of forming new individuals without gametes or the fertilization of an egg by a sperm. Individuals produced this way are identical to the parent and referred to as a clone.
- aster** The star-shaped arrangement of microtubules that is characteristic of a mitotic or meiotic spindle.
- ATP (adenosine triphosphate)** A nucleoside consisting of adenine, ribose, and three phosphate groups that is the main carrier of chemical energy in the cell.
- ATPase** Any enzyme that catalyzes a biochemical reaction by extracting the necessary energy from ATP.
- ATP synthase** A protein located in the inner membrane of the mitochondrion that catalyzes the formation of ATP from ADP and inorganic phosphate using the energy supplied by the electron transport chain.
- autogeneic transplant** A patient receives a transplant of his or her own tissue.
- autosome** Any chromosome other than a sex chromosome.
- axon** A long extension of a neuron's cell body that transmits an electrical signal to other neurons.
- axonal transport** The transport of organelles, such as Golgi vesicles, along an axon to the axonal terminus. Transport also flows from the terminus to the cell body.

bacteria One of the most ancient forms of cellular life (the other is the Archaea). Bacteria are prokaryotes and some are known to cause disease.

bacterial artificial chromosome (BAC) A cloning vector that accommodates DNA inserts of up to 1 million base pairs.

bacteriophage A virus that infects bacteria. Bacteriophages were used to prove that DNA is the cell's genetic material and are now used as cloning vectors.

base A substance that can accept a proton in solution. The purines and pyrimidines in DNA and RNA are organic bases and are often referred to simply as bases.

base pair Two nucleotides in RNA or DNA that are held together by hydrogen bonds. Adenine bound to thymine or guanine bound to cytosine are examples of base pairs.

B cell (B lymphocyte) A white blood cell that makes antibodies and is part of the adaptive immune response.

benign Tumors that grow to a limited size and do not spread to other parts of the body.

beta sheet Common structural motif in proteins in which different strands of the protein run alongside each other and are held together by hydrogen bonds.

biopsy The removal of cells or tissues for examination under a microscope. When only a sample of tissue is removed, the procedure is called an incisional biopsy or core biopsy. When an entire lump or suspicious area is removed, the procedure is called an excisional biopsy. When a sample of tissue or fluid is removed with a needle, the procedure is called a needle biopsy or fine-needle aspiration.

biosphere The world of living organisms.

bivalent A duplicated chromosome paired with its homologous duplicated chromosome at the beginning of meiosis.

blastomere A cell formed by the cleavage of a fertilized egg. Blastomeres are the totipotent cells of the early embryo.

blotting A technique for transferring DNA (Southern blotting), RNA (northern blotting), or proteins (western blotting) from an agarose or polyacrylamide gel to a nylon membrane.

BRCA1 (breast cancer gene 1) A gene on chromosome 17 that may be involved in regulating the cell cycle. A person who inherits an

altered version of the BRCA1 gene has a higher risk of getting breast, ovarian, or prostate cancer.

BRCA2 (breast cancer gene 2) A gene on chromosome 13 that, when mutated, increases the risk of getting breast, ovarian, or prostate cancer.

budding yeast The common name for the baker's yeast *Saccharomyces cerevisiae*, a popular experimental organism that reproduces by budding off a parental cell.

cadherin Belongs to a family of proteins that mediates cell-to-cell adhesion in animal tissues.

calorie A unit of heat. One calorie is the amount of heat needed to raise the temperature of one gram of water by 1°C. Kilocalories (1,000 calories) are used to describe the energy content of foods.

capsid The protein coat of a virus, formed by auto-assembly of one or more proteins into a geometrically symmetrical structure.

carbohydrate A general class of compounds that includes sugars, containing carbon, hydrogen, and oxygen.

carboxyl group A carbon atom attached to an oxygen and a hydroxyl group.

carboxyl terminus The end of a protein containing a carboxyl group.

carcinogen A compound or form of radiation that can cause cancer.

carcinogenesis The formation of a cancer.

carcinoma Cancer of the epithelium, representing the majority of human cancers.

cardiac muscle Muscle of the heart. Composed of myocytes that are linked together in a communication network based on free passage of small molecules through gap junctions.

caspase A protease involved in the initiation of apoptosis.

catabolism Enzyme-regulated breakdown of large molecules for the extraction of chemical-bond energy. Intermediate products are called catabolites.

catalyst A substance that lowers the activation energy of a reaction.

CD28 Cell-surface protein located in T cell membranes, necessary for the activation of T cells by foreign antigens.

cDNA (complementary DNA) DNA that is synthesized from mRNA, thus containing the complementary sequence. cDNA contains coding sequence but not the regulatory sequences that are present in the

genome. Labeled probes are made from cDNA for the study of gene expression.

cell adhesion molecule (CAM) A cell-surface protein that is used to connect cells to each other.

cell body The main part of a cell containing the nucleus, Golgi complex, and endoplasmic reticulum. Used in reference to neurons that have long processes (dendrites and axons) extending some distance from the nucleus and cytoplasmic machinery.

cell coat See **glycocalyx**.

cell-cycle control system A team of regulatory proteins that governs progression through the cell cycle.

cell-division-cycle gene (*cdc* gene) A gene that controls a specific step in the cell cycle.

cell fate The final differentiated state that a pluripotent embryonic cell is expected to attain.

cell-mediated immune response Activation of specific cells to launch an immune response against an invading microbe.

cell nuclear replacement Animal-cloning technique whereby a somatic cell nucleus is transferred to an enucleated oocyte. Synonymous with somatic-cell nuclear transfer.

central nervous system (CNS) That part of a nervous system that analyzes signals from the body and the environment. In animals, the CNS includes the brain and spinal cord.

centriole A cylindrical array of microtubules that is found at the center of a centrosome in animal cells.

centromere A region of a mitotic chromosome that holds sister chromatids together. Microtubules of the spindle fiber connect to an area of the centromere called the kinetochore.

centrosome Organizes the mitotic spindle and the spindle poles. In most animal cells it contains a pair of centrioles.

chiasma (plural chiasmata) An X-shaped connection between homologous chromosomes that occurs during meiosis I, representing a site of crossing-over, or genetic exchange between the two chromosomes.

chromatid A duplicate chromosome that is still connected to the original at the centromere. The identical pair are called sister chromatids.

chromatin A complex of DNA and proteins (histones and nonhistones) that forms each chromosome and is found in the nucleus of all eukaryotes. Decondensed and threadlike during interphase.

chromatin condensation Compaction of different regions of interphase chromosomes that is mediated by the histones.

chromosome One long molecule of DNA that contains the organism's genes. In prokaryotes, the chromosome is circular and naked; in eukaryotes, it is linear and complexed with histone and nonhistone proteins.

chromosome condensation Compaction of entire chromosomes in preparation for cell division.

clinical breast exam An exam of the breast performed by a physician to check for lumps or other changes.

cyclic adenosine monophosphate (cAMP) A second messenger in a cell-signaling pathway that is produced from ATP by the enzyme adenylate cyclase.

cyclin A protein that activates protein kinases (cyclin-dependent protein kinases, or Cdk) that control progression from one state of the cell cycle to another.

cytochemistry The study of the intracellular distribution of chemicals.

cytochrome Colored, iron-containing protein that is part of the electron transport chain.

cytotoxic T cell A T lymphocyte that kills infected body cells.

dendrite An extension of a nerve cell that receives signals from other neurons.

dexrazoxane A drug used to protect the heart from the toxic effects of anthracycline drugs such as doxorubicin. It belongs to the family of drugs called chemoprotective agents.

dideoxy sequencing A method for sequencing DNA that employs dideoxyribose nucleotides.

diploid A genetic term meaning two sets of homologous chromosomes, one set from the mother and the other from the father. Thus diploid organisms have two versions (alleles) of each gene in the genome.

DNA (deoxyribonucleic acid) A long polymer formed by linking four different kinds of nucleotides together like beads on a string. The sequence of nucleotides is used to encode an organism's genes.

DNA helicase An enzyme that separates and unwinds the two DNA strands in preparation for replication or transcription.

DNA library A collection of DNA fragments that are cloned into plasmids or viral genomes.

DNA ligase An enzyme that joins two DNA strands together to make a continuous DNA molecule.

DNA microarray A technique for studying the simultaneous expression of a very large number of genes.

DNA polymerase An enzyme that synthesizes DNA using one strand as a template.

DNA primase An enzyme that synthesizes a short strand of RNA that serves as a primer for DNA replication.

dorsal The backside of an animal. Also refers to the upper surface of anatomical structures, such as arms or wings.

dorsoventral The body axis running from the backside to the frontside or the upperside to the underside of a structure.

double helix The three-dimensional structure of DNA in which the two strands twist around each other to form a spiral.

doxorubicin An anticancer drug that belongs to a family of antitumor antibiotics.

Drosophila melanogaster Small species of fly, commonly called a fruit fly, that is used as an experimental organism in genetics, embryology, and gerontology.

ductal carcinoma in situ (DCIS) Abnormal cells that involve only the lining of a breast duct. The cells have not spread outside the duct to other tissues in the breast. Also called intraductal carcinoma.

dynein A motor protein that is involved in chromosome movements during cell division.

dysplasia Disordered growth of cells in a tissue or organ, often leading to the development of cancer.

ectoderm An embryonic tissue that is the precursor of the epidermis and the nervous system.

electrochemical gradient A differential concentration of an ion or molecule across the cell membrane that serves as a source of potential energy and may polarize the cell electrically.

electron microscope A microscope that uses electrons to produce a high-resolution image of the cell.

- embryogenesis** The development of an embryo from a fertilized egg.
- embryonic stem cell (ES cell)** A pluripotent cell derived from the inner cell mass (the cells that give rise to the embryo instead of the placenta) of a mammalian embryo.
- endocrine cell** A cell that is specialized for the production and release of hormones. Such cells make up hormone-producing tissue such as the pituitary gland or gonads.
- endocytosis** Cellular uptake of material from the environment by invagination of the cell membrane to form a vesicle called an endosome. The endosome's contents are made available to the cell after it fuses with a lysosome.
- endoderm** An embryonic tissue layer that gives rise to the gut.
- endoplasmic reticulum (ER)** Membrane-bounded chambers that are used to modify newly synthesized proteins with the addition of sugar molecules (glycosylation). When finished, the glycosylated proteins are sent to the Golgi apparatus in exocytotic vesicles.
- endothelial cell** A cell that forms the endothelium, a thin sheet of cells lining the inner surface of all blood vessels.
- enhancer** A DNA regulatory sequence that provides a binding site for transcription factors capable of increasing the rate of transcription for a specific gene. Often located thousands of base pairs away from the gene it regulates.
- enveloped virus** A virus containing a capsid that is surrounded by a lipid bilayer originally obtained from the membrane of a previously infected cell.
- enzyme** A protein or RNA that catalyzes a specific chemical reaction.
- epidermis** The epithelial layer, or skin, that covers the outer surface of the body.
- ER signal sequence** The amino terminal sequence that directs proteins to enter the endoplasmic reticulum (ER). This sequence is removed once the protein enters the ER.
- erythrocyte** A red blood cell that contains the oxygen-carrying pigment hemoglobin used to deliver oxygen to cells in the body.
- Escherichia coli* (E. coli)** Rod shape, gram negative bacterium that inhabits the intestinal tract of most animals and is used as an experimental organism by geneticists and biomedical researchers.

euchromatin Lightly staining portion of interphase chromatin, in contrast to the darkly staining heterochromatin (condensed chromatin). Euchromatin contains most, if not all, of the active genes.

eukaryote (eucaryote) A cell containing a nucleus and many membrane-bounded organelles. All life-forms, except bacteria and viruses, are composed of eukaryote cells.

exocytosis The process by which molecules are secreted from a cell. Molecules to be secreted are located in Golgi-derived vesicles that fuse with the inner surface of the cell membrane, depositing the contents into the intercellular space.

exon Coding region of a eukaryote gene that is represented in messenger RNA, and thus directs the synthesis of a specific protein.

expression studies Examination of the type and quantity of mRNA or protein that is produced by cells, tissues, or organs.

fat A lipid material, consisting of triglycerides (fatty acids bound to glycerol), that is stored in adipocytes as an energy reserve.

fatty acid A compound that has a carboxylic acid attached to a long hydrocarbon chain. A major source of cellular energy and a component of phospholipids.

fertilization The fusion of haploid male and female gametes to form a diploid zygote.

fibroblast The cell type that, by secreting an extracellular matrix, gives rise to the connective tissue of the body.

filter hybridization The detection of specific DNA or RNA molecules, fixed on a nylon filter, by incubating the filter with a labelled probe that hybridizes to the target sequence.

fixative A chemical that is used to preserve cells and tissues. Common examples are formaldehyde, methanol, and acetic acid.

flagellum (plural flagella) Whiplike structure found in prokaryotes and eukaryotes that are used to propel cells through water.

fluorescein Fluorescent dye that produces a green light when illuminated with ultraviolet or blue light.

fluorescent dye A dye that absorbs UV or blue light and emits light of a longer wavelength, usually as green or red light.

fluorescent microscope A microscope that is equipped with special filters and a beam splitter for the examination of tissues and cells stained with a fluorescent dye.

- follicle cell** Cells that surround and help feed a developing oocyte.
- G₀** G “zero” refers to a phase of the cell cycle. State of withdrawal from the cycle as the cell enters a resting or quiescent stage. Occurs in differentiated body cells as well as developing oocytes.
- G₁** Gap 1 refers to the phase of the cell cycle that occurs just after mitosis and before the next round of DNA synthesis.
- G₂** Gap 2 refers to the phase of the cell cycle that follows DNA replication and precedes mitosis.
- gap junction** A communication channel in the membranes of adjacent cells that allows free passage of ions and small molecules.
- gastrulation** An embryological event in which a spherical embryo is converted into an elongated structure with a head end, a tail end, and a gut (gastrula).
- gene** A region of the DNA that specifies a specific protein or RNA molecule that is handed down from one generation to the next. This region includes both the coding, noncoding, and regulatory sequences.
- gene regulatory protein** Any protein that binds to DNA and thereby affects the expression of a specific gene.
- gene repressor protein** A protein that binds to DNA and blocks transcription of a specific gene.
- gene therapy** A method for treating disease whereby a defective gene, causing the disease, is either repaired, replaced, or supplemented with a functional copy.
- genetic code** A set of rules that assigns a specific DNA or RNA triplet, consisting of a three-base sequence, to a specific amino acid.
- genome** All of the genes that belong to a cell or an organism.
- genomic library** A collection of DNA fragments, obtained by digesting genomic DNA with a restriction enzyme, that are cloned into plasmid or viral vectors.
- genomics** The study of DNA sequences and their role in the function and structure of an organism.
- genotype** The genetic composition of a cell or organism.
- germ cell** Cells that develop into gametes, either sperm or oocytes.
- glucose** Six-carbon monosaccharide (sugar) that is the principal source of energy for many cells and organisms. Stored as glycogen

in animal cells and as starch in plants. Wood is an elaborate polymer of glucose and other sugars.

glycerol A three-carbon alcohol that is an important component of phospholipids.

glycocalyx A molecular “forest,” consisting of glycosylated proteins and lipids, that covers the surface of every cell. The glycoproteins and glycolipids, carried to the cell membrane by Golgi-derived vesicles, have many functions, including the formation of ion channels, cell-signaling receptors and transporters.

glycogen A polymer of glucose used to store energy in an animal cell.

glycolysis The degradation of glucose with production of ATP.

glycoprotein Any protein that has a chain of glucose molecules (oligosaccharide) attached to some of the amino acid residues.

glycosylation The process of adding one or more sugar molecules to proteins or lipids.

glycosyl transferase An enzyme in the Golgi complex that adds glucose to proteins.

Golgi complex (Golgi apparatus) Membrane-bounded organelle in eukaryote cells that receives glycoproteins from the ER, which are modified and sorted before being sent to their final destination. The Golgi complex is also the source of glycolipids that are destined for the cell membrane. The glycoproteins and glycolipids leave the Golgi by exocytosis. This organelle is named after the Italian histologist Camillo Golgi, who discovered it in 1898.

granulocyte A type of white blood cell that includes the neutrophils, basophils, and eosinophils.

growth factor A small protein (polypeptide) that can stimulate cells to grow and proliferate.

haploid Having only one set of chromosomes. A condition that is typical in gametes, such as sperm and eggs.

HeLa cell A tumor-derived cell line, originally isolated from a cancer patient in 1951. Currently used by many laboratories to study the cell biology of cancer and carcinogenesis.

helix-loop-helix A structural motif common to a group of gene regulatory proteins.

helper T cell A type of T lymphocyte that helps stimulate B cells to make antibodies directed against a specific microbe or antigen.

hemoglobin An iron-containing protein complex, located in red blood cells that picks up oxygen in the lungs and carries it to other tissues and cells of the body.

hemopoiesis Production of blood cells, occurring primarily in the bone marrow.

hepatocyte A liver cell.

heterochromatin A region of a chromosome that is highly condensed and transcriptionally inactive.

histochemistry The study of chemical differentiation of tissues.

histology The study of tissues.

histone Small nuclear proteins, rich in the amino acids arginine and lysine, that form the nucleosome in eukaryote nuclei, a beadlike structure that is a major component of chromatin.

HIV The human immunodeficiency virus that is responsible for AIDS.

homolog One of two or more genes that have a similar sequence and are descended from a common ancestor gene.

homologous Organs or molecules that are similar in structure because they have descended from a common ancestor. Used primarily in reference to DNA and protein sequences.

homologous chromosomes Two copies of the same chromosome, one inherited from the mother and the other from the father.

hormone A signaling molecule, produced and secreted by endocrine glands. Usually released into general circulation for coordination of an animal's physiology.

housekeeping gene A gene that codes for a protein that is needed by all cells, regardless of the cell's specialization. Genes encoding enzymes involved in glycolysis and the Krebs cycle are common examples.

hybridization A term used in molecular biology (recombinant DNA technology) meaning the formation of a double-stranded nucleic acid through complementary base-pairing. A property that is exploited in filter hybridization, a procedure that is used to screen gene libraries and to study gene structure and expression.

hydrophilic A polar compound that mixes readily with water.

hydrophobic A nonpolar molecule that dissolves in fat and lipid solutions but not in water.

hydroxyl group (-OH) Chemical group consisting of oxygen and hydrogen that is a prominent part of alcohol.

image analysis A computerized method for extracting information from digitized microscopic images of cells or cell organelles.

immunofluorescence Detection of a specific cellular protein with the aid of a fluorescent dye that is coupled to an antibody.

immunoglobulin (Ig) An antibody made by B cells as part of the adaptive immune response.

incontinence Inability to control the flow of urine from the bladder (urinary incontinence) or the escape of stool from the rectum (fecal incontinence).

in situ hybridization A method for studying gene expression, whereby a labeled cDNA or RNA probe hybridizes to a specific mRNA in intact cells or tissues. The procedure is usually carried out on tissue sections or smears of individual cells.

insulin Polypeptide hormone secreted by β (beta) cells in the vertebrate pancreas. Production of this hormone is regulated directly by the amount of glucose that is in the blood.

interleukin A small protein hormone, secreted by lymphocytes, to activate and coordinate the adaptive immune response.

interphase The period between each cell division, which includes the G_1 , S, and G_2 phases of the cell cycle.

intron A section of a eukaryotic gene that is noncoding. It is transcribed, but does not appear in the mature mRNA.

in vitro Refers to cells growing in culture, or a biochemical reaction occurring in a test tube (Latin for “in glass”).

in vivo A biochemical reaction, or a process, occurring in living cells or a living organism (Latin for “in life”).

ion An atom that has gained or lost electrons, thus acquiring a charge. Common examples are Na^+ and Ca^{++} ions.

ion channel A transmembrane channel that allows ions to diffuse across the membrane and down their electrochemical gradient.

Jak-STAT signaling pathway One of several cell-signaling pathways that activates gene expression. The pathway is activated through cell-surface receptors and cytoplasmic Janus kinases (Jaks), and signal transducers and activators of transcription (STATs).

- karyotype** A pictorial catalog of a cell's chromosomes, showing their number, size, shape, and overall banding pattern.
- keratin** Proteins produced by specialized epithelial cells called keratinocytes. Keratin is found in hair, fingernails, and feathers.
- kinesin** A motor protein that uses energy obtained from the hydrolysis of ATP to move along a microtubule.
- kinetochore** A complex of proteins that forms around the centromere of mitotic or meiotic chromosomes, providing an attachment site for microtubules. The other end of each microtubule is attached to a chromosome.
- Krebs cycle (citric acid cycle)** The central metabolic pathway in all eukaryotes and aerobic prokaryotes, discovered by the German chemist Hans Krebs in 1937. The cycle oxidizes acetyl groups derived from food molecules. The end products are CO_2 , H_2O , and high-energy electrons, which pass via NADH and FADH₂ to the respiratory chain. In eukaryotes, the Krebs cycle is located in the mitochondria.
- labeling reaction** The addition of a radioactive atom or fluorescent dye to DNA or RNA for use as a probe in filter hybridization.
- lagging strand** One of the two newly synthesized DNA strands at a replication fork. The lagging strand is synthesized discontinuously, and therefore, its completion lags behind the second, or leading, strand.
- lambda bacteriophage** A viral parasite that infects bacteria. Widely used as a DNA cloning vector.
- leading strand** One of the two newly synthesized DNA strands at a replication fork. The leading strand is made by continuous synthesis in the 5' to 3' direction.
- leucine zipper** A structural motif of DNA binding proteins, in which two identical proteins are joined together at regularly spaced leucine residues, much like a zipper, to form a dimer.
- leukemia** Cancer of white blood cells.
- lipid bilayer** Two closely aligned sheets of phospholipids that form the core structure of all cell membranes. The two layers are aligned such that the hydrophobic tails are interior, while the hydrophilic head groups are exterior on both surfaces.

liposome An artificial lipid bilayer vesicle used in membrane studies and as an artificial gene therapy vector.

locus A term from genetics that refers to the position of a gene along a chromosome. Different alleles of the same gene occupy the same locus.

long-term potentiation (LTP) A physical remodeling of synaptic junctions that receive continuous stimulation.

lymphocyte A type of white blood cell that is involved in the adaptive immune response. There are two kinds of lymphocytes: T lymphocytes and B lymphocytes. T lymphocytes (T cells) mature in the thymus and attack invading microbes directly. B lymphocytes (B cells) mature in the bone marrow and make antibodies that are designed to immobilize or destroy specific microbes or antigens.

lysis The rupture of the cell membrane followed by death of the cell.

lysosome Membrane-bounded organelle of eukaryotes that contains powerful digestive enzymes.

macromolecule A very large molecule that is built from smaller molecular subunits. Common examples are DNA, proteins, and polysaccharides.

magnetic resonance imaging (MRI) A procedure in which radio waves and a powerful magnet linked to a computer are used to create detailed pictures of areas inside the body. These pictures can show the difference between normal and diseased tissue. MRI makes better images of organs and soft tissue than other scanning techniques, such as CT or X-ray. MRI is especially useful for imaging the brain, spine, the soft tissue of joints, and the inside of bones. Also called nuclear magnetic resonance imaging.

major histocompatibility complex Vertebrate genes that code for a large family of cell-surface glycoproteins that bind foreign antigens and present them to T cells to induce an immune response.

malignant Refers to the functional status of a cancer cell that grows aggressively and is able to metastasize, or colonize, other areas of the body.

mammography The use of X-rays to create a picture of the breast.

MAP-kinase (mitogen-activated protein kinase) A protein kinase that is part of a cell-proliferation-inducing signaling pathway.

M-cyclin A eukaryote enzyme that regulates mitosis.

meiosis A special form of cell division by which haploid gametes are produced. This is accomplished with two rounds of cell division but only one round of DNA replication.

melanocyte A skin cell that produces the pigment melanin.

membrane The lipid bilayer, and the associated glycocalyx, that surrounds and encloses all cells.

membrane channel A protein complex that forms a pore or channel through the membrane for the free passage of ions and small molecules.

membrane potential A buildup of charged ions on one side of the cell membrane establishes an electrochemical gradient that is measured in millivolts (mV). An important characteristic of neurons as it provides the electric current, when ion channels open, that enable these cells to communicate with each other.

mesoderm An embryonic germ layer that gives rise to muscle, connective tissue, bones, and many internal organs.

messenger RNA (mRNA) An RNA transcribed from a gene that is used as the gene template by the ribosomes, and other components of the translation machinery, to synthesize a protein.

metabolism The sum total of the chemical processes that occur in living cells.

metaphase The stage of mitosis at which the chromosomes are attached to the spindle but have not begun to move apart.

metaphase plate Refers to the imaginary plane established by the chromosomes as they line up at right angles to the spindle poles.

metaplasia A change in the pattern of cellular behavior that often precedes the development of cancer.

metastasis Spread of cancer cells from the site of the original tumor to other parts of the body.

methyl group (-CH₃) Hydrophobic chemical group derived from methane. Occurs at the end of a fatty acid.

micrograph Photograph taken through a light, or electron, microscope.

micrometer (μm or micron) Equal to 10⁻⁶ meters.

microtubule A fine cylindrical tube made of the protein tubulin, forming a major component of the eukaryote cytoskeleton.

millimeter (mm) Equal to 10⁻³ meters.

mitochondrion (plural mitochondria) Eukaryote organelle, formerly free-living, that produces most of the cell's ATP.

mitogen A hormone or signaling molecule that stimulates cells to grow and divide.

mitosis Division of a eukaryotic nucleus. From the Greek *mitos*, meaning “a thread,” in reference to the threadlike appearance of interphase chromosomes.

mitotic chromosome Highly condensed duplicated chromosomes held together by the centromere. Each member of the pair is referred to as a sister chromatid.

mitotic spindle Array of microtubules, fanning out from the polar centrioles and connecting to each of the chromosomes.

molecule Two or more atoms linked together by covalent bonds.

monoclonal antibody An antibody produced from a B cell–derived clonal line. Since all of the cells are clones of the original B cell, the antibodies produced are identical.

monocyte A type of white blood cell that is involved in the immune response.

motif An element of structure or pattern that may be a recurring domain in a variety of proteins.

M phase The period of the cell cycle (mitosis or meiosis) when the chromosomes separate and migrate to the opposite poles of the spindle.

multi-pass transmembrane protein A membrane protein that passes back and forth across the lipid bilayer.

mutant A genetic variation within a population.

mutation A heritable change in the nucleotide sequence of a chromosome.

myelin sheath Insulation applied to the axons of neurons. The sheath is produced by oligodendrocytes in the central nervous system and by Schwann cells in the peripheral nervous system.

myeloid cell White blood cells other than lymphocytes.

myoblast Muscle precursor cell. Many myoblasts fuse into a syncytium, containing many nuclei, to form a single muscle cell.

myocyte A muscle cell.

NAD (nicotine adenine dinucleotide) Accepts a hydride ion (H^-), produced by the Krebs cycle, forming NADH, the main carrier of electrons for oxidative phosphorylation.

NADH dehydrogenase Removes electrons from NADH and passes them down the electron transport chain.

nanometer (nm) Equal to 10^{-9} meters or 10^{-3} microns.

natural killer cell (NK cell) A lymphocyte that kills virus-infected cells in the body. It also kills foreign cells associated with a tissue or organ transplant.

neuromuscular junction A special form of synapse between a motor neuron and a skeletal muscle cell.

neuron A cell specially adapted for communication that forms the nervous system of all animals.

neurotransmitter A chemical released by neurons at a synapse that transmits a signal to another neuron.

non-small-cell lung cancer A group of lung cancers that includes squamous cell carcinoma, adenocarcinoma, and large cell carcinoma. The small cells are endocrine cells.

northern blotting A technique for the study of gene expression. Messenger RNA (mRNA) is fractionated on an agarose gel and then transferred to a piece of nylon filter paper (or membrane). A specific mRNA is detected by hybridization with a labeled DNA or RNA probe. The original blotting technique invented by E. M. Southern inspired the name.

nuclear envelope The double membrane (two lipid bilayers) enclosing the cell nucleus.

nuclear localization signal (NLS) A short amino acid sequence located on proteins that are destined for the cell nucleus after they are translated in the cytoplasm.

nucleic acid DNA or RNA, a macromolecule consisting of a chain of nucleotides.

nucleolar organizer Region of a chromosome containing a cluster of ribosomal RNA genes that gives rise to the nucleolus.

nucleolus A structure in the nucleus where ribosomal RNA is transcribed and ribosomal subunits are assembled.

nucleoside A purine or pyrimidine linked to a ribose or deoxyribose sugar.

nucleosome A beadlike structure, consisting of histone proteins.

nucleotide A nucleoside containing one or more phosphate groups linked to the 5' carbon of the ribose sugar. DNA and RNA are nucleotide polymers.

nucleus Eukaryote cell organelle that contains the DNA genome on one or more chromosomes.

oligodendrocyte A myelinating glia cell of the vertebrate central nervous system.

oligo labeling A method for incorporating labeled nucleotides into a short piece of DNA or RNA. Also known as the random-primer labeling method.

oligomer A short polymer, usually consisting of amino acids (oligopeptides), sugars (oligosaccharides), or nucleotides (oligonucleotides). Taken from the Greek word *oligos*, meaning “few” or “little.”

oncogene A mutant form of a normal cellular gene, known as a proto-oncogene, that can transform a cell to a cancerous phenotype.

oocyte A female gamete or egg cell.

operator A region of a prokaryote chromosome that controls the expression of adjacent genes.

operon Two or more prokaryote genes that are transcribed into a single mRNA.

organelle A membrane-bounded structure, occurring in eukaryote cells, that has a specialized function. Examples are the nucleus, Golgi complex, and endoplasmic reticulum.

osmosis The movement of solvent across a semipermeable membrane that separates a solution with a high concentration of solutes from one with a low concentration of solutes. The membrane must be permeable to the solvent but not to the solutes. In the context of cellular osmosis, the solvent is always water, the solutes are ions and molecules, and the membrane is the cell membrane.

osteoblast Cells that form bones.

ovulation Rupture of a mature follicle with subsequent release of a mature oocyte from the ovary.

oxidative phosphorylation Generation of high-energy electrons from food molecules that are used to power the synthesis of ATP from ADP and inorganic phosphate. The electrons are eventually transferred to oxygen to complete the process. Occurs in bacteria and mitochondria.

p53 A tumor-suppressor gene that is mutated in about half of all human cancers. The normal function of the *p53* protein is to block passage through the cell cycle when DNA damage is detected.

parthenogenesis A natural form of animal cloning whereby an individual is produced without the formation of haploid gametes and the fertilization of an egg.

pathogen An organism that causes disease.

PCR (polymerase chain reaction) A method for amplifying specific regions of DNA by temperature cycling a reaction mixture containing the template, a heat-stable DNA polymerase, and replication primers.

peptide bond The chemical bond that links amino acids together to form a protein.

pH Measures the acidity of a solution as a negative logarithmic function (p) of H^+ concentration (H). Thus a pH of 2.0 (10^{-2} molar H^+) is acidic, whereas a pH of 8.0 (10^{-8} molar H^+) is basic.

phagocyte A cell that engulfs other cells or debris by phagocytosis.

phagocytosis A process whereby cells engulf other cells or organic material by endocytosis. A common practice among protozoans and cells of the vertebrate immune system. (Derived from the Greek word *phagein*, "to eat.")

phenotype Physical characteristics of a cell or organism.

phospholipid The kind of lipid molecule used to construct cell membranes. Composed of a hydrophilic head-group, phosphate, glycerol, and two hydrophobic fatty acid tails.

phosphorylation A chemical reaction in which a phosphate is covalently bonded to another molecule.

photoreceptor A molecule or cell that responds to light.

photosynthesis A biochemical process in which plants, algae, and certain bacteria use energy obtained from sunlight to synthesize macromolecules from CO_2 and H_2O .

phylogeny The evolutionary history of an organism, or group of organisms, often represented diagrammatically as a phylogenetic tree.

pinocytosis A form of endocytosis whereby fluid is brought into the cell from the environment.

placebo An inactive substance that looks the same, and is administered in the same way, as a drug in a clinical trial.

plasmid A minichromosome, often carrying antibiotic-resistant genes, that occurs naturally among prokaryotes. Used extensively as a DNA cloning vector.

platelet A cell fragment, derived from megakaryocytes and lacking a nucleus, that is present in the bloodstream and is involved in blood coagulation.

ploidy The total number of chromosomes (n) that a cell has. Ploidy is also measured as the amount of DNA (C) in a given cell relative to a haploid nucleus of the same organism. Most organisms are diploid, having two sets of chromosomes, one from each parent, but there is great variation among plants and animals. The silk gland of the moth *Bombyx mori*, for example, has cells that are extremely polyploid, reaching values of 100,000C. Flowers are often highly polyploid, and vertebrate hepatocytes may be 16C.

point mutation A change in DNA, particularly in a region containing a gene, that alters a single nucleotide.

polyploid Possessing more than two sets of homologous chromosomes.

portal system A system of liver vessels that carries liver enzymes directly to the digestive tract.

probe Usually a fragment of a cloned DNA molecule that is labeled with a radioisotope or fluorescent dye and used to detect specific DNA or RNA molecules on Southern or northern blots.

promoter A DNA sequence to which RNA polymerase binds to initiate gene transcription.

prophase The first stage of mitosis. The chromosomes are duplicated and beginning to condense but are attached to the spindle.

protein A major constituent of cells and organisms. Proteins, made by linking amino acids together, are used for structural purposes and regulate many biochemical reactions in their alternative role as enzymes. Proteins range in size from just a few amino acids to more than 200.

protein glycosylation The addition of sugar molecules to a protein.

proto-oncogene A normal gene that can be converted to a cancer-causing gene (oncogene) by a point mutation or through inappropriate expression.

protozoa Free-living, single-cell eukaryotes that feed on bacteria and other microorganisms. Common examples are *Paramecium* and *Amoeba*. Parasitic forms are also known that inhabit the digestive and urogenital tract of many animals, including humans.

purine A nitrogen-containing compound that is found in RNA and DNA. Two examples are adenine and guanine.

pyrimidine A nitrogen-containing compound found in RNA and DNA. Examples are cytosine, thymine, and uracil (RNA only).

radioactive isotope An atom with an unstable nucleus that emits radiation as it decays.

randomized clinical trial A study in which the participants are assigned by chance to separate groups that compare different treatments; neither the researchers nor the participants can choose which group. Using chance to assign people to groups means that the groups will be similar and that the treatments they receive can be compared objectively. At the time of the trial, it is not known which treatment is best.

reagent A chemical solution designed for a specific biochemical or histochemical procedure.

recombinant DNA A DNA molecule that has been formed by joining two or more fragments from different sources.

regulatory sequence A DNA sequence to which proteins bind that regulate the assembly of the transcriptional machinery.

replication bubble Local dissociation of the DNA double helix in preparation for replication. Each bubble contains two replication forks.

replication fork The Y-shaped region of a replicating chromosome. Associated with replication bubbles.

replication origin (origin of replication, ORI) The location at which DNA replication begins.

respiratory chain (electron transport chain) A collection of iron- and copper-containing proteins, located in the inner mitochondrion membrane, that utilize the energy of electrons traveling down the chain to synthesize ATP.

restriction enzyme An enzyme that cuts DNA at specific sites.

restriction map The size and number of DNA fragments obtained after digesting with one or more restriction enzymes.

retrovirus A virus that converts its RNA genome to DNA once it has infected a cell.

reverse transcriptase An RNA-dependent DNA polymerase. This enzyme synthesizes DNA by using RNA as a template, the reverse of the usual flow of genetic information from DNA to RNA.

ribosomal RNA (rRNA) RNA that is part of the ribosome and serves both a structural and functional role, possibly by catalyzing some of the steps involved in protein synthesis.

ribosome A complex of protein and RNA that catalyzes the synthesis of proteins.

rough endoplasmic reticulum (rough ER) Endoplasmic reticulum that has ribosomes bound to its outer surface.

Saccharomyces Genus of budding yeast that are frequently used in the study of eukaryote cell biology.

sarcoma Cancer of connective tissue.

Schwann cell Glia cell that produces myelin in the peripheral nervous system.

screening Checking for disease when there are no symptoms.

senescence Physical and biochemical changes that occur in cells and organisms with age.

signal transduction A process by which a signal is relayed to the interior of a cell where it elicits a response at the cytoplasmic or nuclear level.

smooth muscle cell Muscles lining the intestinal tract and arteries. Lacks the striations typical of cardiac and skeletal muscle, giving it a smooth appearance when viewed under a microscope.

somatic cell Any cell in a plant or animal except those that produce gametes (germ cells or germ cell precursors).

somatic cell nuclear transfer Animal cloning technique whereby a somatic cell nucleus is transferred to an enucleated oocyte. Synonymous with cell nuclear replacement.

Southern blotting The transfer of DNA fragments from an agarose gel to a piece of nylon filter paper. Specific fragments are identified by hybridizing the filter to a labeled probe. Invented by the Scottish scientist E. M. Southern in 1975.

stem cell Pluripotent progenitor cell, found in embryos and various parts of the body, that can differentiate into a wide variety of cell types.

steroid A hydrophobic molecule with a characteristic four-ringed structure. Sex hormones, such as estrogen and testosterone, are steroids.

- structural gene** A gene that codes for a protein or an RNA. Distinguished from regions of the DNA that are involved in regulating gene expression but are noncoding.
- synapse** A neural communication junction between an axon and a dendrite. Signal transmission occurs when neurotransmitters, released into the junction by the axon of one neuron, stimulate receptors on the dendrite of a second neuron.
- syncytium** A large multinucleated cell. Skeletal muscle cells are syncytiums produced by the fusion of many myoblasts.
- syngeneic transplants** A patient receives tissue or an organ from an identical twin.
- tamoxifen** A drug that is used to treat breast cancer. Tamoxifen blocks the effects of the hormone estrogen in the body. It belongs to the family of drugs called antiestrogens.
- T cell (T lymphocyte)** A white blood cell involved in activating and coordinating the immune response.
- telomere** The end of a chromosome. Replaced by the enzyme telomerase with each round of cell division to prevent shortening of the chromosomes.
- telophase** The final stage of mitosis in which the chromosomes decondense and the nuclear envelope reforms.
- template** A single strand of DNA or RNA whose sequence serves as a guide for the synthesis of a complementary, or daughter, strand.
- therapeutic cloning** The cloning of a human embryo for the purpose of harvesting the inner cell mass (ES cells).
- topoisomerase** An enzyme that makes reversible cuts in DNA to relieve strain or to undo knots.
- transcription** The copying of a DNA sequence into RNA, catalyzed by RNA polymerase.
- transcriptional factor** A general term referring to a wide assortment of proteins needed to initiate or regulate transcription.
- transfection** Introduction of a foreign gene into a eukaryote cell.
- transfer RNA (tRNA)** A collection of small RNA molecules that transfer an amino acid to a growing polypeptide chain on a ribosome. There is a separate tRNA for amino acid.
- transgenic organism** A plant or animal that has been transfected with a foreign gene.

trans-Golgi network The membrane surfaces where glycoproteins and glycolipids exit the Golgi complex in transport vesicles.

translation A ribosome-catalyzed process whereby the nucleotide sequence of an mRNA is used as a template to direct the synthesis of a protein.

transposable element (transposon) A segment of DNA that can move from one region of a genome to another.

ultrasound (ultrasonography) A procedure in which high-energy sound waves (ultrasound) are bounced off internal tissues or organs producing echoes that are used to form a picture of body tissues (a sonogram).

umbilical cord blood stem cells Stem cells, produced by a human fetus and the placenta, that are found in the blood that passes from the placenta to the fetus.

vector A virus or plasmid used to carry a DNA fragment into a bacterial cell (for cloning) or into a eukaryote to produce a transgenic organism.

vesicle A membrane-bounded bubble found in eukaryote cells. Vesicles carry material from the ER to the Golgi and from the Golgi to the cell membrane.

virus A particle containing an RNA or DNA genome surrounded by a protein coat. Viruses are cellular parasites that cause many diseases.

western blotting The transfer of protein from a polyacrylamide gel to a piece of nylon filter paper. Specific proteins are detected with labeled antibodies. The name was inspired by the original blotting technique invented by E. M. Southern.

yeast Common term for unicellular eukaryotes that are used to brew beer and make bread. Bakers yeast, *Saccharomyces cerevisiae*, is also widely used in studies on cell biology.

zygote A diploid cell produced by the fusion of a sperm and egg.

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INDEX



Italic page numbers
indicate illustrations.

A

Abbe, Ernst 127
aberration
 chromatic 126–127, 128
 spherical 126–127, 128
absorptive cells 101, 102
acetylcholine (ACh) 121
actin 106, 107
actin filament 39, 43
adaptive immune response 110, 111
adenine 7, 9, 30, 64
adenosine diphosphate (ADP) 22
adenosine monophosphate (AMP) 22
adenosine triphosphate. *See* ATP
agarose gel electrophoresis 135, 136
a gene 84, 85
alanine 32
aldehydes 133
algae, green 95
alpha-actinin 107
Altman, Sidney 11
Alzheimer's disease xvi, xvii, 120
amino acid(s) 4
 coding for 31, 32
 as neurotransmitters 121–122
 in protein 7
 structure of 5, 6
ammonia 3–4
amoeba 92–95, 93, 94
AMP. *See* adenosine monophosphate
amphetamines 121
anabolism 21
anaphase
 in meiosis 71, 72, 73
 in mitosis 68, 69–70
“animalcules” 17
anthrax 35, 37
antibiotic-resistant bacteria 20
antigen 111
apochromatic objectives 127–129
Archaea 15, 16
arginine 32, 48
asparagine 32
aspartic acid 32
atmosphere xv, 3–4
ATP (adenosine triphosphate) 5, 22
 phosphate bonds of 22, 24

 production of 23, 24–26, 42
autotrophs 38
axon 113, 114
 connection between dendrite and 114, 118, 119
 myelination of 118–120, 120

B

BAC. *See* bacterial artificial chromosome
bacillus 18
Bacillus amyloliquefaciens 134
Bacillus anthracis 35, 37
bacteria. *See* prokaryote(s)
bacterial artificial chromosome (BAC) 76–78, 77
bacteriophage 137–139
*Bam*HI 134
basal cells 100, 101
base-flippers 88–89
base pair 9, 10
basophils 109
B cell (B lymphocyte) 109, 109
 β -galactosidase 84

big bang 1–2
 bile 104
 binary fission 67
 binocular microscope
 129, 132
 biotechnology xi–xii
 bioterrorism 37
 B lymphocyte. *See* B cell
 brain, neurons in 44,
 122, 124, 125
 brush-border cells. *See*
 absorptive cells

C

cadherin 97, 97
Caenorhabditis elegans
 78
 calcium 107
 cAMP. *See* cyclic
 adenosine monophos-
 phate
 cancer xvi, xvii
 cancer research 83–84
 CAP. *See* catabolite
 activator protein
 CapZ 106, 107
 carbon 2
 carboxyl group 5, 7
 cardiac muscle
 104–105, 107–108
 cartilaginous fish 93
 catabolism
 definition of 21
 enzymes for 24
 catabolite activator
 protein (CAP) 84, 85
 cDNA (complementary
 DNA) 139
 Cech, Thomas 11
 cell(s). *See also* specific
 cells
 classification of
 15–16, 16

human 98–108, 99
 molecules of 5–7, 6
 cell cycle 60–74, 61
 cell junctions 96–98,
 97
 cell membrane
 in eukaryotes 39
 in prokaryotes 18,
 19
 cell-to-cell communi-
 cation (cell signaling)
 xv, xvii, 15, 52–57
 cellular colonies 93, 95
 cellular metabolism
 21
 cellular slugs 92–95,
 93, 94
 cellulose 5
 cell wall 19, 27
 central nervous system
 (CNS) 121, 122
 centromere 67
 centrosome 39, 42,
 67–70, 68
 ceramide 50, 51
Chlamydomonas 95
 chromatic aberration
 126–127, 128
 chromatin 45, 47
 chromatin compaction
 45–47, 46
 chromosome(s). *See*
also meiosis; mitosis
 in eukaryotes
 40–41, 43, 44, 45,
 46, 47, 64, 67–74
 homologous 72
 human 45, 69, 72,
 73–74
 in prokaryotes 18,
 19, 20, 45, 64
 cilia 21, 42
 citric acid cycle. *See*
 Krebs cycle

Clark's solution 132,
 133
 clones 77, 78
 cloning xvi. *See also*
 DNA cloning
 cnidarians 93, 96
 CNS. *See* central
 nervous system
 cocaine 121
 coccus 18
 codon 30, 31, 32
 colonies, cellular 93,
 95
 complementary DNA.
See cDNA
 compound microscope
 126
 connective tissue 98,
 99
 connexin 97
 contig 77, 78
 contractile vacuoles
 42
 copper 2
 cuboidal basal cells
 100, 101
 cyanobacteria 33–34,
 34
 cyclic adenosine
 monophosphate
 (cAMP) 95
 cysteine 32
 cysts 35, 36
 cytochrome 23
 cytochrome oxidase
 23
 cytoplasm
 in eukaryotes 39,
 43
 histology of 133
 in prokaryotes 18,
 19, 20
 cytoplasmic bridges
 95, 96, 98

cytosine 7, 30
 coding for lysine 29
 deamination of 88,
 89
 pairing with guanine
 9, 64
 cytoskeleton 39, 40, 43

D

dandruff 100
 deamination 88, 89
 dendrite 113, 113–114,
 119
 dendritic cells 109, 111
 deoxyadenine triphos-
 phate (dATP) 139
 deoxy-cytosine
 triphosphate (dCTP)
 140
 deoxyribonucleic acid.
See DNA
 Department of Energy
 76
 depurination 88, 89
 desmosomes 96–98, 97
Dictyostelium 92–95,
 93, 94
 di-deoxynucleotides
 141, 141
 digestive tract, epithe-
 lium of 101–104, 102
 diploid 69, 70, 104
 disaccharides 9
 DNA (deoxyribonucleic
 acid) 4, 5
 auto-assembly of 9
 damage to 88–89,
 89
 detecting, with Feul-
 gen reaction 133
 noncoding 83
 replication of 9,
 62–67, 65

vs. RNA 28
 for storing genes
 28, 29
 structure of 7, 8, 62,
 63, 64
 DNA cloning 135–137,
 138, 139–140
 DNA helicase 64–66,
 65
 DNA labeling 139–140
 DNA library 137–139.
See also recombinant
 DNA technology
 DNA ligase 134–135,
 135
 DNA-modifying
 enzymes 134,
 134–135
 DNA polymerase 66,
 139, 140
 DNA primase 64–66,
 65
 DNA sequencing 140,
 140–142, 141, 142,
 143
 DOE. *See* Department
 of Energy
 dopamine 79, 121–122
 double helix 8, 9, 63

E

Earth
 creation of 2
 prebiotic xv, 2,
 2–15
 echinoderms 93, 96
E. coli. See Escherichia
coli
EcoRI 134, 137
 egg 70
 electron transport
 chain. *See* respiratory
 chain

embryo, development
 of 96
 endocrine system
 55–56
 endocytosis 59
 endogenous opioids
 122
 endoplasmic reticulum
 (ER) 39
 of epithelial cells
 101, 102, 104
 function of 41–42,
 49–52
 proteins sent to 43
 traffic in and around
 44, 59
 enzyme(s) 5
 for catabolism 24
 DNA-modifying
 134, 134–135
 extracting energy
 21
 lysosomal 59
 protein 25, 28, 29
 restriction 134, 135,
 137
 enzyme-linked receptor
 57
 eosin 133
 eosinophils 109
 epithelial cells
 100–104
 in liver 103, 104
 from pancreas 40
 epithelium 98, 99,
 100–104
 of digestive tract
 101–104, 102
 structure of 100,
 100–101
 ER. *See* endoplasmic
 reticulum
Escherichia coli (E. coli)
 19, 76, 134, 137

eukaryote(s) xv, 38–59
 acquiring energy 38
 cell-to-cell communication of 52–57
 duplication of
 genome in 64
 evolution of 15, 16,
 38–40, 58, 59
 gene expression control in 85–88, 86
 genes of 80–83, 81,
 82
 glycocalyx of 49–52,
 50, 51, 59
 lysosomes of 39, 42,
 58–59
 multicellular 92–111
 nucleus of 39,
 40–41, 43, 45–48
 protein synthesis in
 48–49
 reproduction of 60,
 67–74
 structure of 39,
 40–44
 symbionts of 58
 exon 80, 81
 exon shuffling 82, 82
 eyepiece, of microscope
 126, 127

F

fat(s) 4
 as building materials
 27
 as energy reserve
 7–9
 fatty acid(s) 5
 in phospholipids
 7–9
 structure of 6, 7
 Feulgen reaction 133

filipodia 100, 101
 filter hybridization 143
 flagellum (flagella) 21
 fluorescein 130
 fluorescent dye 129
 fluorescent in situ
 hybridization (FISH)
 145
 fluorescent microscope
 129–130, 131, 145
 formalin 132–133
 fruit fly, genes of 78

G

G₀ 60, 61
 G₁ 60, 61
 G₂ 60, 61
 G4 genome 76
 GABA 121
 galactose 84
 gamete 70
 Gap 1. *See* G₁
 Gap 2. *See* G₂
 gap junctions 97, 98
 gel electrophoresis 135,
 136
 gene(s) 31, 75–91
 eukaryote 80–83,
 81, 82
 prokaryote 79–80,
 80
 gene content 78
 gene expression, control of 84–88
 in eukaryotes
 85–88, 86
 in prokaryotes 84,
 85, 87
 gene organization
 79–80
 gene origins 78–79
 gene swapping 79

gene therapy 91
 genetic code 31, 32
 genetic recombination
 71, 72
 genome xvi, 20. *See*
 also Human Genome
 Project
 duplication of 64
 transposable 90–91
 genomic library
 137–139
 genomic research
 83–84
 germ cell 70
 glucose
 as building material
 27
 as source of energy
 22–24
 structure of 5, 6
 z gene coding for 84
 glucose transporters 25
 glutamate receptors 121
 glutamic acid 32
 glutamine 32, 121
 glycerol, structure of 5,
 6
 glycine 29, 32, 121
 glycocalyx
 of eukaryotes 42, 44,
 49–52, 50, 51, 59
 of prokaryotes
 25–26, 26
 glycogen 24, 27
 glycolipids 41–42, 50,
 51
 glycolysis 24, 25
 glycoproteins
 cell-surface 44
 in glycocalyx 25, 26,
 50, 51
 Golgi vesicles carrying 53

transport of 41–42, 51–52
 types of 52–56
 glycosylation 52
 goblet cells 101–103, 102
 Golgi complex (Golgi apparatus) 39
 of epithelial cells 101, 102, 104
 function of 41–42, 49–52
 proteins sent to 43
 traffic in and around 44, 59
 Golgi vesicles 39
 carrying glycoproteins 52, 53
 exocytosis of 118
 function of 42
 lysosomal enzymes in 59
 neurotransmitters in 119
 gonad 70
Gonium 93, 95
 G-protein 54–55, 56
 Gram, Hans Christian 32–33
 Gram stain 33
 granulocytes 108–111, 109, 110
 green algae 95
 guanine 7, 9, 30, 64
 guanosine triphosphate (GTP) 54

H

Haemophilus influenzae 134
 haploid 70, 73–74, 104
 helium 1–2

hemotoxylin 133
 heterotrophs 38
 hierarchical shotgun sequencing 76–78, 77
 high-resolution microscope 17
HindII 134
 histidine 32
 histochemistry 130–133
 histology 130–133
 histone 45, 46, 47
 human cells 98–108, 99
 human chromosomes 45, 69, 72, 73–74
 Human Genome Organization (HUGO) 76
 Human Genome Project 75, 76–83
 Huxley, Sir Andrew 107
 hybridization 9
 hydrogen 1–2
 hydrophilic headend 12, 13
 hydrophobic fatty acid tail 7–9, 12, 13
 hydrophobic hydrocarbon tail 7
 hydroxyl group (-OH) 7
 hypnosis 113

I

immune system, cells of 108–111
 immunizing serum 111
 innate immune response 109–111, 110

insertional mutagenesis 82–83, 90
 interphase 60, 61
 intron 80, 81, 139
 ion channels 54–56, 55
 ligand-gated 54, 55, 55, 114–118, 115, 117
 voltage-gated 54, 55, 55, 114–118, 116, 117
 ion pumps 100
 iron 2
 isoleucine 32

J

jumping genes 78, 90–91

K

keratin 100, 101
 keratinization 100, 101
 kinetochore 67, 69
 Krebs cycle (citric acid cycle) 23, 24–25

L

Lac operon 84, 85, 87
Lactobacillus 34
 lactose 25, 84
 lagging strand 66
 lambda bacteriophage 137–139
 Langerhans cells 109, 111
 leading strand 66
 Leeuwenhoek, Antonie van 17, 126
 lenses, of microscope 126, 127, 127

leucine 32
 life, origin of 1–16
 ligand 114
 ligand-gated ion channels 54, 55, 55,
 114–118, 115, 117
 light microscopy
 126–130, 127
 lipid bilayer 13–15, 14
 liver cells 103, 104
 lumen 96
 lymphocytes 108–111,
 109, 110
 lysine 29, 32, 48
 lysosomal enzymes 59
 lysosome 39, 42, 58–59

M

macromolecule(s)
 auto-assembly of 9
 structure of 7–9, 8
 macronucleus 42
 macrophages 108, 109
 major histocompatibility complex (MHC)
 110
 maltose 25
 MAO. *See* monoamine oxidase
 Margulis, Lynn 58
 maturation-promoting factor (MPF) 60–61
 McClintock, Barbara 90
 meiosis 70–74, 71
 meiosis I 71, 72
 meiosis II 72–74, 73
 melanocyte 100, 101
 membrane depolarization 117
 membrane hybridization 143, 144
 memory 112–113, 122
 Mereschovsky, Konstantin 58
 mescaline 121
 messenger RNA (mRNA) 5, 29, 30, 79
 as copy of gene 31, 43
 synthesis of 48–49
 metaphase
 in meiosis 71, 73
 in mitosis 68, 69
 methane 3–4
 methionine 32
 MHC. *See* major histocompatibility complex
 microscope(s)
 binocular 129, 132
 compound 126, 129
 fluorescent
 129–130, 131, 145
 high-resolution 17
 light 126–130, 127
 monocular 129, 130, 130
 for viewing prokaryotes 17, 20–21
 microtubules 39, 42, 69, 70, 118, 119
 microvilli 101
 Miller, Stanley 4
 “mitochondrial Eve” 58
 mitochondrion (mitochondria)
 ATP production by 23, 24–26, 42
 in eukaryotes 39, 41, 42
 origin of 58
 mitosis 67–70, 68
 mitotic spindle 68, 69

molecule(s) 4
 mollusks 93, 96
 Monera 15
 monoamine oxidase (MAO) 79
 monoamines 121
 monocular microscope 129, 130, 130
 monocytes 108–111, 109, 110
 monosaccharides 8, 9
 MPF. *See* maturation-promoting factor
 mRNA. *See* messenger RNA
 multicellularity 92–111
 muscle 98, 99, 104–108
 muscle contraction 107–108
 mutagenesis 82–83, 90
 myelin 119
 myelination 118–120
 myelin sheath 119, 120
 myofibril 105, 105
 myosin 106, 107
 myxobacteria 34, 34–36, 36, 92

N

NAD (nicotine adenine dinucleotide) 23
 NADH dehydrogenase 23
 National Research Council 76
 natural killer cell (NK cell) 108–109, 109
 natural selection 61
Neisseria gonorrhoeae 35

nervous system,
 evolution of 96
 nervous tissue 98, 99
 neural circuit 122–125,
 123
 neural stimulus 107
 neuron(s) xv–xvi,
 112–125
 anatomy of 113,
 113–114
 in brain 44, 122,
 124, 125
 importance of 112
 myelination of
 118–120, 120
 signal transduction
 in 114–118, 117
 synapse in 114, 114,
 118, 119
 neuropeptides 122
 neurotransmitters 117,
 118, 119, 121–122
 new biology xi–xii
 nicotine adenine dinu-
 cleotide. *See* NAD
 nitrogen 2
 NK cell. *See* natural
 killer cell
 NLS. *See* nuclear local-
 ization signal
 Nobel Prize, in medi-
 cine, for Barbara
 McClintock 90
 Nomarski's interference
 129
 noncoding DNA 83
 norepinephrine (NE)
 79, 121
 northern blotting
 143–145, 144
 nuclear localization
 signal (NLS) 48
 nuclear pore 47

nucleic acid(s)
 auto-assembly of 9
 carrying negative
 charge 135
 and protein 4–5
 structure of 7, 8
 nucleolus 39, 41, 47
 nucleosome 45
 nucleotide(s) 5
 base pairing 9
 damage to 88, 89
 linkage of 62,
 62–64, 64
 in nucleic acids 7
 structure of 6, 7
 nucleus 39, 43, 45–48
 components of
 40–41
 histology of 133
 traffic in 47

O

oceans, formation of
 2–3
 octoploid cells 104
 oligodendrocytes
 119–120, 120
 oligo labeling 140
 oligosaccharides 9
 operator 84, 85
 organelle xvi
 origin of life 1–16
 oxygen
 lack of 3
 need for 4

P

pancreas, epithelial cells
 from 40
 paneth cells 102,
 103–104

Paramecium candatum
 42
 Parkinson's disease xvi,
 120, 121–122
 PCR (polymerase chain
 reaction) 145–146
 peptide bond 7
 peripheral nervous
 system (PNS) 121
 permease 84
 peroxisome 39, 42
 phagocyte 108, 109
 phagocytosis 59,
 109–111, 110
 phase contrast 129
 phenylalanine 32, 121
 phosphate 5, 6
 phosphate bonds 22,
 24
 phosphokinase 45
 phospholipid(s)
 fats converted to
 27
 structure of 7–9, 8
 phospholipid bilayer
 19
 phospholipid bubbles
 12, 13
 photosynthesis 24, 38
 phytoplankton 33–34
 plasmid 18, 20
 PNS. *See* peripheral
 nervous system
 polyacrylamide gel
 135, 141, 141–142
 polymerase chain
 reaction. *See* PCR
 polysaccharides 7, 8,
 9
 portal system 104
 prebiotic Earth xv, 2,
 2–15
 prickle cells 100, 101

- prokaryote(s) xv,
17–37
 acquiring energy
 21–26, 33
 antibiotic-resistant
 20
 binary fission in
 67
 classification of
 32–37
 discovery of 17
 duplication of
 genome in 64
 evolution of 15, 16
 gene expression con-
 trol in 84, 85, 87
 genes of 79–80, 80
 genetic information
 of 28–32
 microscopy of 17,
 20–21
 mitochondria origi-
 nating with 58
 molecule model of
 20
 motile 21
 non-motile 21
 protein synthesis in
 19, 27, 28, 29
 repair in 26–27
 reproduction of 60,
 67
 scanning electron
 micrograph of 19
 structure of 18,
 19–21
prokaryote ribosomes
 19–20
proline 32
promoter 84–87, 85, 86
prophase
 in meiosis 71, 73
 in mitosis 67, 68
- protein(s)
 as building materials
 27
 and nucleic acids
 4–5
 structure of 7, 8
 synthesis of
 in eukaryotes
 48–49
 in prokaryotes
 19, 27, 28, 29
 ribosomes for
 19, 39, 42,
 48–49
 RNA for 27, 28,
 29
protein encoding genes
 79
protein enzymes 25,
 28, 29
protein plaque 96–97,
 97
protista 92
protoplasm. *See*
 cytoplasm
Providencia situarii
 134
pseudomonads 34, 34
PstI 134
purine 6, 7, 9, 88
Purkinje cell 122
pyrimidine 6, 7, 9
- R**
- random primer 140
receptors 54–55, 56,
 57
recombinant DNA
 technology xi–xii,
 xvi, 133–146
 DNA cloning in
 135–137, 138
- DNA-modifying
 enzymes in 134,
 134–135
DNA sequencing in
 140, 140–142, 141,
 142, 143
fluorescent in situ
 hybridization in
 145
gel electrophoresis
 in 135, 136
genomic and DNA
 libraries in
 137–139
labeling cloned
 DNA in 139–140
northern blotting in
 143–145, 144
polymerase chain
 reaction in
 145–146
 Southern blotting in
 142–143
regulatory proteins 85,
 87
replication 9, 10, 11, 44
replication fork 66
repressor protein 84,
 85
reproduction. *See also*
 cell cycle
 by binary fission 67
 of eukaryotes 60,
 67–74
 of prokaryotes 60,
 67
resolution 127–128
respiratory chain 23,
 24–25, 42
restriction enzyme
 134, 135, 137
reverse transcriptase
 11, 29

R group 5
 ribonucleic acid. *See* RNA
 ribose, structure of 5–7, 6
 ribosomal RNA (rRNA) 29, 31
 ribosomes 39, 42
 components of 48–49
 prokaryote 19–20
 ribozymes 11, 27, 28
 RNA (ribonucleic acid) 4. *See also* messenger RNA
 auto-assembly of 9
 vs. DNA 28
 in protein synthesis 27, 28, 29
 replication of 9, 10, 11
 ribosomal 29, 31
 for storing genes 27, 28, 29
 structure of 7, 8
 RNA (ribonucleic acid) world 11, 27–28
 roundworm, genes of 78
 Royal Society of London 17
 rRNA. *See* ribosomal RNA

S

Salmonella 35, 37
 Sanger, Fred 76, 140
 Sanger Institute 83
 Sanger sequencing reaction 140–142, 141
 sarcomere 106, 107

scanning electron micrograph (SEM) of bacteria 19
 of human chromosomes 69
 Schiff's reagent 133
 Schwann cell 119, 120, 120
 sebaceous glands 100
 self-hybridization 9
 SEM. *See* scanning electron micrograph
 serine 32
 serotonin 79, 122
 shorelines 11–12
 signal transduction 114–118, 117
 sister chromatids 67, 68
 skeletal muscle 105, 105–107
 slugs, cellular 92–95, 93, 94
 smooth muscle 105, 108, 108
 solar system, formation of 2
 somatic cell 70
 Southern, E. M. 142–143
 Southern blotting 142–143
 sperm 70
 S phase 60
 spherical aberration 126–127, 128
 spindle 60, 68, 69
 spirillum 18, 34
 sponges 93, 95–96
 spores 35, 36, 37, 92, 95
 squamous cells 100, 100–101

Staphylococcus 35, 37
 storms, on prebiotic Earth xv, 2, 2–4
Streptococcus 35, 37
 streptomycin 137
 substantia nigra 122
 sugar polymers 7, 8, 9
 sugars 4
 structure of 5–7, 6
 super cell 59
 symbiogenesis theory 58
 symbiosis 57–58
 synapse 114, 118
 synaptic gap 118, 119
 synaptic junction 114, 118, 119, 122

T

T cell (T-lymphocyte) 108–109, 109
 telophase
 in meiosis 71, 73
 in mitosis 68, 70
 TEM. *See* transmission electron micrograph
 tetraploid 104
 threonine 32
 thymine 7, 9, 30, 31, 64
 tight junctions 96, 97
 T-lymphocyte. *See* T cell
 transacetylase 84
 transcriptase 139
 transcription 29, 30, 44
 transcription factor 84, 85, 86, 87
 translation 28, 29
 transmission electron micrograph (TEM), of housefly brain 41

transport bubbles 44
 transporters 52–54
 transposable element
 (transposon) 90–91
 transposable genome
 90–91
 tryptophan 32
 tube, of microscope
 126, 127
 tyrosine 32, 121

U

University of
 California, Berkeley
 58
 University of
 Cambridge 76
 University of Chicago
 4

University of Colorado
 11
 uracil 7, 30, 31
 cytosine converted
 to 88, 89
 pairing with adenine
 9
 Urey, Harold 4
 Urey-Miller experiment
 3, 4

V

valine 32
 vegetative cells 35–36,
 36, 37
 vertebrates 93, 96
Vibrio cholerae 35
 virus, genome of
 90–91

voltage 116
 voltage-gated ion
 channels 54, 55, 55,
 114–118, 116, 117
Volvox 93, 95

W

white blood cells
 108–111, 109

Y

Yale University 11
y gene 84, 85

Z

Z disc 106, 107
z gene 84, 85